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Final SRS Report

Project : Digital Farming Assistant

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Table Of Contents

Contents	Page No.
Introduction.....	4
Purpose	4
Scope	5
Expected End –Users	5
Short User Story.....	6-8
Features & Subfeatures.....	9
Assumptions and Constraints	9
Feasibility Analysis	10
Summary of Stakeholder Meetings	12
App Review Summary: Analysis Of Related Projects.....	13
Survey Questionnaire.....	14-16
Project Charter	17
Requirements Modeling And Estimation	18
Works Breakdown Structure(WBS).....	18
WBS Description	19
Cost Estimation	20-22
Use Case Diagram	23
Diagram (Level-0)	24

Diagram (Level-1)	25-26
Diagram (Level-1.1)	27-28
Diagram (Level-1.2)	29-30
Diagram (Level-1.3)	31-32
Diagram (Level-1.4)	33-34
Diagram (Level-1.5)	35-36
Diagram (Level-1.6)	37-38
Diagram (Level-1.7)	39-40
Diagram (Level-1.8)	41
Activity Diagram	42
Diagram (Level-1)	43-44
Diagram (Level-1.1)	45-46
Diagram (Level-1.2)	47-48
Diagram (Level-1.3)	49-50
Diagram (Level-1.4)	51-52
Diagram (Level-1.5)	53-54
Diagram (Level-1.6)	55-56
Diagram (Level-1.7)	57-58
Swimlane Diagram	59
Diagram (Level-1.1)	59
Diagram (Level-1.2)	60
Diagram (Level-1.3)	61

Diagram (Level-1.4)	62
Diagram (Level-1.5)	63
Diagram (Level-1.6)	64
Diagram (Level-1.7)	65
Data Based Modeling	66
Data Object Identification	66-68
Relationship Between Data Objects.....	69-72
ER Diagram.....	73
Data Flow Diagram	74
Diagram (Level-0)	75
Diagram (Level-1)	76
Class Based Modeling	77
Noun List	78-80
Verb List	80-81
General Classification	82-84
Selection Criteria	85-86
Class Cards	87-100
CRC Diagram	101
ID - 1	101
ID - 2	102
ID - 3	103
ID - 4	104
ID - 5	105
ID - 6	106
ID - 7	107
ID - 8	108
ID - 9	109
ID - 10	109
ID - 11	110
ID - 12	111
ID - 13	112
ID - 14	113
ID - 14	113
ID - 14	113
Main CRC.....	114
Sequence Diagram	115-116

Introduction :

The Software Requirement Specification (SRS) for the Digital Farming Assistant outlines the functional and non-functional requirements of the system, serving as a formal agreement between stakeholders and the development team. This document introduces the project in the context of the agricultural market, where farmers increasingly rely on digital tools to improve productivity, reduce risks, and access timely information.

Agriculture remains the backbone of many economies, yet farmers often face challenges such as unpredictable weather, crop diseases, and limited access to expert advice. The Digital Farming Assistant is designed to bridge this gap by offering a mobile-based solution that integrates crop health monitoring, weather forecasting, crop knowledge libraries, and community support.

This SRS provides a structured design blueprint that ensures all stakeholders including farmers, agricultural experts, developers, and project sponsors share a common understanding of the system's objectives and features. By documenting requirements clearly, the SRS reduces ambiguity, supports efficient development, and ensures that the final product meets both user needs and market expectations.

Project Purpose :

Every day, farmers stand in their fields making decisions that shape their harvests and their livelihoods. Some choices are simple, but many are uncertain — when to water, what to plant, how to protect crops from disease, or whether the weather will favor their efforts. The Digital Farming Assistant was created to stand beside them in these moments, not as a complicated tool, but as a trusted companion.

Instead of struggling with complex logins or unfamiliar systems, farmers can enter with just their name, phone number, and preferred language technology brought closer to them in the simplest way possible. When crops show signs of trouble, they can capture an image and receive clear health checks and treatment suggestions, protecting both their harvest and their livelihood.

As the skies change, the assistant delivers real-time forecasts and farm planning advice, guiding farmers on when to irrigate, when to harvest, and even which crops might thrive next season. Knowledge is no longer locked away in books or distant experts it flows openly through a crop library and a community forum, where farmers share experiences, ask questions, and learn from one another. Also, because farming often happens far from strong internet connections, the assistant works offline too — saving crop profiles and queuing images until the network returns.

In short, the Digital Farming Assistant is more than an app. It is a partner in the farming journey, helping farmers save time, reduce risks, and grow with confidence, every single day.

Project Scope :

The Digital Farming Assistant aims to empower farmers with smart tools, but it operates within clear boundaries. It won't support offline or SMS access, advanced equipment like drones, financial services, or multilingual features beyond the primary language. Personalized advice based on soil or location is also out of scope. The project targets a 6–9 month timeline, focusing first on core features like crop health detection, weather forecasts, and a crop library, with the community forum arriving later. Due to limited time, user testing will be minimal. Budget constraints mean relying on open-source tools and free-tier services, with limited spending on cloud infrastructure, premium APIs, and expert content creation. Technically, crop health accuracy depends on image quality and training data, while weather forecasts rely on third-party APIs. Forum moderation will be basic due to resource limits. Despite these constraints, the assistant is designed to deliver meaningful support to farmers through a streamlined, data-driven platform.

Expected End Users:

- ❖ Local farmers (small to medium-scale) and agricultural extension officers
- ❖ Agronomists, crop experts, Students and researchers in agriculture

Short User Story :

A dedicated Farmer, logs into the Digital Farming Assistant using his Account ID. His Name, Phone Number, and Language Preference were set during Registration Date, and his Authentication Method is simple , just simple password, and his phone number. His Profile Image appears on the dashboard, and his Offline Access Enabled flag ensures he can use the app even in low-connectivity zones. His Location ID is synced to provide localized insights, and his Crop history, Diagnosis history, Forum Posts, and Saved Crop **Profiles** are stored under his Farmer ID.

In the background, the Admin team ,identified by Admin ID manages the system. Each admin has a Name, Email, and Phone Number, and their Role (e.g., Moderator or SuperAdmin) defines their access level. They monitor System Update Logs, track Authentication Logs, handle Forum Moderation Actions, and manage User Management Actions. Their Last Login timestamp ensures accountability.

Rafiq heads to his field and captures an Image of his crops using his smartphone. The image is tagged with an Image ID, linked to his FarmerID, and includes a Timestamp, File Path, and Crop Type. The system detects Symptoms Detected and stores the Field Location. Since the farmer is offline, the Upload Status is marked as “queued” and the Offline Flag is set to true.

The Image Analysis Service, part of the AI Assistant, processes the image. The assistant, identified by Assistant ID, uses Tool Used (e.g., ML model) to generate a ResponseContent with a ConfidenceScore. The Service Type is “ImageAnalysis,” and the Request ID links the session. The result is tied to a Linked Diagnosis ID.

A new Diagnosis is created with a unique DiagnosisID, linked to the Image ID, Farmer ID, and Crop ID. It detects a Disease — say, a fungal rot and records the Disease Detected, Severity Level, Accuracy Score, and Description. The Timestamp logs the moment of analysis, and the system provides a Recommendation. An Agronomist, identified by Agronomist ID, later verifies the result. Their Name, Phone Number, Expertise Area (e.g., pest control), and Verified Status are stored. They add Recommendations Given and **Forum Comments** to the system. Their Profile Image, Location, and Language Preference help tailor their interface.

The Disease entity includes its Name, Symptoms, SeverityScale, AffectedCrops, and TreatmentOptions. The system flags a Pest Outbreak Flag, estimates the Rot Percentage Range, and logs the Detection Accuracy and Prevention Tips.

Meanwhile, the Weather API feeds Weather Data into the app. Each record has a Weather Data ID, Location, Date, Temperature, Humidity, Rainfall, Wind Speed, and Forecast Type (daily/weekly). The Source is OpenWeatherMap, and each entry has a Timestamp. If a storm is

predicted, a Weather Alert is triggered with an AlertType, and a Notification is sent. The Notification ID, Recipient ID, Type (e.g., WeatherAlert), Message Content, Timestamp, ReadStatus, and Delivery Method (App, Email, SMS) are all logged.

The AI Assistant also provides Guidance on Cultivation. Rafiq's Cultivation ID links to his Farmer ID and Crop ID, with attributes like Start Date, End Date, Field Location, Irrigation Schedule, Harvesting Date, Soil Type, and Yield Estimate. The assistant helps him Schedule irrigation based on soil moisture and forecast data.

Curious about new crops, Rafiq explores the Crop Library. Each Crop has a Crop ID, Name, Season, Soil Type, Growth Duration, Common Diseases, Recommended Fertilizers, Recommended Pesticides, Tips, and an Image Reference. He filters using Keyword Tags and views Crop Profiles with Profile ID, Planting Method, Soil Requirement, Fertilizer Recommendation, Pesticide Recommendation, Field Layout Guide, and Offline Saved Flag. The Last Viewed Date and Created By (Farmer or Agronomist) are tracked.

In the evening, Rafiq joins the Community Forum. He creates a Post ID with an Image Attached, writes Post Content, and adds a Timestamp. Other farmers and agronomists leave Comments, click Likes, and reply. Each post is tagged with KeywordTags, marked with a Searchable Flag, and linked to a Forum ID.

Behind the scenes, the Authentication system logs each Auth ID, User ID, Phone Number, OTP Status, Last Login Timestamp, Device Info, Language Preference, Forgot Password Flag, and Auth Method.

By the end of the day, Rafiq feels supported. His Account — with its Account ID, User Type, Name, Phone Number, Language Preference, Registration Date, Profile Image, Last Login, Offline Access Enabled, and Authentication Method — connects him to a system that understands his needs.

Features and sub features	Description
Login System Sub-features includes : i) name ii) phone number iii) preferred language	This is our first and small features and Simple authentication using name and phone number only (no email and password, no otp complexity). Also, they can choose the language they want to use.
Crop health Checker Sub-features includes : i) Image Capture & Upload ii) Diagnosis Summary iii) Offline Image Queue iv) Diagnosis History	Uses trained machine learning models to analyze crop images and detect the percentage of rot, pest attack, or disease, then provides recommendations for treatment or care. They can click the pictures before and upload it in the offline image queue because of internet connectivity and also check previous images and feedbacks in history.
Weather Forecast Sub-features includes : i) daily and weekly forecast view ii) Location-Based Forecast iii) Farm Planning advisory	Displays real-time and weekly weather forecasts using integrated APIs(Open Weather Map) which will help farmers decide irrigation or harvesting schedules. Also, weather alerts (heavy rainfall,storm) will be there through notifications. Location-based forecast details will also be there to give accurate insights to farmers and Farm Planning advisory based on the upcoming weather suggest which crop will be better to grow now or should crop be given water, irrigation advices etc
Crop Library Sub-feature includes : i) Filter-Based Search ii) Crop Profiles (Planting methods,Soil requirements,recommend fertilizers etc) iii) Offline Access	A searchable database of crops with planting methods, soil requirements, fertilizers, and pest control guidelines., filter-based search by seasons and soils, moreover saving option for most viewed crops which enables offline accessibility.

Community Forum Sub-feature includes : i) post with image ii) comment iii) like iv) search questions by keyword	A discussion board where farmers can ask questions, share experiences, and get expert suggestions. Here farmers can post anything with image, comment section is also here where they can receive guidance by the fellow farmers, like system is here to show the relevant feedback and search by keyword option helps to find existing problem and its solutions.
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Assumptions and Constraints

Assumptions: In this project, it is assumed that most users have smartphones or at least a basic internet connection. This allows them to access the application and its online features easily. Reliable weather API services are also available to provide accurate and timely weather updates for the farmers. It is further assumed that farmers are able to capture clear images of their crops using their mobile phones, which will help the AI system detect crop diseases or problems effectively. The initial AI model is trained with a diverse and regionally relevant dataset, ensuring that the system can recognize common crops and conditions found in different areas.

Constraints: Despite the advantages, there are several limitations. The AI system may face reduced accuracy for certain crops due to limited training data. The application depends on an internet connection for real-time weather updates and access to the community forum, which may be difficult in areas with poor connectivity. Another major constraint is the limited technical literacy among rural farmers, which may affect how quickly they adopt and use the technology. Finally, storing large numbers of user-uploaded crop images could become difficult, so cloud storage integration will be necessary to manage data efficiently.

Feasibility Analysis:

1. Legal Feasibility: The system is designed to comply with national and international data protection and privacy laws. It collects only minimal personal information, such as the user's name and phone number, ensuring user confidentiality. Data is securely stored using encrypted databases, and user consent is taken before any data collection. Additionally, the project adheres to ethical standards related to AI usage and data sharing, reducing any risk of legal complications.

2. Technical Feasibility: The proposed system is technically feasible using current and accessible technologies. The AI and machine learning components can be implemented using Python frameworks such as TensorFlow or PyTorch. The mobile application can be developed using Flutter or React Native, allowing compatibility across Android and iOS devices. For storage and real-time data management, Firebase or MongoDB can be integrated as cloud databases. Weather and image recognition APIs are readily available and can be connected to the system for accurate predictions and analysis. Proper testing and optimization will ensure stable performance even in low-network areas.

3. Scheduling Feasibility: The Digital Farming Assistant is feasible within a **6–7 month development cycle**, provided the project is efficiently planned and executed. The extended timeline allows for deeper feature integration, better testing, and user feedback incorporation. A proposed schedule is as follows:

Month 1: Requirement Gathering & System Design Define functional and non-functional requirements, finalize tech stack, and design system architecture and UI wireframes.

Month 2-3: AI/ML Model Development & Mobile App Interface Creation collect data for crop health detection models and develop core mobile app screens including login, image upload, and dashboard.

Month 4: API Integration & Database Setup Integrate OpenWeatherMap and other APIs, build crop library backend, and set up secure data storage.

Month 5: Community Forum & Language Support Implementation Develop posting, commenting, and liking features; integrate Bangla and English language support and chatbot assistant.

Month 6: Testing, Debugging & Optimization Conduct unit, integration, and usability testing; fix bugs and optimize performance across devices.

Month 7: Final Deployment & User Feedback Collection Launch MVP, onboard test users, and gather feedback for future improvements.

4. Economic Feasibility: The cost of developing and maintaining the system is relatively low compared to the benefits it offers to farmers and agricultural communities. Most of the tools and platforms used (Python, Flutter, Firebase) are open-source or have affordable plans. The project can also attract funding or sponsorship from government agricultural departments, NGOs, or agro tech companies. In the long run, the system can generate value by improving crop yield predictions and reducing losses, making it a cost-effective investment. Hence, we estimated that about 1.2 lakh cash will be needed to provide a smooth and effective test phase for our project. With the available budget, the project can focus on building a strong foundation rather than a fully advanced system. A portion of the funds will be used to collect and label an initial image dataset of common crops showing both healthy and diseased conditions — a key investment for future AI improvements. The chatbot will go beyond basic Q&A, incorporating guided logic from the crop library to help diagnose crop issues through interactive questioning. A simple prototype AI model may also be developed using transfer learning for one crop and one disease as a proof of concept, though not ready for full deployment. Additional resources will be dedicated to improving the user interface for better accessibility and ensuring a reliable backend infrastructure to support stable performance of the community forum, crop library, and user data management.

5. Operational Feasibility: The system can operate efficiently in real-world conditions with minimal technical support. It is user-friendly and designed to function smoothly on basic smartphones with internet access. Local language options can be added to improve accessibility for rural farmers. The platform's maintenance can be handled by a small technical team, and cloud-based systems will ensure data availability and reliability. With proper training and awareness, farmers can easily adopt the solution, leading to long-term operational success.

Summary of Stakeholder Meetings and Engagement Strategy :

As part of our stakeholder engagement strategy, we launched an outreach campaign to academic experts across Bangladesh. We emailed over 40 professors, associate professors, and assistant professors from various agricultural universities. Our aim was to invite informed voices to contribute to the upcoming project meeting. Dr. Swapon Kumar Roy from IUBAT responded positively and showed interest. Rajesh Chakrabarty and Abdul Awal Chowdhury from Sher-e-Bangla Agricultural University also replied with enthusiasm. Their engagement added meaningful value to our initiative. We extended invitations to research bodies as well, including the Computer & GIS Unit of BARC. They responded with interest, aligning well with the technical aspects of our project. However, time constraints and scheduling conflicts limited our options. After evaluating availability and relevance, we chose to proceed with Rajesh Chakrabarty sir for an online meeting. His expertise and responsiveness made him an ideal participant. This outreach helped us identify key stakeholders for collaboration. We view this as a foundation for future engagement.

App Review Summary: Lessons from Related Projects

As part of our research and development, we analyzed four prominent agricultural apps in Bangladesh: Krishi Apps, Khamari, iFarmer, and Cropwise Grower. The objective was to evaluate existing solutions and identify features relevant to our project. Krishi Apps focus on government services for farmers, while Khamari assists with crop selection and fertilizer recommendations based on land conditions. iFarmer operates as a market-driven platform, connecting farmers with buyers and supporting agricultural commerce. Cropwise Grower emerged as the most aligned with our concept due to its image-based diagnostics, allowing users to photograph affected plants and receive treatment suggestions. This feature bridges the gap between field observation and expert advice. However, the app lacks a community forum for agricultural stakeholders to share insights and collaborate. Its diagnostic database is limited, primarily supporting rice, with inadequate coverage for crops like wheat, potato, and chili pepper. We also encountered login issues, particularly with OTP verification, which may affect user accessibility. Despite its strengths, Cropwise Grower requires broader crop support, enhanced user experience, and community-driven features. These findings will inform the development of our own solution, tailored to the diverse needs of Bangladesh's agricultural sector.

SURVEY QUESTIONNAIRE

1. Consider our target users are farmers, From your experience and analysis what should be core feature which cannot be ignored at any cost for our mobile app?

→ Precised Weather forecast system

2. For Crop security and succesfull crop product which will be better between day to day and weekly weather forecast?

→ If a crop is 70% grown and heavy rain is expected within 15 days, weekly forecasts help plan the harvest. Daily forecasts, meanwhile, reveal if it'll rain today, allowing the crop to grow as much as possible before the storm. That's why both daily and weekly weather updates are crucial.

3. How much does the humidity,wind speed or in general temperature information matters for farmers?

→ Massive wind speed may uproot young paddy plants, hamper soil quality, as well as because of high humidity crops are affected by fungal blight diseases.

4. Can you give us idea about a crop library (details showing), what kind of details can we show?

→ Details about the most cultivated crops, soils in different sectors of bangladesh (30 sector) , pesticide, fertilizer etc

5. As you said, we should focus on the most cultivated crops , what kind of details can we share there?

→ Brieft discussion on most cultivated crops, their seasons, soil they can be cultivated at, diseases, proper and accurate guideline to cultivate them.

6. So, as you stated about the seasons and soil, so we can use them as filters for farmers to search for crops?

→ Use seasons,soil as well as cultivation technique to filter crop search, use icons if possible to make farmers like easier. Also in the guideline is arificial irrigation techniques.

7. You also mentioned about ‘artificial irrigation and precision agriculture’ how can we implement these details in our app’s crop library?

→ This is not my expertise but based on my experience I would say – “Use perfect and accurate guidelines for irrigation, focus on quality over quantity, keep the tradeoff as much balanced as possible. Attach detailed information from recent proved research “.

Before any other query we shared our interest which is including a feature where we will implement AI/ML integration for detecting crop rotting percentage and eventually generate a proper feedback.

8. Sir, we know this is not the area of your expertise but based on our interest, what do you think about this feature? Can we successfully implement it in our app?

→ The implementation is quite possible but the accuracy of detecting both rotting and pure crop should be high. As I earlier stated precision is the key and here try a few more combinations other than rice.

9. So, as you mentioned about accuracy and detailing, we should go beyond common diseases?

→ Not really, as common diseases should be the primary concern here but don’t work on only one crop as others do. Keep variety if possible. Example : potatoe’s blast disease, leaf rust etc.

10. As, you have pointed out about a ill crop influencing others, What threshold of infection should we recommend for immediate elimination?

→ No one can answer that question, it depends on many factors. but as Diseases like blast spreads faster, so other crops will get affected soon, so here the measures should be taken as fast as possible. It depends on the crop, its ability, disease, weather, health etc.

11. How can the feature of marketplace can help users?

→ Yes, it can help the user but I would recommend you to use if only you are confident and can actually provide the details about most accurate prices. Usually it is not possible in Bangladesh due to dishonest brokers, hidden cost (charging extra fare from delivery to everything). So eventually farmers won't get the most accurate news.

12. What if we can implement info like price of crops and a forecast of price drop or rise in our app ?

→ Usually it is not possible to show accurate prices in Bangladesh due to dishonest brokers, hidden cost (charging extra fare from delivery to everything). So eventually farmers won't get the most accurate price for their crops as it is a chain system, everyone wants to get benefitted, even if there is direct link between market and farmers, there will be hidden cost in between.

13. We saw other apps has login features as like user have to put an email and password, so should we add this feature in our app?

→ Definitely, you can use it, but try to keep it as simple as you can. Otherwise farmers will find it difficult to sign up

14. I personally analyzed few apps that has some unique features like app suggesting users what type of product he should focus, what is the profitable product this year, is this necessary to implement those features?

→ For your basic app system, as it's your first time, you should not get into that, if you have that big of a data analyst team then you guys can try. Otherwise, I would say not to get into that sector so early.

Project Charter :

Project Charter			
Project Title	Digital Farming Assistant		
Start Date	6/10/2025	Project Manager	Mashrul Hossain
End Date	18/5/2026	Sponsor	East West University
Project Description And Need			
The Digital Farming Assistant aims to enhance agricultural efficiency and productivity. It will help increase crop yield by providing timely weather forecasts to alert farmers about upcoming natural disasters, and by offering guidance to protect crops from various threats. Also by the integration of AI and ML , users would be able to detect severity of rotten crops and will receive guidance based on that. In addition, the system will include a community forum where farmers can share their problems, experiences, and creative ideas with others and knowledge exchange within the farming community.			
Project Scope		Risk And Issues	
1. Exclude offline or SMS-based access, advanced equipment integration (e.g., drones), and financial or trading services. 2. Limit features to one primary language without multilingual support. 3. Complete development within a 6–9 month timeframe, prioritizing essential features first. 4. Utilize open-source tools and free-tier services due to budget constraints. 5. Conduct limited user testing because of time and resource limitations.		1. Unauthorized access or data breach due to weak security in the login system. 2. Limited accuracy of agricultural and weather data. 3. Low user adoption due to limited digital literacy among farmers. 4. Poor internet connectivity in rural areas.	
Deliverables		Assumptions	
1. Prepare the system design and development plan for the Digital Farming Assistant. 2. Development of the application with all core modules (weather, crop recommendation, community forum, and login system) 3. Testing and debugging of the system to ensure functionality and reliability. 4. Preparation of user manual and project documentation.		1. All farmers will have access to smartphones and internet connectivity. 2. All users will actively use the system for agricultural guidance and information sharing. 3. All weather and agricultural data sources will remain accurate and regularly updated. 4. All farmers will understand and use the system primarily in Bangla language. 5. All system modules (forecasting, crop recommendation, and forum) will function as intended during the project timeline.	
Budget			
420,000			
Milestone	Target Date of completion		Actual Date
Project prototype	10/5/2026		18/5/2026
HR Consultants	Dr. Rajesh Chakraborty	Sponsor	East West University
Team Members	Team Name	chasha-e-tech	
	Utsab Roy , Mashrul Hossain , Fahim Shahriar , MD. Afridi , Mazharul Islam Mridul		

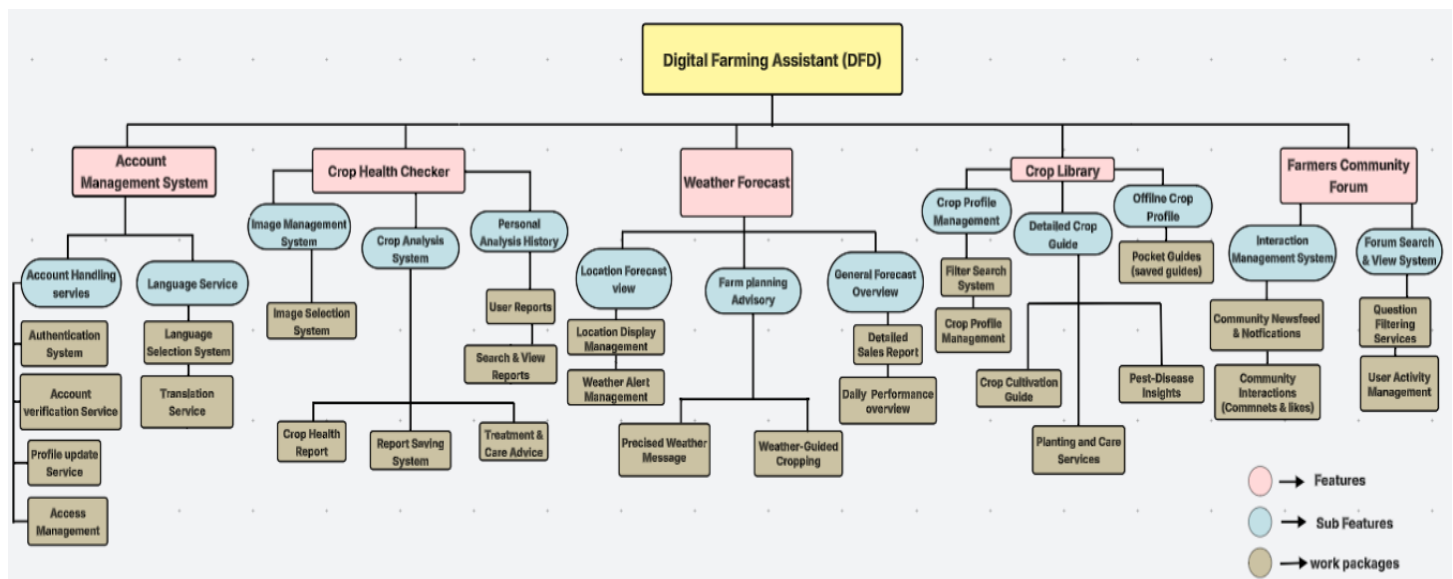
Requirements Modeling :

Works Breadown Structure Diagram (WBS)

A Work Breakdown Structure (WBS) diagram is a visual representation that organizes a project into smaller, more manageable components. It takes the overall scope of work and breaks it down into hierarchical levels, starting from the main deliverable and moving into sub-deliverables, tasks, and work packages. By showing this structure in a diagram, it becomes easier to see how different parts of the project connect, how responsibilities can be assigned, and how progress can be tracked.

The purpose of a WBS diagram is to provide clarity and control over complex projects. It helps everyone involved understand exactly what needs to be done, how the work is divided, and where their role fits within the bigger picture. It reduces ambiguity by defining tasks clearly, supports better estimation of time and resources, and ensures that nothing important is overlooked. For stakeholders, it acts as a communication tool that aligns expectations, while for project managers, it serves as a roadmap to plan, monitor, and deliver the project successfully. In essence, the WBS diagram transforms a large, overwhelming project into a structured plan that is easier to execute and manage.

WBS diagram :



WBS Description :

The Work Breakdown Structure (WBS) for the Digital Farming Assistant organizes the system into five major functional areas, each representing a core feature of the mobile application. These features are broken down into sub-features and further into work packages, allowing for a clear and manageable view of the project's scope and deliverables.

The first major feature is the Account Management System, which handles user access and personalization. It includes two sub-systems: Account Handling Services and Language Service. Account Handling covers the Authentication System, Account Verification, Profile Update, and Access Management, ensuring secure and user-friendly login and profile control. The Language Service includes Language Selection and Translation Service, allowing users to interact with the app in their preferred language, which is especially important for rural accessibility.

Next is the Crop Health Checker, a feature that empowers farmers to monitor crop conditions using AI. It consists of three sub-systems: Image Management, Crop Analysis, and Personal Analysis History. The Image Management System includes Image Selection, enabling users to capture or upload crop images. The Crop Analysis System generates a Crop Health Report using machine learning models. Personal Analysis History allows users to access User Reports, Search & View Reports, receive Treatment & Care Advice, and use the Report Saving System for offline access and future reference.

The third feature is the Weather Forecast module, which supports farm planning through real-time weather data. It includes Location Forecast View, Farm Planning Advisory, and General Forecast Overview. Location Forecast View manages Location Display and Weather Alerts. Farm Planning Advisory provides Precised Weather Messages and Weather-Guided Cropping suggestions. The General Forecast Overview includes a Detailed Sales Report and Daily Performance Overview, helping farmers align their activities with weather trends.

The Crop Library is the fourth feature, offering a rich database of crop-related knowledge. It includes Crop Profile Management, Detailed Crop Guide, and Offline Crop Profile. Crop Profile Management supports Filter Search and profile access. The Detailed Crop Guide includes Crop Cultivation Guide, Planting and Care Services, and Pest-Disease Insights. Offline Crop Profile enables Pocket Guides, allowing farmers to save and access crop information without internet connectivity.

Finally, the Farmers Community Forum fosters collaboration and peer support. It includes Interaction Management System and Forum Search & View System. Interaction Management handles the *Community Newsfeed*, *Notifications*, and Community Interactions such as *Comments* and *Likes*. The Forum Search & View System includes Question Filtering Services and User Activity Management, helping users find relevant discussions and track their contributions. Together, this WBS provides a comprehensive blueprint for development, ensuring that each feature is broken into actionable components. It supports clear task assignment, progress tracking, and stakeholder alignment, making the Digital Farming Assistant both technically robust and user-centric.

Cost estimation of DFD:

Our project time range is 6 months.

The team consists of approximately five Developers, for each developer paid per month 15000tk.

totaling 75,000 BDT per month for salary.

The estimated annual cost for Domain, cloud hosting, and development tools is around 60,000 BDT.

So, for each month $(60,000 \div 12) = 5,000$ BDT

For our project time range we need,

Cost of Development tools = 35,000 BDT

The overall project cost is approximately 465,000 BDT.

The average developer salary per day is about 500 BDT.

$$\text{Three point Average} = \frac{\textit{Pessimistic} + \textit{Most Likely} + \textit{Optimistic}}{3}$$

$$\text{Pert Estimation} = \frac{\textit{Pessimistic} + 4 \textit{ Most Likely} + \textit{Optimistic}}{3}$$

Sl. no.	Work package	Optimistic	Realistic (Most Likely)	Pessimistic	3 points estimation	PERT estimation
1	Sign up	12,000	15,000	20,000	15666.67	15333.33
2	User's profile	12,000	16,000	19,000	15666.67	15833.33
3	Account Verification	20,000	24,000	27,500	23833.33	23916.67
4	Choose language	6,000	7,000	9,000	7333.33	7166.67
5	Translate text	10,000	12,000	16,000	12666.67	12333.33
6	Upload Image	7,000	9,000	10,000	8666.67	8833.33
7	Pick from Gallery	7,000	10,000	13,500	10166.67	10083.33
8	View crop health report	22,000	27,000	33,500	27500	27250
9	Add to My Records	5,000	8,000	10,000	7666.67	7833.33
10	Treatment and Care Advice	15,000	17,000	20,000	17333.33	17166.67
11	User reports	7,500	9,500	10,000	9000	9250
12	Search and View reports	11,000	13,000	16,000	13333.33	13166.67
13	My Farm location	13,500	15,000	17,000	15166.67	15083.33
14	Weather alerts	14,000	15,000	20,000	16333.33	15666.67
15	Precise weather message	10,500	12,000	13,000	11833.33	11916.67
16	Weather-Base Crop Planning	15,000	19,000	24,500	19500	19250
17	Today's weather data	13,000	15,500	19,000	15833.33	15666.67

18	Farmer-Focused Weather Data	10,000	11,000	15,000	12000	11500
19	Search and Filter	7,500	8,500	10,000	8666.67	8583.33
20	Crop Profile	15,500	18,000	20,000	17833.33	17916.67
21	Full Crop Cultivation Guide	10,000	13,000	17,000	13333.34	13166.67
22	Planting and Care Instructions	10,000	14,000	17,000	13666.67	13833.33
23	Common Pests & Diseases	15,000	17,500	20,000	17500	17500
24	Pocket guide (Saved guides)	16,500	20,000	23,000	19833.33	19916.67
25	Community News Feed & notifications	21,000	23,000	24,500	22833.33	22916.67
26	Community Interactions (Comments & Likes)	14,000	16,000	19,500	16500	16250
27	Search Questions	5,000	7,000	8,500	6833.34	6916.67
28	My Activity	13,000	15,500	17,000	15166.67	15333.33
	Sum:	338000	407,500	489,500	411,666.68	409,583.34

Use Case Diagram :

A use case diagram is a visual representation in system analysis that shows how different users, known as actors, interact with a system to achieve specific goals. It belongs to the Unified Modeling Language (UML) and focuses on the functional requirements of a system rather than its technical details. In the diagram, actors are placed outside the system boundary, while use cases—depicted as ovals—are inside, connected by lines that illustrate interactions.

The primary purpose of a use case diagram is to capture and communicate what the system should do from the user's perspective. It defines the scope of the system by clarifying which functions are included and which are not. This makes it easier for stakeholders, both technical and non-technical, to understand the system's intended behavior. By presenting requirements in a simple, intuitive way, the diagram bridges communication gaps between developers, analysts, and clients.

Beyond communication, use case diagrams serve as a foundation for system design and testing. They help identify actors, their roles, and the tasks they need to perform, which is essential for planning security, access control, and workflow. They also guide developers in structuring features and provide testers with clear scenarios to validate. In short, a use case diagram is a high-level map of system functionality, ensuring that everyone involved shares a common understanding of how the system will support its users.

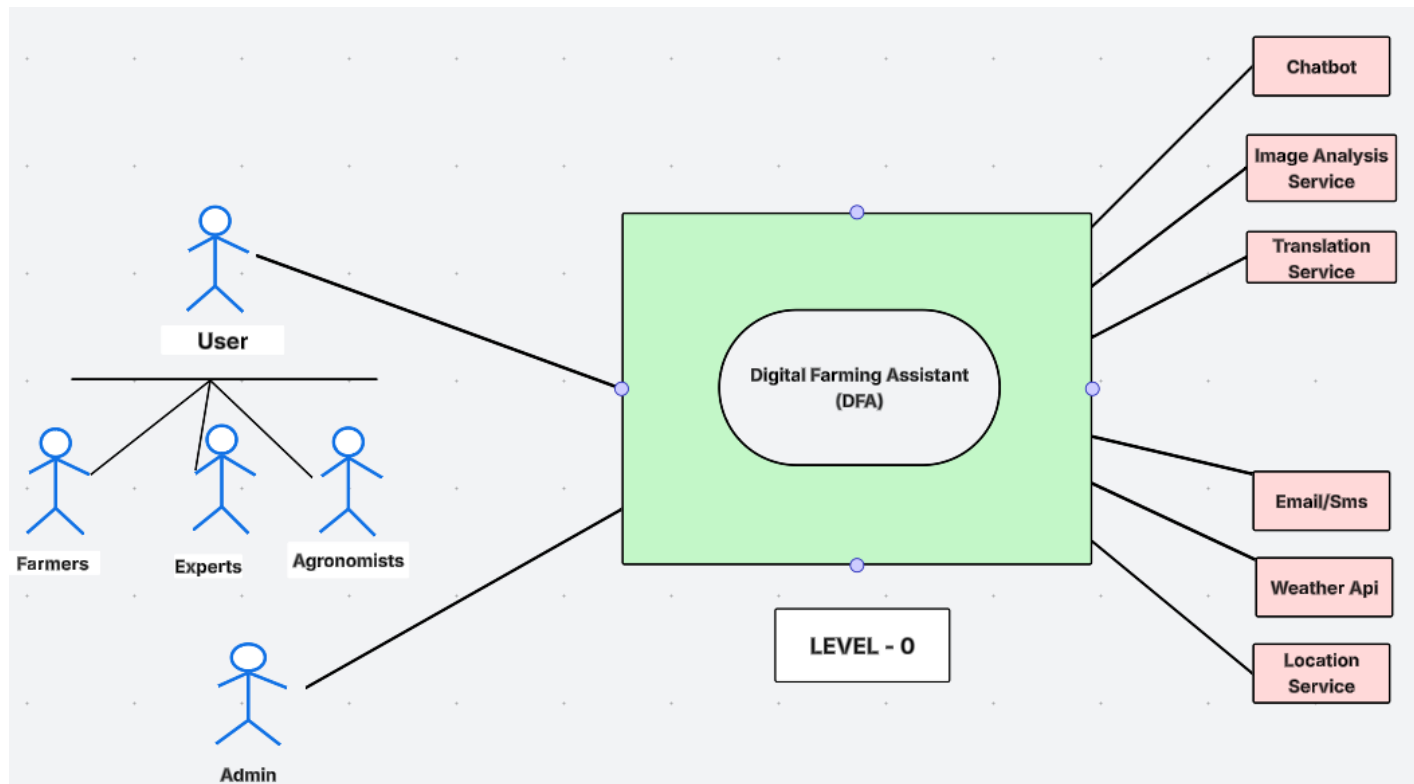
Use Case Diagram :

Level : 0

Name : Digital Farming Assistant (DFA)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : Chatbot,Image Analysis Service,Translation Service,Email/sms, Weather Api, Location Service

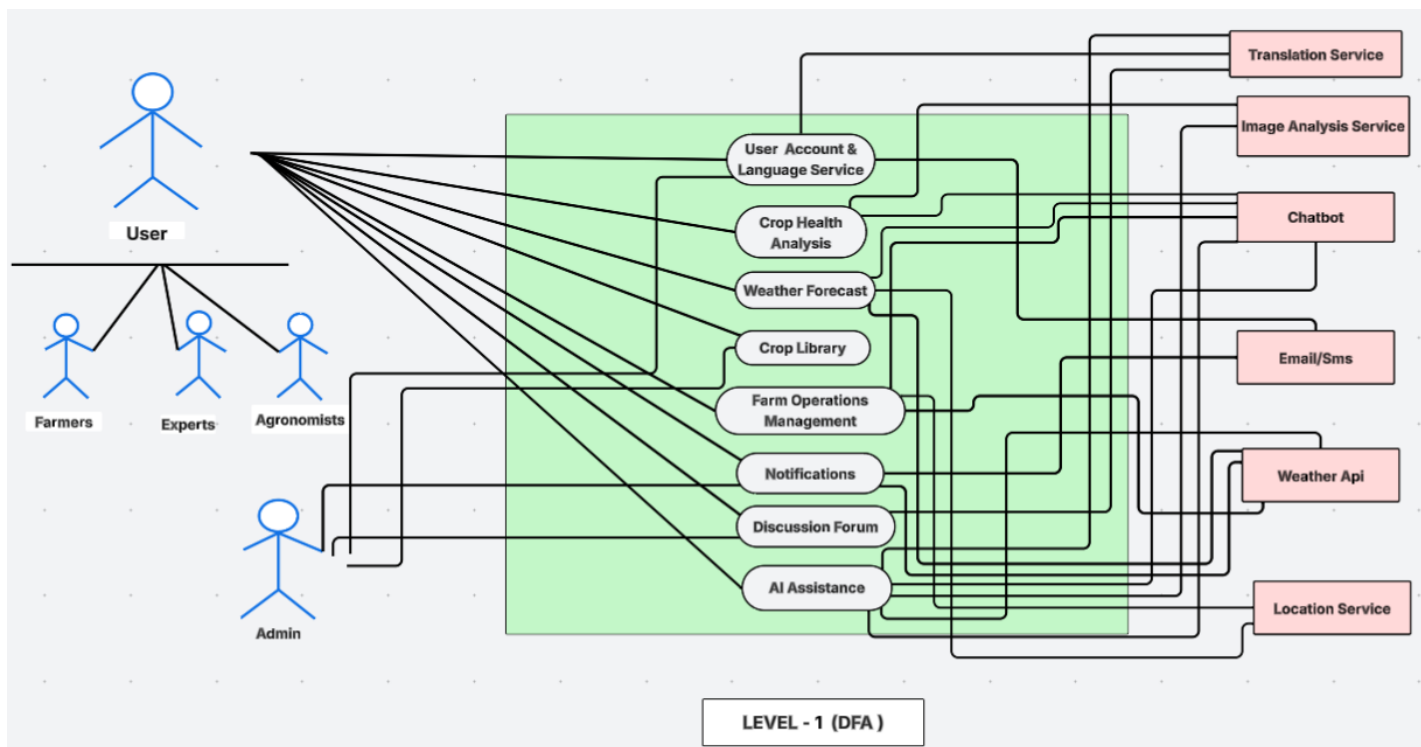


Level : 1

Name : Digital Farming Assistant (DFA)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : Chatbot,Image Analysis Service,Translation Service,Email/sms, Weather Api, Location Service



Level – 1 Description :

The Level 1 system architecture diagram of the Digital Farming Assistant (DFA) presents a structured overview of how different user types interact with the platform's core services and external systems.

On the left, the diagram identifies four key user roles: Farmers, Experts, Agronomists, and Admins. These users access the system through the central set of **core services**, which are grouped into eight functional modules:

- User Account & Language Service manages registration, authentication, profile updates, and language preferences.
- Crop Health Analysis allows users to upload images, receive AI-generated health reports, and view treatment advice.
- Weather Forecast provides location-based forecasts, alerts, and cropping recommendations.
- Crop Library offers searchable crop profiles, cultivation guides, and offline access to saved information.
- Farm Operations Management supports planning tasks like irrigation scheduling and harvesting timelines.
- Notifications deliver alerts and updates, including weather warnings and diagnosis results.
- Discussion Forum enables community interaction through posts, comments, and likes.
- AI Assistance powers intelligent features like chatbots, image analysis, and personalized recommendations.

These core services are connected to several external **systems** shown on the right:

- Translation Service ensures multilingual support.
- Image Analysis Service processes crop images using machine learning.
- Chatbot provides conversational guidance.
- Email/SMS handles communication and alerts.
- Weather API supplies real-time weather data.
- Location Service enables precise forecasting and advisory functions.

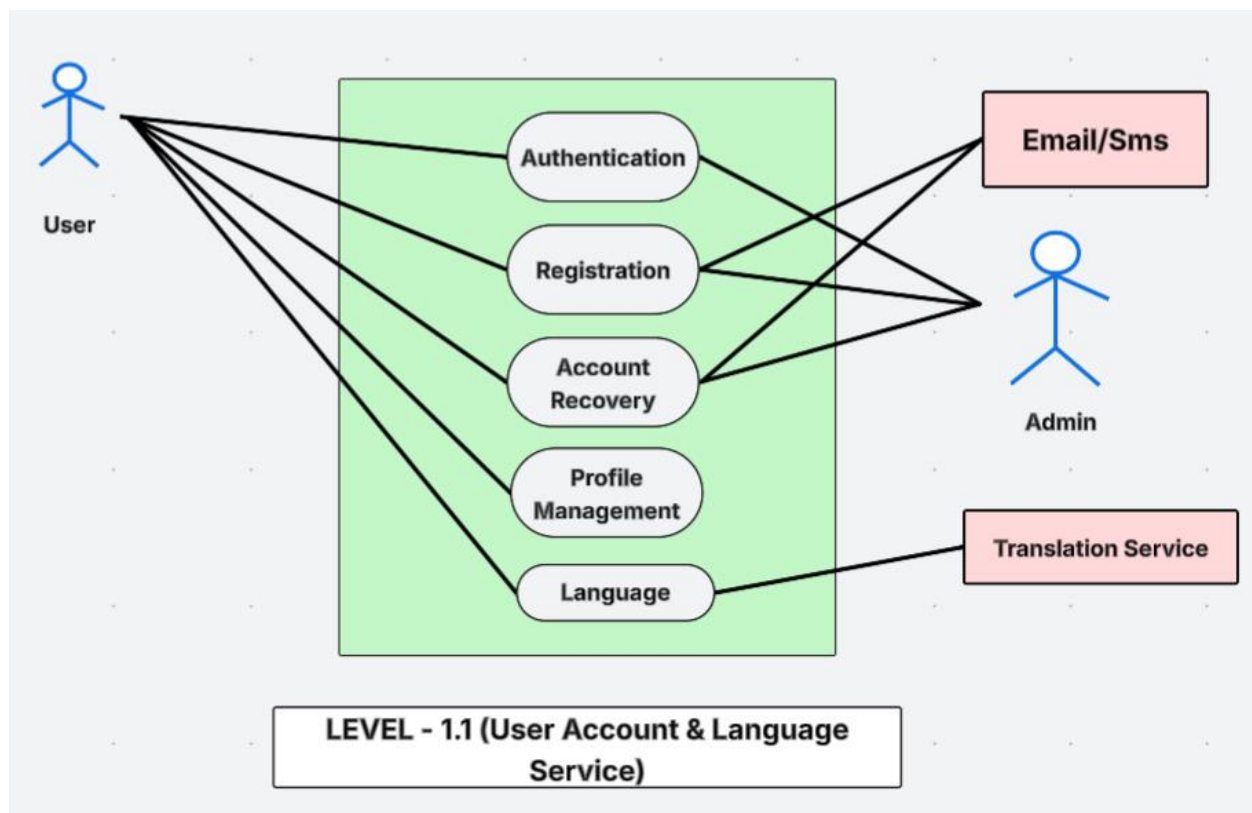
The diagram illustrates a highly integrated ecosystem where users interact with intelligent modules that rely on external APIs and services. It reflects a scalable and user-centric design, ensuring that farmers and experts can make informed decisions through a seamless digital experience.

Level : 1.1

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : Translation Service,Email/sms



Level – 1.1 Description :

This Level 1.1 system architecture diagram outlines the User Account & Language Service module of the Digital Farming Assistant. It shows how a User interacts with five key components: Authentication, Registration, Account Recovery, Profile Management, and Language.

The Authentication, Registration, and Account Recovery services are connected to both the Admin and an external Email/SMS system, indicating that user verification, password recovery, and notifications are managed through admin oversight and communication channels.

The Language component links to a Translation Service, enabling users to select their preferred language and receive translated content throughout the app.

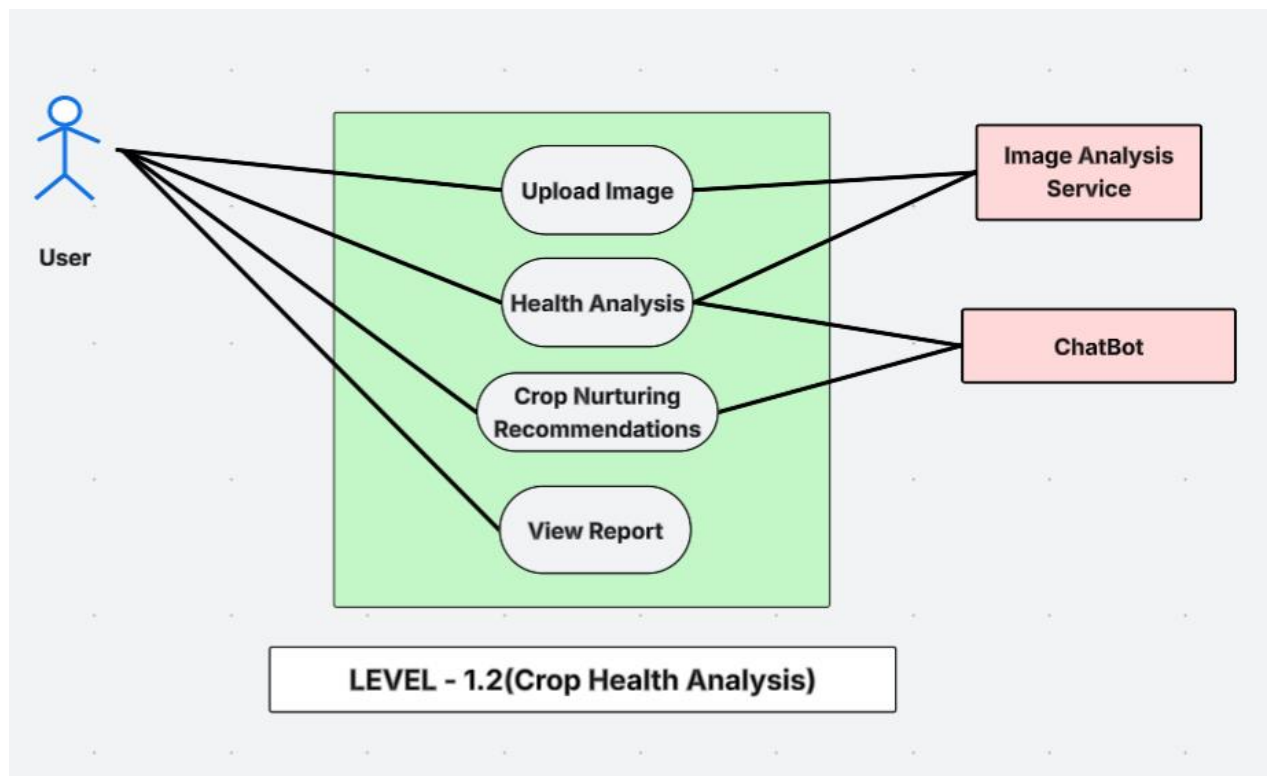
Overall, the diagram highlights how user identity, access, and personalization are securely managed, with support from external services and administrative control.

Level : 1.2

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists

Secondary Actor : Chatbot,Image Analysis Service



Level – 1.2 Description :

This Level 1.2 diagram presents the Crop Health Analysis process within the Digital Farming Assistant. It shows how a user engages with four key modules: Upload Image, Health Analysis, Crop Nurturing Recommendations, and View Report.

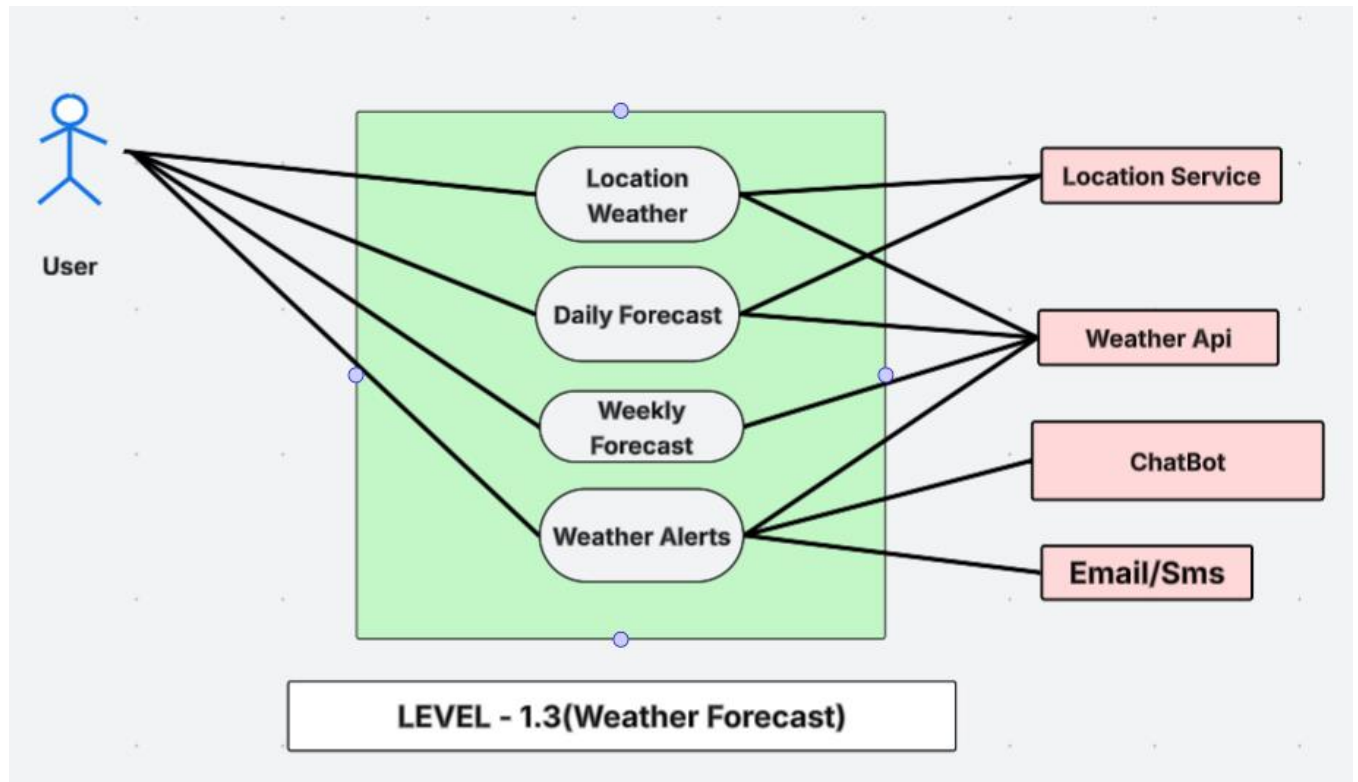
The Upload Image module enables users to submit crop photos, which are processed by the Image Analysis Service to identify symptoms and assess crop condition. The Health Analysis module uses both the Image Analysis Service and the ChatBot to interpret the image data and generate a diagnosis. Following this, the Crop Nurturing Recommendations module offers tailored treatment and care suggestions through the ChatBot. Finally, the View Report module lets users access saved reports and track their crop health history over time.

Level : 1.3

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists

Secondary Actor : Chatbot,Email/sms, Weather Api, Location Service



Level – 1.3 Description :

This Level 1.3 diagram outlines the Weather Forecast module of the Digital Farming Assistant. A user interacts with four components: Location Weather, Daily Forecast, Weekly Forecast, and Weather Alerts.

Location Weather connects to both the Location Service and Weather API to provide area-specific data. Daily and Weekly Forecast modules rely on the Weather API for real-time and extended weather updates. The Weather Alerts module integrates with the Weather API, ChatBot, and Email/SMS to deliver timely warnings and advisory messages.

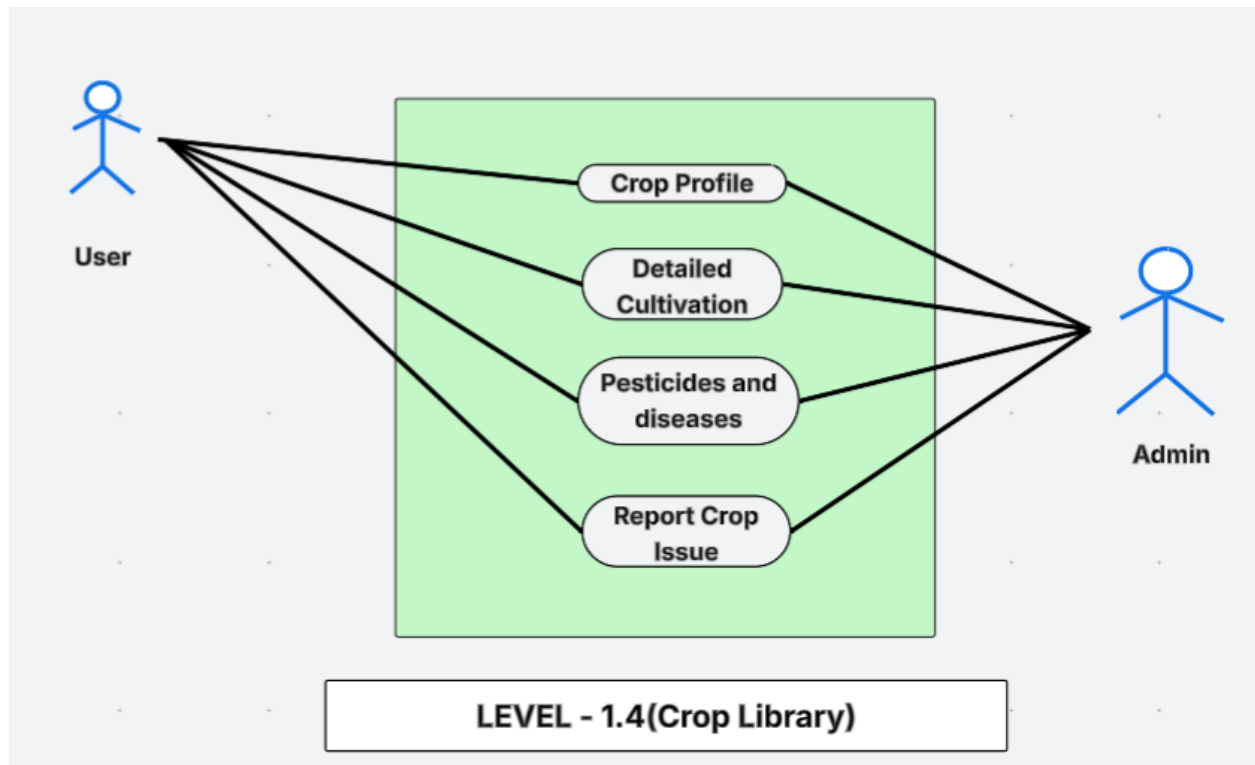
Together, these components ensure users receive accurate, location-based weather insights and alerts to support farm planning and decision-making.

Level : 1.3

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : None



Level – 1.4 Description :

This Level 1.4 diagram illustrates the Crop Library module of the Digital Farming Assistant. Both the user and the admin interact with four key components: Crop Profile, Detailed Cultivation, Pesticides and Diseases, and Report Crop Issue.

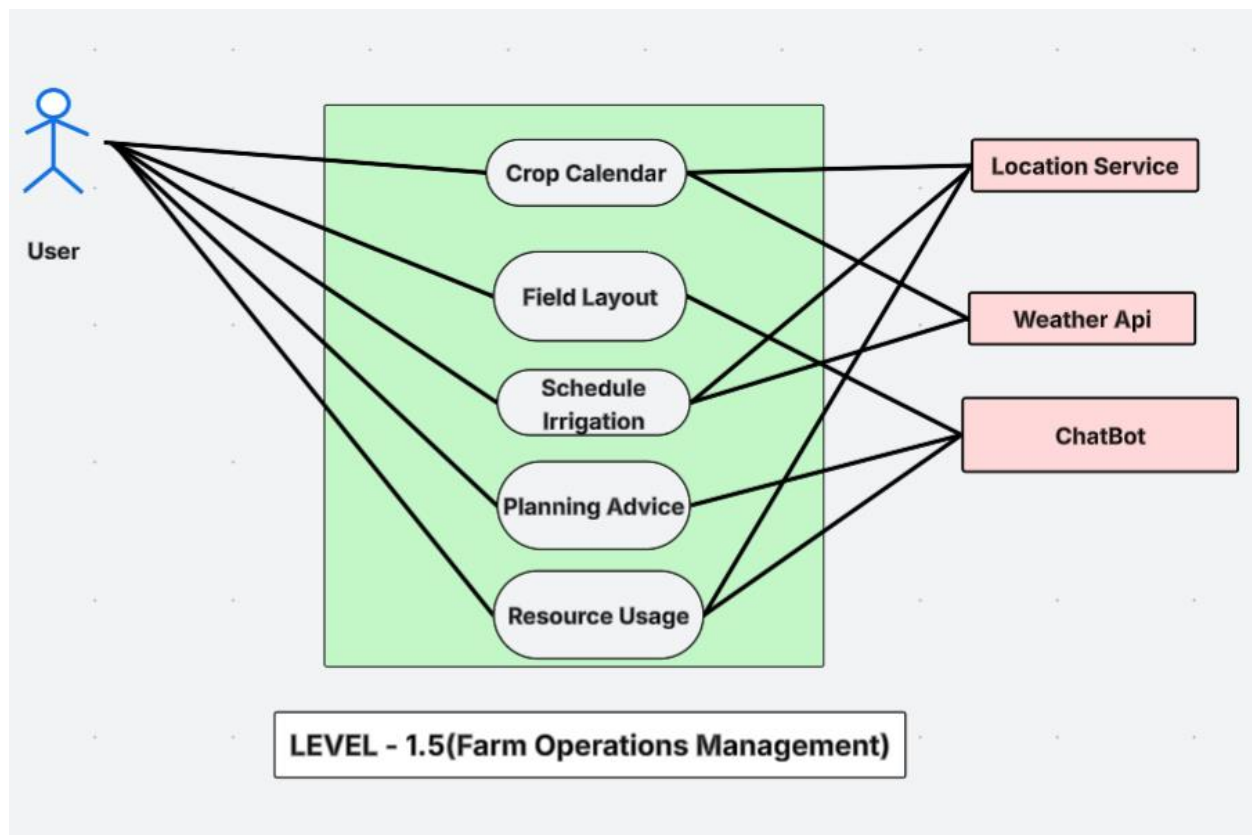
Users can access crop information, cultivation guides, and pest-related insights, while admins help manage and verify the content. The Report Crop Issue feature allows users to flag problems, which admins can review and respond to. This shared structure ensures accurate, up-to-date crop data and collaborative problem-solving.

Level : 1.5

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers, Experts, Agronomists

Secondary Actor : Chatbot, Weather Api, Location Service



Level – 1.5 Description :

This Level 1.5 diagram shows the Farm Operations Management module of the Digital Farming Assistant. A user interacts with five components: Crop Calendar, Field Layout, Schedule Irrigation, Planning Advice, and Resource Usage.

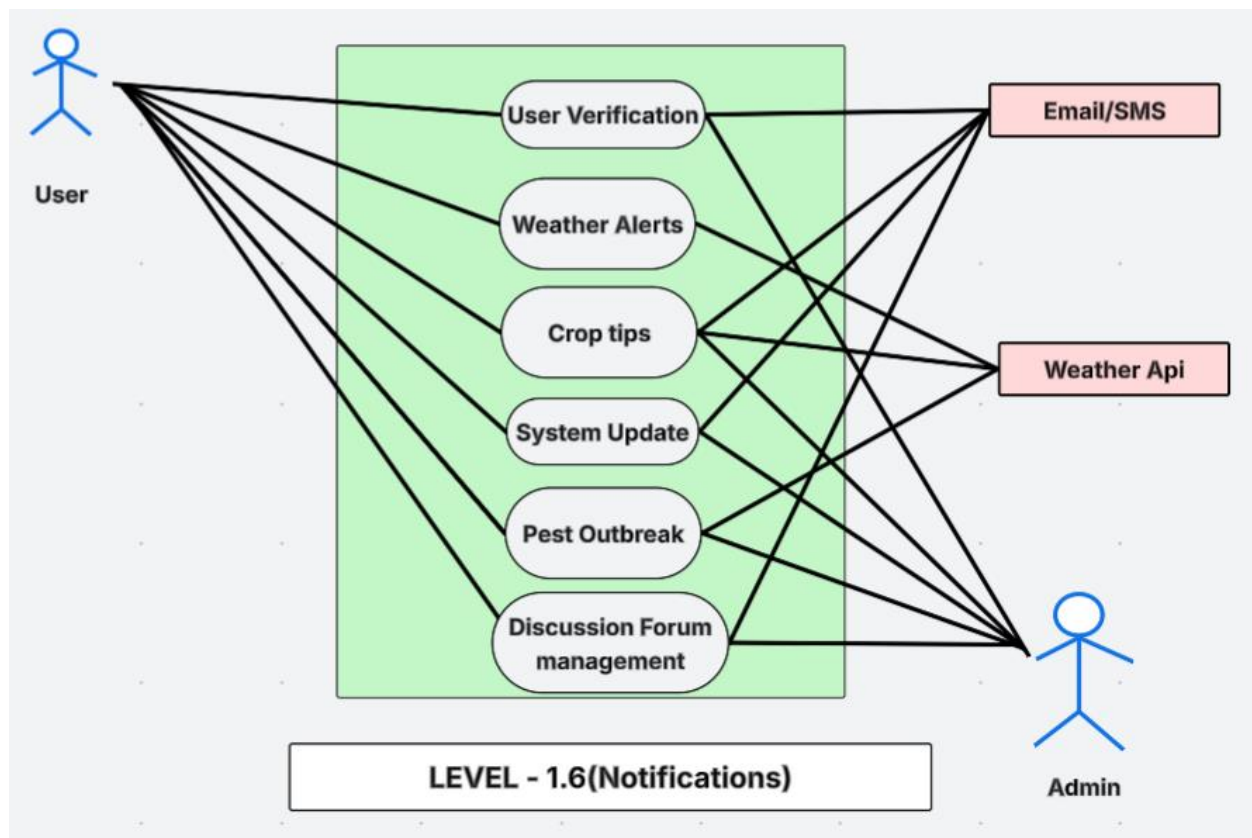
These components connect to external services like Location Service, Weather API, and ChatBot to provide location-based planning, weather-informed scheduling, and interactive guidance. The system helps users organize field activities, optimize irrigation, and manage resources efficiently.

Level : 1.6

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : Email/sms, Weather Api



Level – 1.6 Description :

This Level 1.6 diagram outlines the Notifications module of the Digital Farming Assistant. It shows how both the user and admin interact with six key components: User Verification, Weather Alerts, Crop Tips, System Update, Pest Outbreak, and Discussion Forum Management.

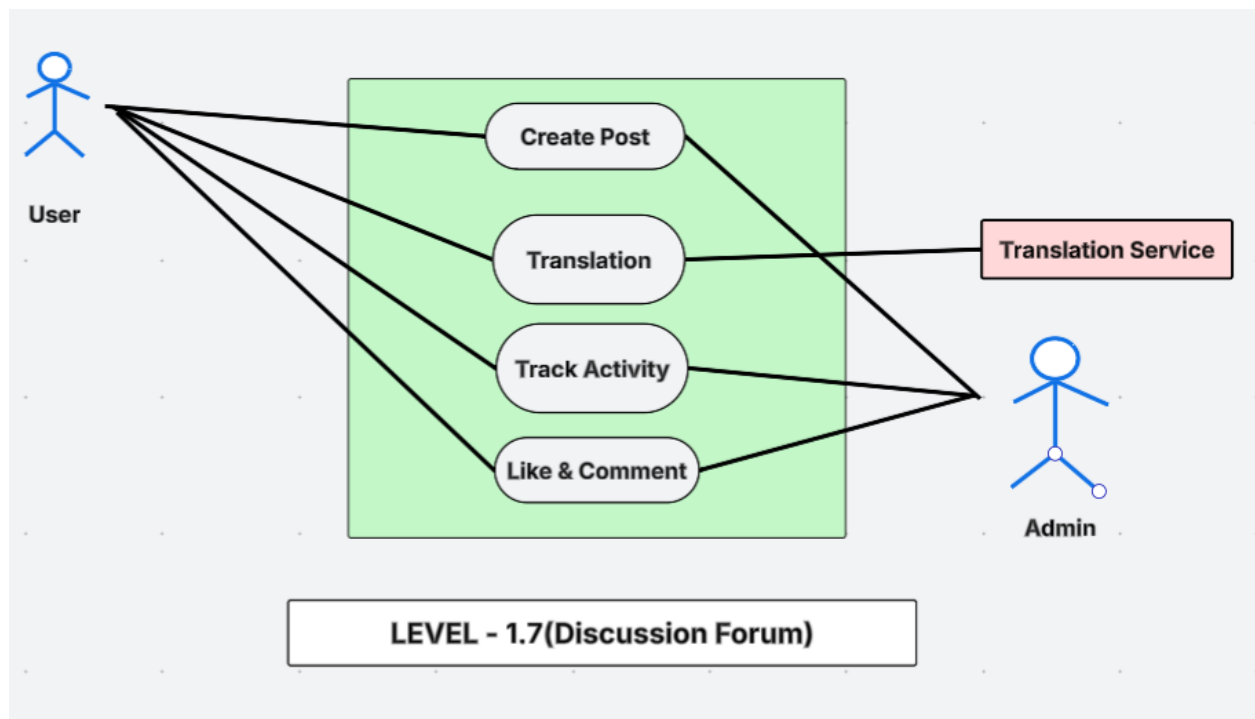
Notifications are delivered through external services like Email/SMS and Weather API, ensuring timely updates and alerts. The system supports communication for verification, weather warnings, crop care, and forum activity, helping users stay informed and responsive.

Level : 1.7

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists,Admin

Secondary Actor : Translation Service



Level – 1.7 Description :

This Level 1.7 diagram presents the Discussion Forum module of the Digital Farming Assistant. A user interacts with four main functions: Create Post, Translation, Track Activity, and Like & Comment.

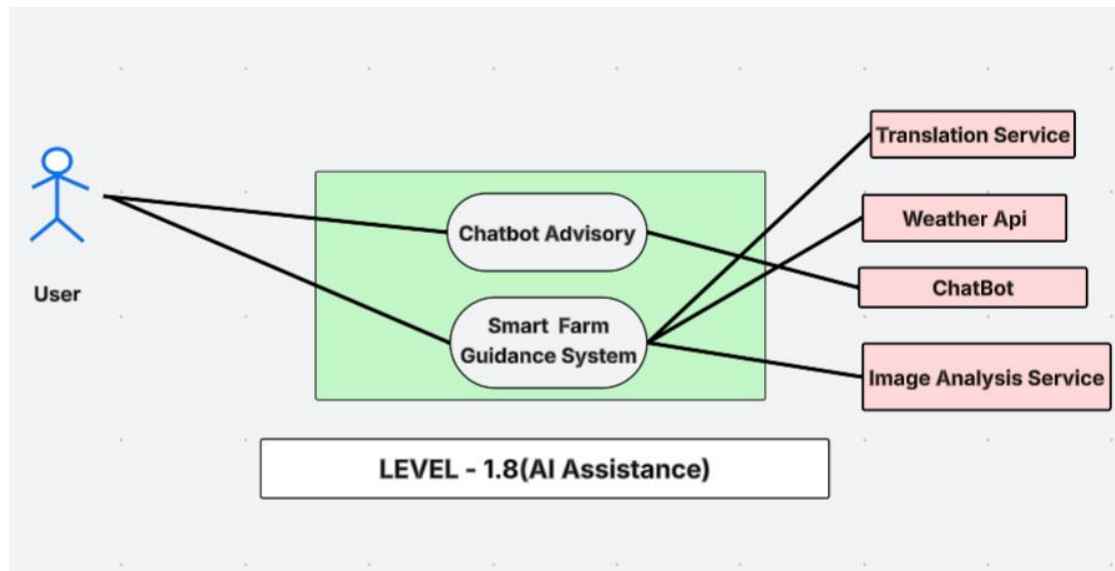
The Translation function connects to an external Translation Service to support multilingual communication. An admin also has access to all four functions, enabling moderation, oversight, and support for user interactions. The system promotes inclusive and active community engagement among farmers and experts.

Level : 1.8

Name : Digital Farming Assistant (DFD)

Primary Actor : Farmers,Experts,Agronomists,

Secondary Actor : Chatbot,Image Analysis Service,Translation Service, Weather Api



Level – 1.8 Description :

This Level 1.8 diagram outlines the AI Assistance module of the Digital Farming Assistant. A user interacts with two main components: Chatbot Advisory and Smart Farm Guidance System.

These components connect to external services including Translation Service, Weather API, ChatBot, and Image Analysis Service. Together, they provide personalized advice, weather-informed decisions, and image-based crop insights, enhancing the user's farming experience through intelligent automation.

Activity Diagram :

An activity diagram is a type of visual representation used in Unified Modeling Language (UML) to describe the flow of activities or actions within a system or process. It shows how tasks are carried out, the order in which they occur, and the conditions that guide transitions between them. By mapping out the sequence of steps, decisions, and parallel processes, it provides a clear picture of how a system behaves dynamically rather than just structurally.

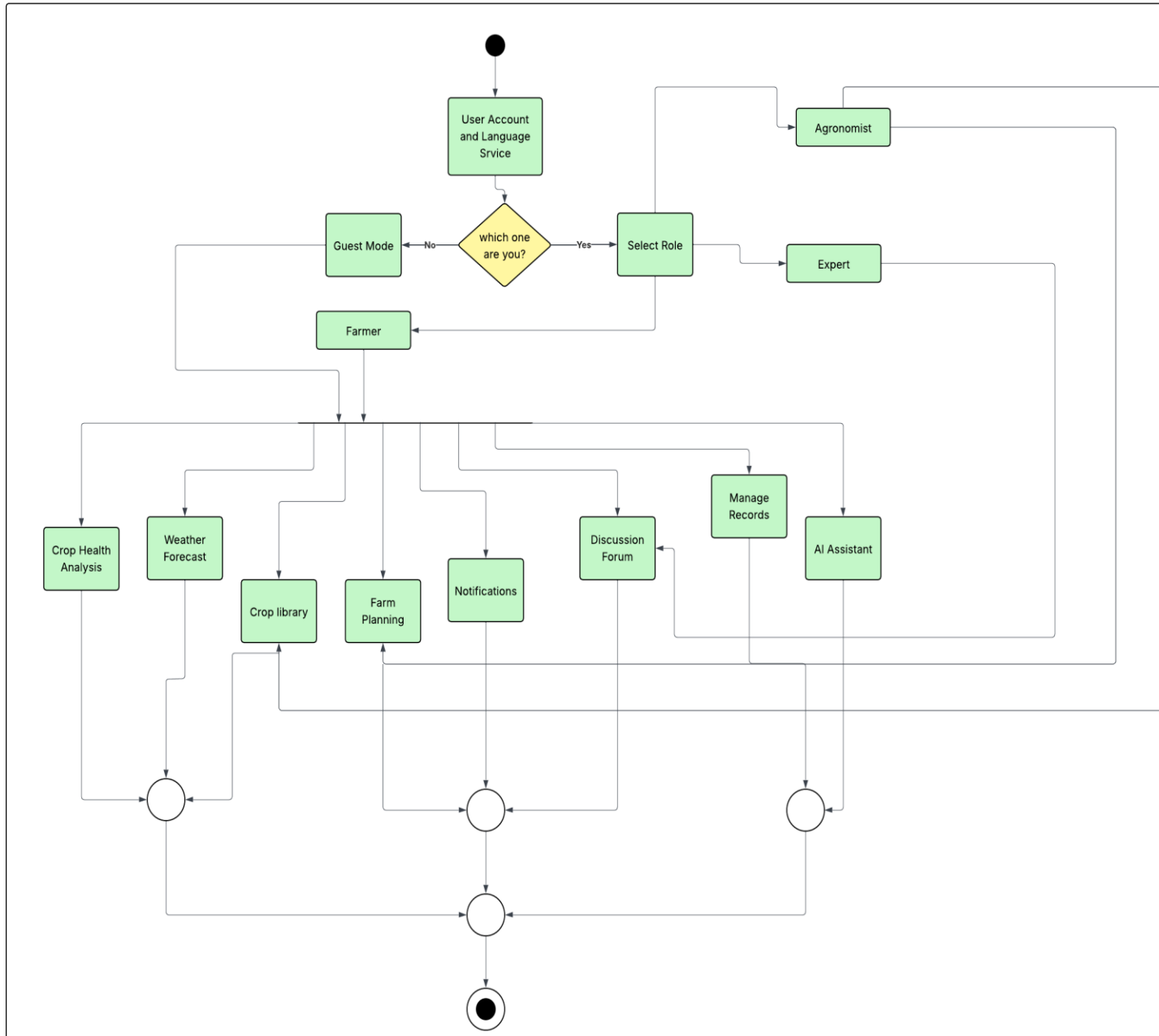
The purpose of an activity diagram is to help stakeholders understand workflows, identify bottlenecks, and clarify responsibilities. It is often used during system analysis and design to model business processes, use cases, or detailed operations. Developers and analysts rely on it to communicate complex logic in a way that is easy to follow, ensuring that requirements are captured accurately and that the system's behavior is well understood before implementation.

Activity Diagram Level: 1

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1

Digital Farming Assistant



Activity Diagram Level: 1 (Description)

This flowchart provides a structured overview of the Digital Farming Assistant system, focusing on how users enter the platform and access its core services. It begins with the User Account and Language **Service**, which handles login, registration, and language preferences. From there, users are prompted with a decision point: “which one are you?”—leading to two pathways.

One path is Guest Mode, which directly assigns the user the role of Farmer. The other path is Select Role, allowing users to choose between Agronomist and Expert. This branching ensures that users receive role-specific access and functionality tailored to their expertise and needs.

Once the role is determined, the Farmer role connects to a comprehensive suite of services, each represented by a green box. These include:

- **Crop Health Analysis** for diagnosing crop conditions using image-based AI tools
- **Weather Forecast** for location-specific updates and planning
- **Crop Library** offering cultivation guides and pest insights
- **Farm Planning** to manage irrigation, scheduling, and field layout
- **Notifications** for alerts on weather, pests, and system updates
- **Discussion Forum** for community interaction and expert advice
- **Manage Records** to track farming activities and history
- **AI Assistant** for smart, conversational guidance and recommendations
-

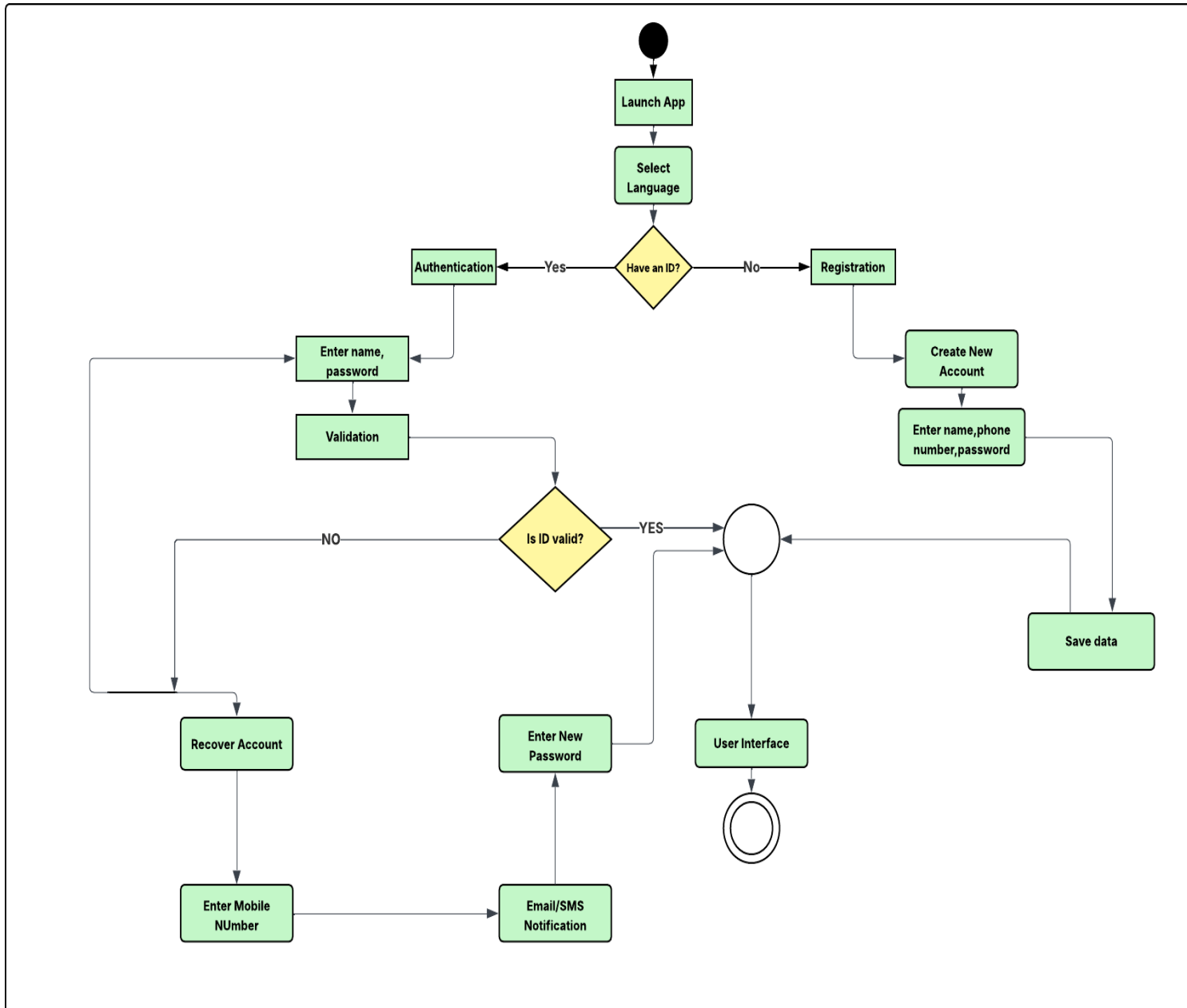
Each service is interconnected, forming a unified digital ecosystem that supports farmers in decision-making, learning, and collaboration. The diagram emphasizes user flow, role-based access, and the integration of intelligent tools to enhance agricultural productivity.

Activity Diagram Level: 1.1

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.1

User Account and Language service



Activity Diagram Level: 1.1 (Description)

This flowchart illustrates the user authentication and registration process for a mobile application, guiding users from the moment they launch the app to accessing its main interface. The journey begins with the user launching the app and selecting a preferred language, ensuring accessibility and personalization from the start.

Next, the user encounters a decision point: whether they already have an ID. If the answer is yes, they proceed to the authentication step, where they enter their name and password. The system then validates these credentials. If the ID is valid, the user gains access to the main interface. If not, the flow redirects to the account recovery process. Here, the user enters their mobile number and receives a notification via email or SMS. They are then prompted to set a new password, which is validated again before granting access.

If the user does not have an ID, they follow the registration path. This involves creating a new account by entering their name, phone number, and password. Once the data is saved, the system validates the new ID. If successful, the user is granted access to the interface.

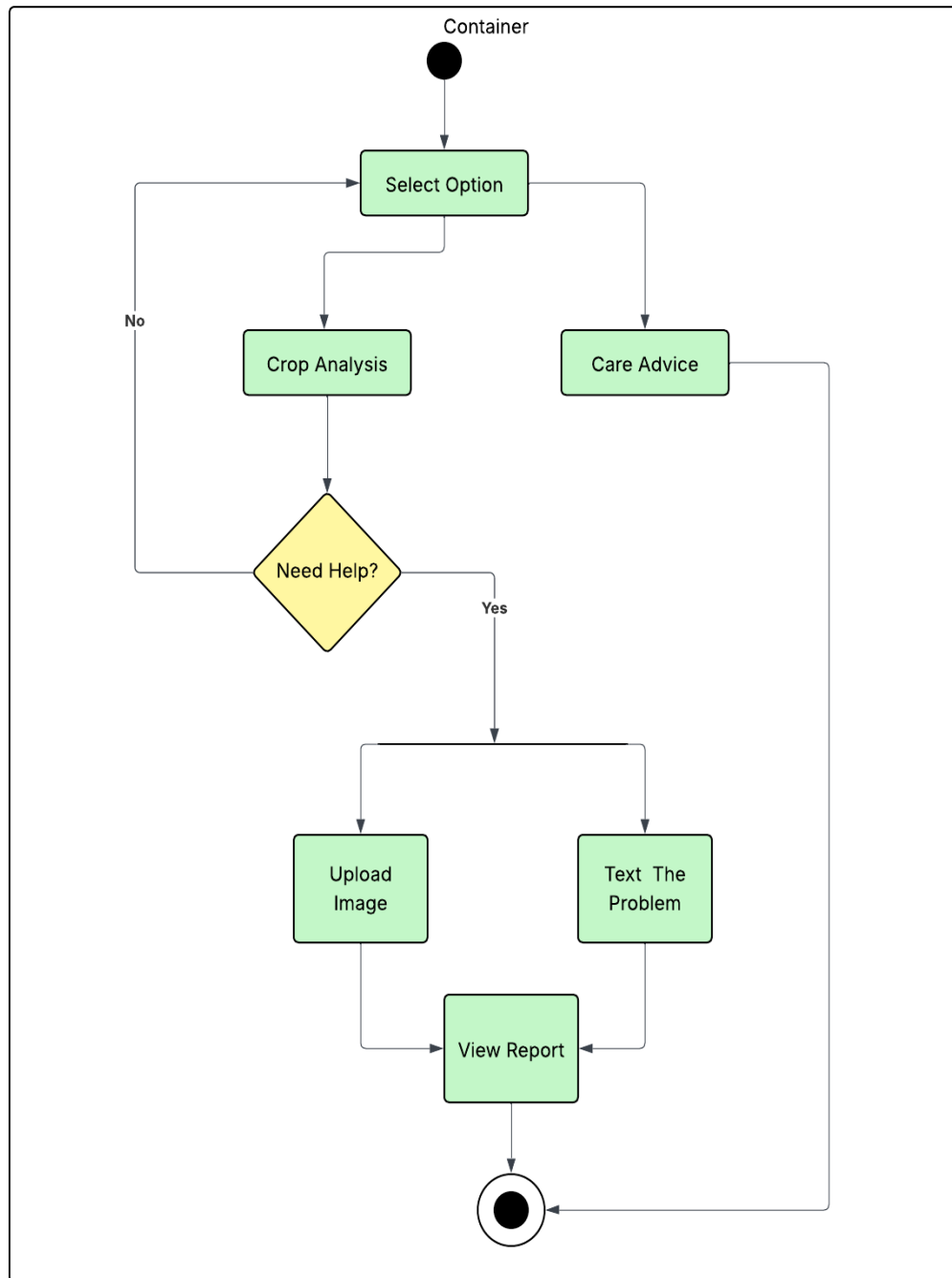
This structured flow ensures that both new and returning users can securely access the system. It includes robust validation and recovery mechanisms to handle forgotten credentials, while also streamlining the registration process for first-time users. The use of decision points and external services like email/SMS enhances security and user experience. This model is particularly effective for platforms like the Digital Farming Assistant, where reliable access and multilingual support are essential for diverse user groups such as farmers, agronomists, and experts

Activity Diagram Level: 1.2

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.2

Crop Health Analysis



Activity Diagram Level: 1.2 (Description)

This flowchart titled “Crop Health Analysis” outlines a structured decision-making process within a digital farming system, guiding users through crop diagnosis and care advisory steps. The process begins with a starting point labeled “Container,” which serves as the entry node for the crop health module. From there, the user is prompted to “Select Option,” leading to two distinct paths: “Crop Analysis” and “Care Advice.”

If the user selects “Crop Analysis,” the system presents a decision point labeled “Need Help?” This step determines whether the user requires assistance in diagnosing crop issues. If the answer is “No,” the process loops back, allowing the user to continue independently or reselect options. If the answer is “Yes,” the system offers two support methods: “Upload Image” or “Text The Problem.” These options reflect the system’s flexibility in handling both visual and textual input. Uploading an image allows the platform to use AI-based image analysis to detect symptoms, while texting the problem enables users to describe issues manually, which may be interpreted by a chatbot or expert system.

Both support paths converge at the “View Report” stage, where the user receives a diagnosis or advisory output. This report may include health status, treatment suggestions, and care recommendations. The process ends at a terminal node, indicating completion of the analysis cycle.

The alternative path, “Care Advice,” likely bypasses the diagnostic phase and directly provides nurturing tips, preventive measures, or seasonal care strategies based on crop type or user queries.

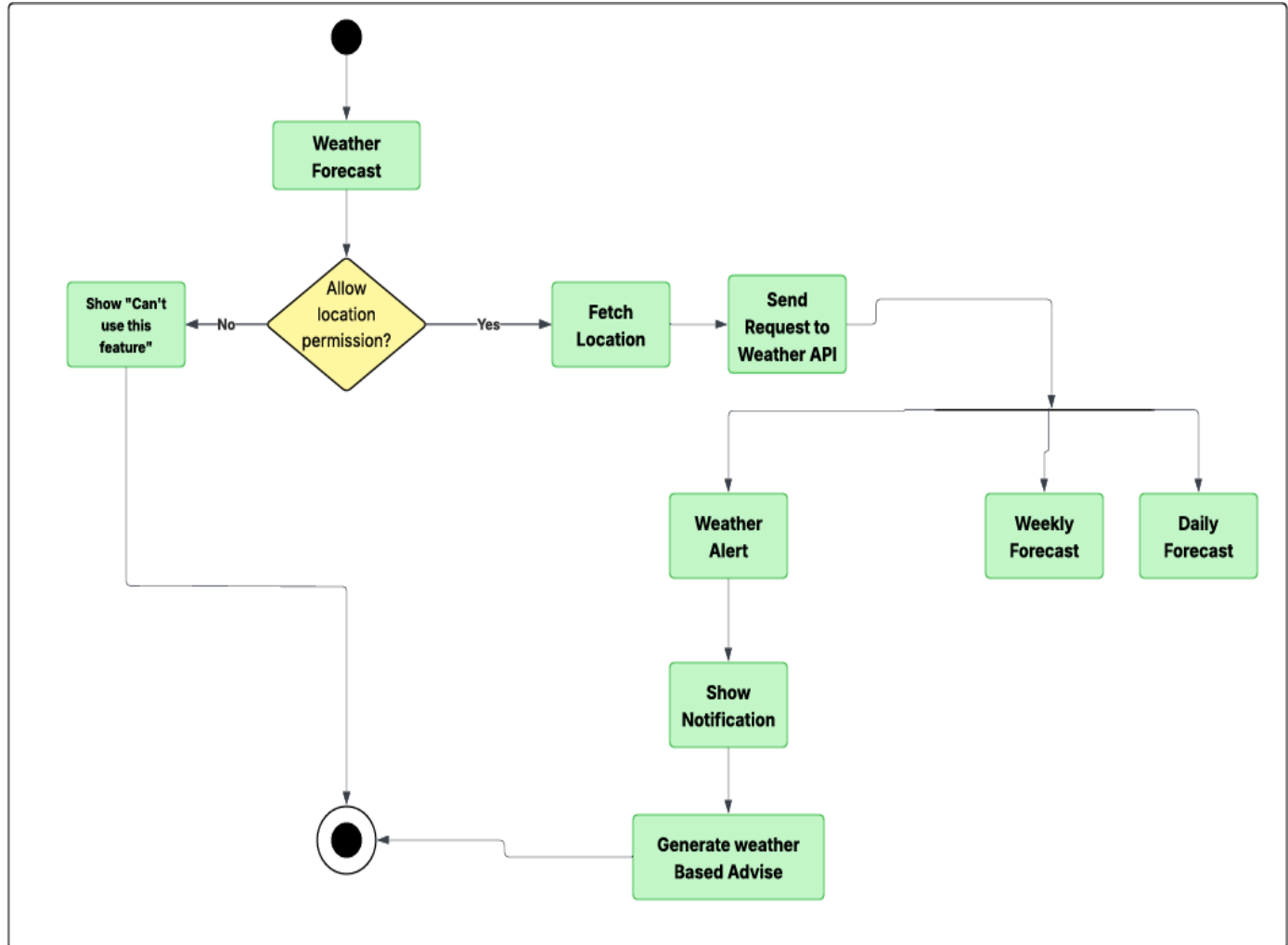
This flowchart emphasizes user autonomy while offering intelligent support when needed. It integrates decision points and multiple input modes to accommodate diverse user preferences and technical capabilities. The structure is particularly useful for farmers who may prefer quick advice or need detailed analysis depending on the situation. By combining image-based diagnostics and text-based queries, the system ensures accessibility and precision in crop health management.

Activity Diagram Level: 1.3

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.3

Weather Forecast



Activity Diagram Level: 1.3 (Description)

This flowchart outlines the logic behind a location-based weather forecasting feature within a digital farming or mobile application. The process begins with a user initiating a Weather Forecast request. The system immediately checks whether location permission has been granted, which is essential for delivering accurate, localized weather data.

If the user denies location permission, the system responds with a message stating “Can’t use this feature”, and the process ends. This ensures that users are aware of the limitation and that the system does not proceed without necessary data access.

If the user allows location permission, the system proceeds to fetch the location using device sensors or location services. Once the location is obtained, the system sends a request to a Weather API, which acts as the external data source for real-time weather information.

The response from the Weather API can lead to three possible outcomes:

1. Weather Alert – If severe or unusual weather conditions are detected, the system triggers a notification to inform the user. Additionally, it generates weather-based advice, which may include safety tips, crop protection measures, or irrigation adjustments.
2. Weekly Forecast – This provides a broader view of upcoming weather trends, helping users plan agricultural activities like planting, harvesting, or pesticide application over several days.
3. Daily Forecast – This offers short-term insights, useful for immediate decisions such as watering schedules or fieldwork timing.

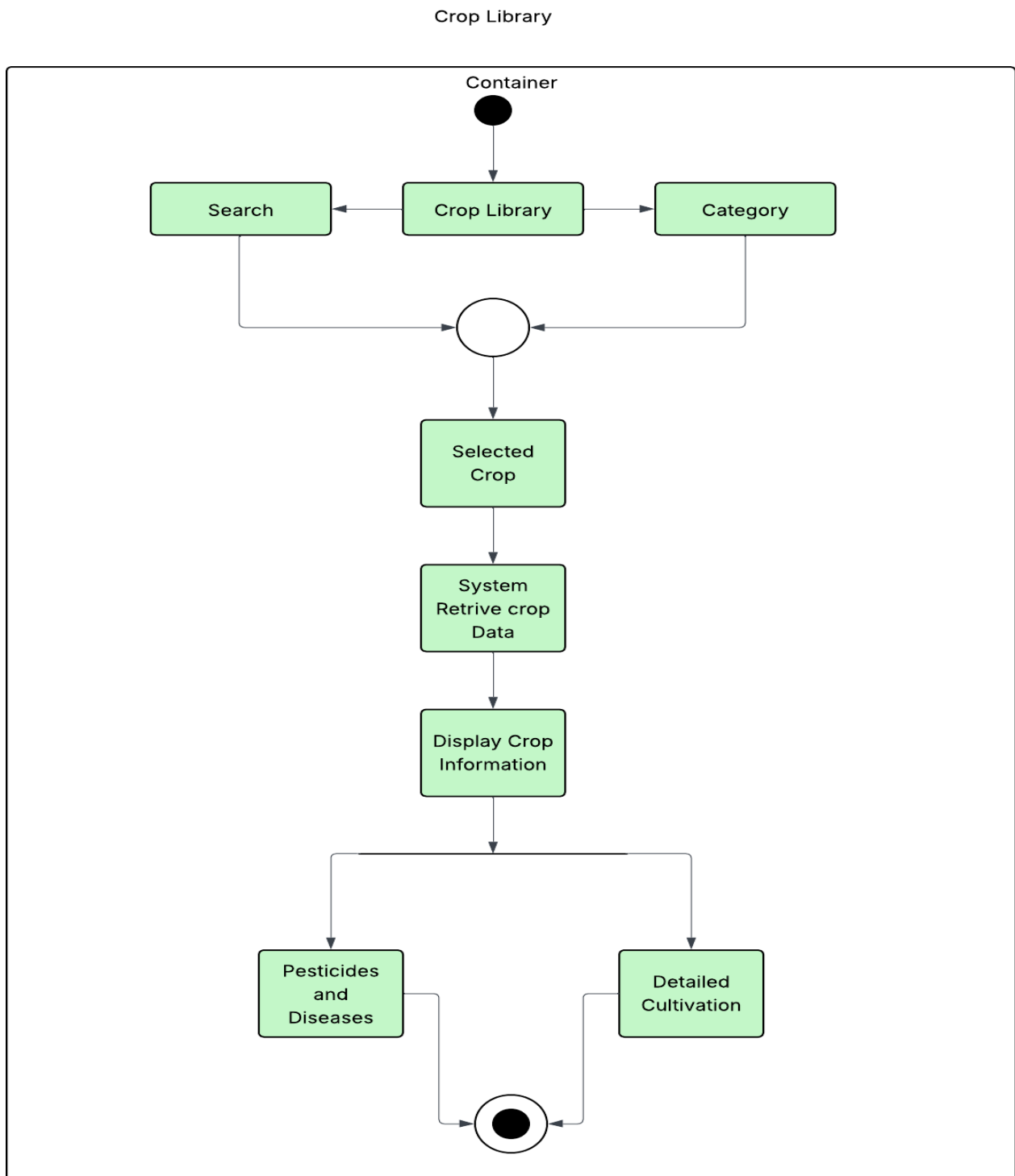
The flowchart uses decision diamonds and process rectangles to clearly represent conditional logic and sequential actions. It emphasizes user consent, external data integration, and adaptive responses based on weather conditions.

This structure is particularly valuable in agricultural applications, where timely and location-specific weather data can significantly impact productivity and risk management. By incorporating permission checks, API integration, and conditional outputs, the system ensures both user control and intelligent automation.

Activity Diagram Level: 1.4

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.4



Activity Diagram Level: 1.4 (Description)

This flowchart titled Crop Library illustrates the structured process of accessing crop-related information within a digital farming system. The flow begins at a starting point labeled Container, which serves as the entry node for the crop library module. From this point, the user is presented with three initial options: Search, Crop Library, and Category. These options represent different ways a user might begin exploring crop data, whether by typing a specific query, browsing the full library, or filtering by predefined categories.

Once an option is selected, the flow proceeds to a decision point that results in a Selected Crop. This step indicates that the user has identified a specific crop of interest, either through direct search or by navigating through categories. The system then performs a retrieval operation, labeled System Retrieve Crop Data, which accesses the relevant database or backend service to gather detailed information about the chosen crop.

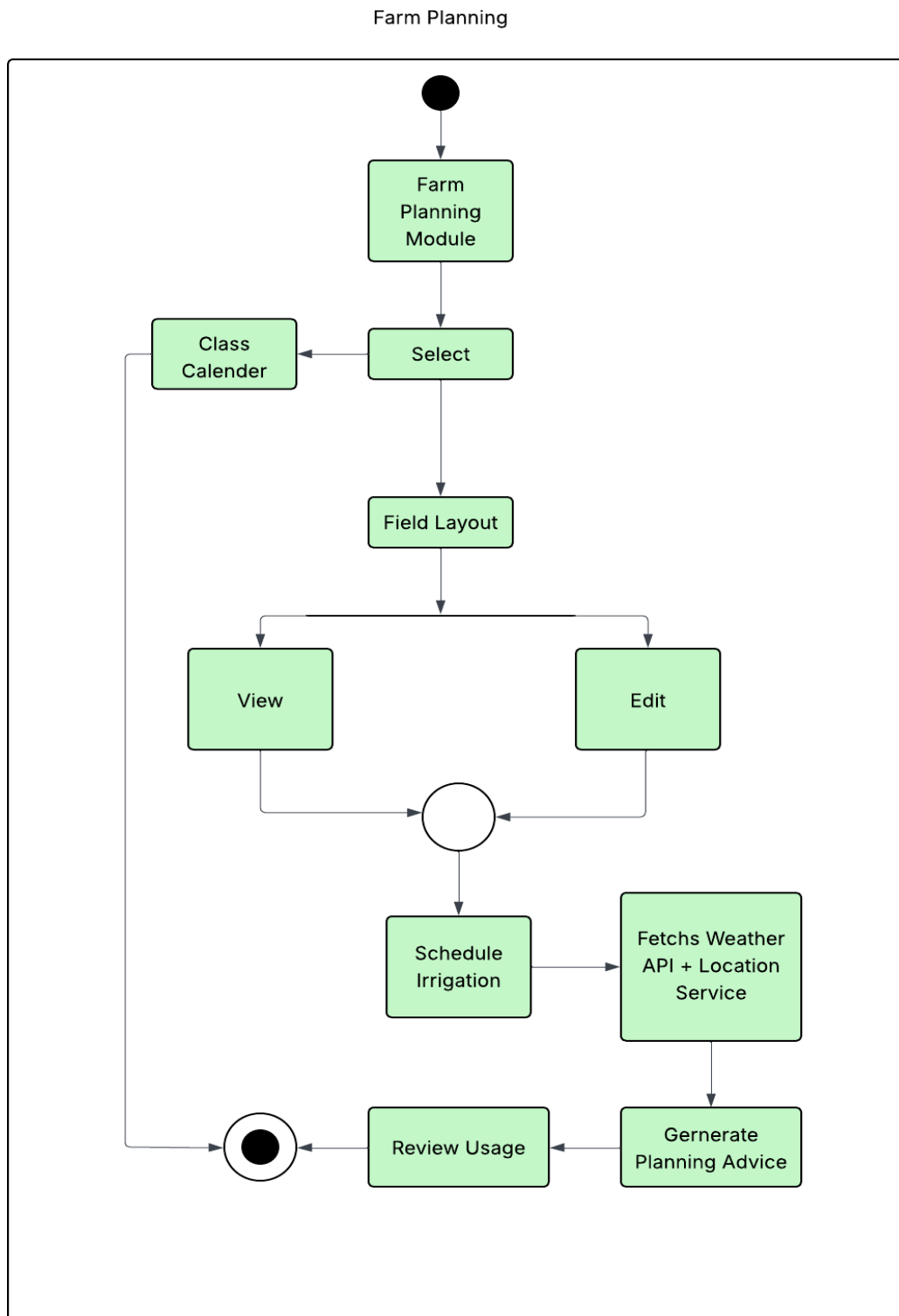
Following data retrieval, the system moves to Display Crop Information, where the user is presented with comprehensive details about the selected crop. This may include growth requirements, soil preferences, climate suitability, and other agronomic data. From this display point, the user has two further options for deeper exploration: Pesticides and Diseases or Detailed Cultivation. The Pesticides and Diseases path provides insights into common threats to the crop and recommended treatments or preventive measures. The Detailed Cultivation path offers step-by-step guidance on how to grow and manage the crop effectively, including planting schedules, irrigation needs, and harvesting techniques.

Both paths lead to an end node, signifying the completion of the information retrieval process. This flowchart highlights the system's ability to support both general exploration and targeted learning, making it useful for farmers, agronomists, and agricultural students. By offering multiple entry points and branching into specialized content areas, the Crop Library module ensures that users can access relevant and actionable information in a logical and user-friendly manner.

Activity Diagram Level: 1.5

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.5 (Farm Operations Management)



Activity Diagram Level: 1.5 (Description)

This flowchart titled Farm Planning outlines the structured process of managing agricultural activities within a digital farming system. The flow begins with a start node, leading into the Farm Planning Module, which serves as the central hub for organizing field operations. From this module, the user is prompted to select specific planning tasks, initiating a sequence of actions that guide them through layout design, irrigation scheduling, and resource management.

The first major component is Field Layout, where users can view and edit the configuration of their farmland. This allows them to define plot boundaries, assign crop zones, and organize spatial arrangements for efficient cultivation. The ability to view and edit ensures flexibility, enabling users to adapt plans based on seasonal changes or crop rotation strategies.

Following layout configuration, the flow reaches a decision point that leads to Schedule Irrigation. This module integrates with external services such as the Weather API and Location Service to fetch real-time environmental data. By analyzing weather forecasts and geographic conditions, the system helps users determine optimal irrigation schedules, reducing water waste and improving crop health.

Once irrigation is scheduled, the system proceeds to Generate Planning Advice. This step synthesizes data from previous modules and external inputs to offer tailored recommendations. These may include planting timelines, fertilization plans, or pest control strategies, all aligned with current weather and field conditions.

The next step is Review Usage, where users can assess resource consumption and operational efficiency. This module provides insights into water usage, labor allocation, and input costs, helping farmers make informed decisions and adjust plans as needed.

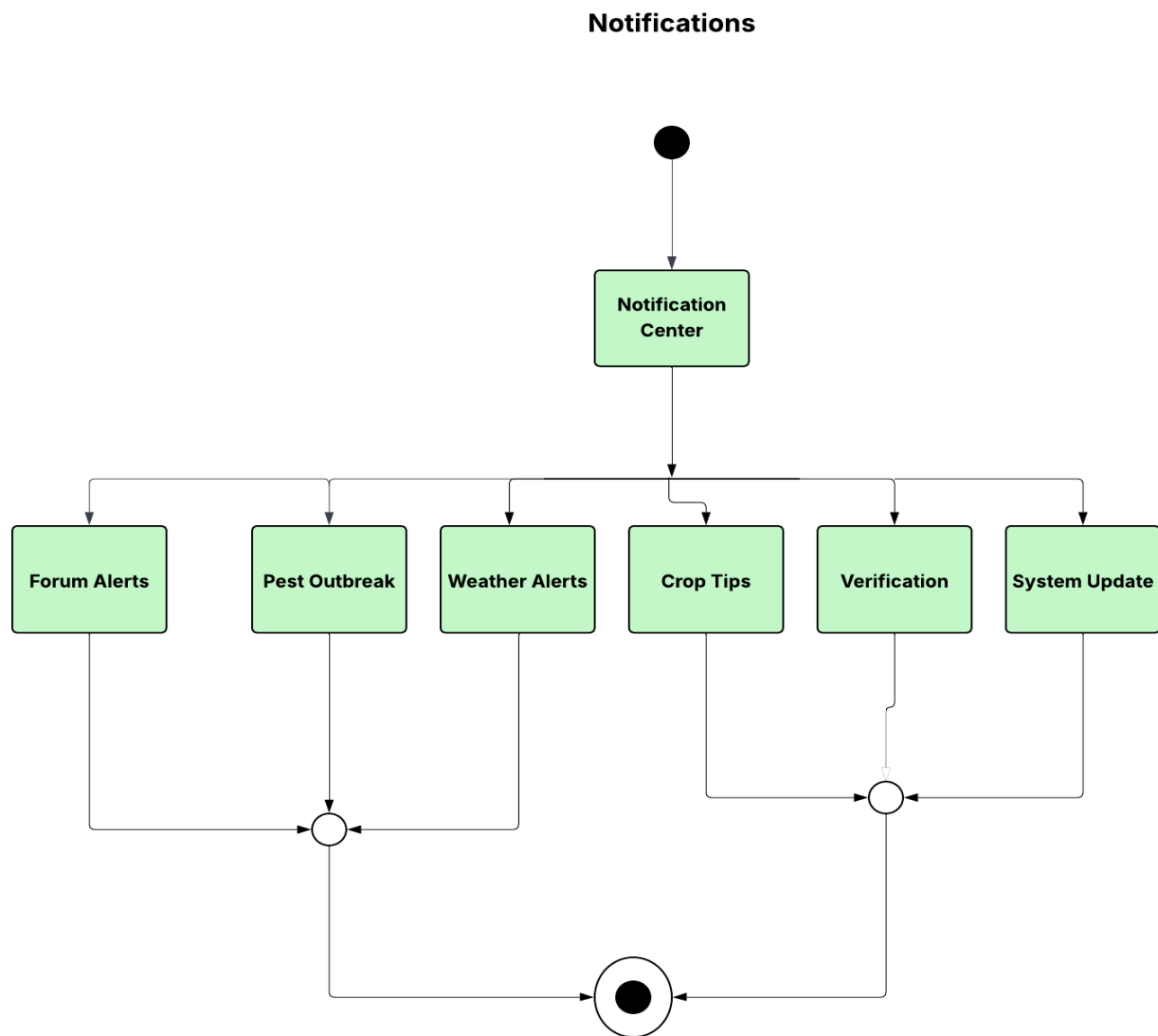
The process concludes at an end node, signifying the completion of the planning cycle. An additional component, Class Calendar, is included to support scheduling and coordination, possibly linking farm tasks with training sessions or seasonal events.

Overall, the flowchart presents a logical and data-driven approach to farm planning. It combines user input with automated analysis, ensuring that decisions are both practical and responsive to environmental factors. This structure supports efficient resource management and strategic planning, making it a valuable tool for modern agriculture.

Activity Diagram Level: 1.6

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.6



Activity Diagram Level: 1.6 (Description)

This flowchart titled Notifications presents a structured overview of how different types of alerts are managed within a centralized notification system. The process begins at a starting node and flows into the Notification Center, which acts as the core module responsible for distributing various types of messages to users. From this central hub, the system branches out into six distinct notification categories: Forum Alerts, Pest Outbreak, Weather Alerts, Crop Tips, Verification, and System Update.

Forum Alerts are likely triggered by activity within the discussion forum, such as new posts, replies, or mentions. These alerts help users stay engaged with community interactions and ensure timely responses to shared content. Pest Outbreak notifications are critical for agricultural users, providing early warnings about pest activity that could threaten crops. These alerts may be based on regional data or user-submitted reports and are essential for preventive action.

Weather Alerts inform users about significant weather changes, such as storms, droughts, or temperature shifts. These notifications are typically generated using external weather data and help farmers adjust their field activities accordingly. Crop Tips offer users practical advice on cultivation, fertilization, irrigation, or harvesting. These tips may be seasonal or triggered by specific crop conditions, supporting better farm management.

Verification notifications are related to user account activities, such as confirming identity, resetting passwords, or validating changes in profile information. These alerts ensure secure access and maintain system integrity. System Update notifications inform users about changes in the platform, such as new features, maintenance schedules, or policy updates. Keeping users informed about system changes helps maintain transparency and usability.

Each of these six notification types flows into two circular nodes, which likely represent processing or delivery stages. These nodes consolidate the information before directing it to the final output, represented by a larger circle at the bottom center of the diagram. This final node may symbolize the user interface or the endpoint where notifications are displayed or sent via email, SMS, or in-app messages.

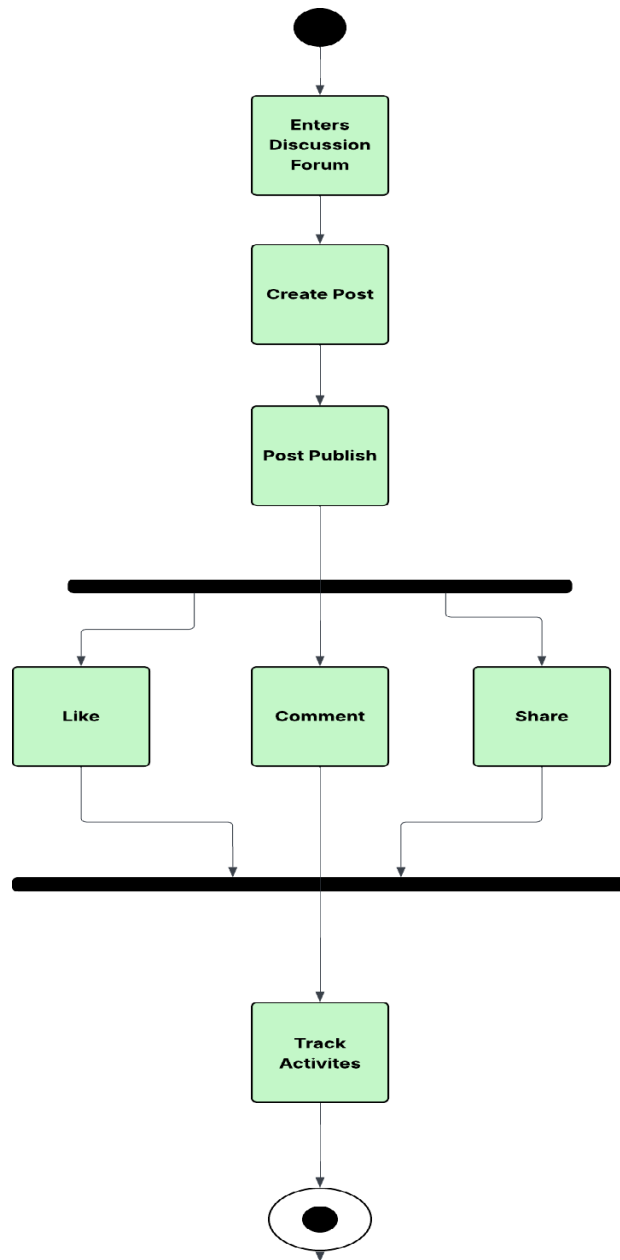
Overall, the flowchart emphasizes modularity, clarity, and responsiveness. It ensures that users receive timely and relevant information across multiple domains, supporting informed decision-making and active engagement within the digital farming ecosystem. I can also help you convert this into a use case diagram or notification handling scenario for documentation.

Activity Diagram Level: 1.7

Name: Digital Farming Assistant

Reference: Use Case Diagram Level – 1.7 (Discussion Forum)

Discussion Forum



Activity Diagram Level: 1.7 (Description)

This flowchart titled Discussion Forum outlines the user interaction process within a digital community platform, specifically designed to facilitate agricultural dialogue and knowledge sharing. The sequence begins when a user enters the discussion forum, marking the starting point of engagement. This entry allows the user to explore existing conversations or initiate new ones.

The first active step is to create a post. This involves composing a message, question, or informational content that the user wishes to share with the community. The post may include text, images, or links, depending on the platform's capabilities. Once the post is ready, the user proceeds to publish it, making it visible to other members of the forum.

After the post is published, it becomes interactive. Other users can engage with the content in three primary ways: liking, commenting, and sharing. The like function allows users to express appreciation or agreement with the post. Commenting enables users to respond, ask follow-up questions, or add insights, fostering a collaborative environment. Sharing allows users to distribute the post to other platforms or within the forum, increasing its visibility and reach.

All these interactions are tracked under the final step labeled Track Activities. This module records user engagement metrics such as the number of likes, comments, and shares. It may also log user participation patterns, helping administrators monitor forum health and identify active contributors. Tracking activities is essential for maintaining transparency, encouraging meaningful participation, and supporting moderation efforts.

This flowchart presents a streamlined and intuitive process for community interaction. It emphasizes user-generated content and peer engagement, which are critical for building a vibrant and informative discussion space. The structure supports both individual expression and collective learning, making it suitable for platforms like the Digital Farming Assistant, where farmers, agronomists, and experts can exchange ideas, solve problems, and share experiences.

Swimlane Diagram :

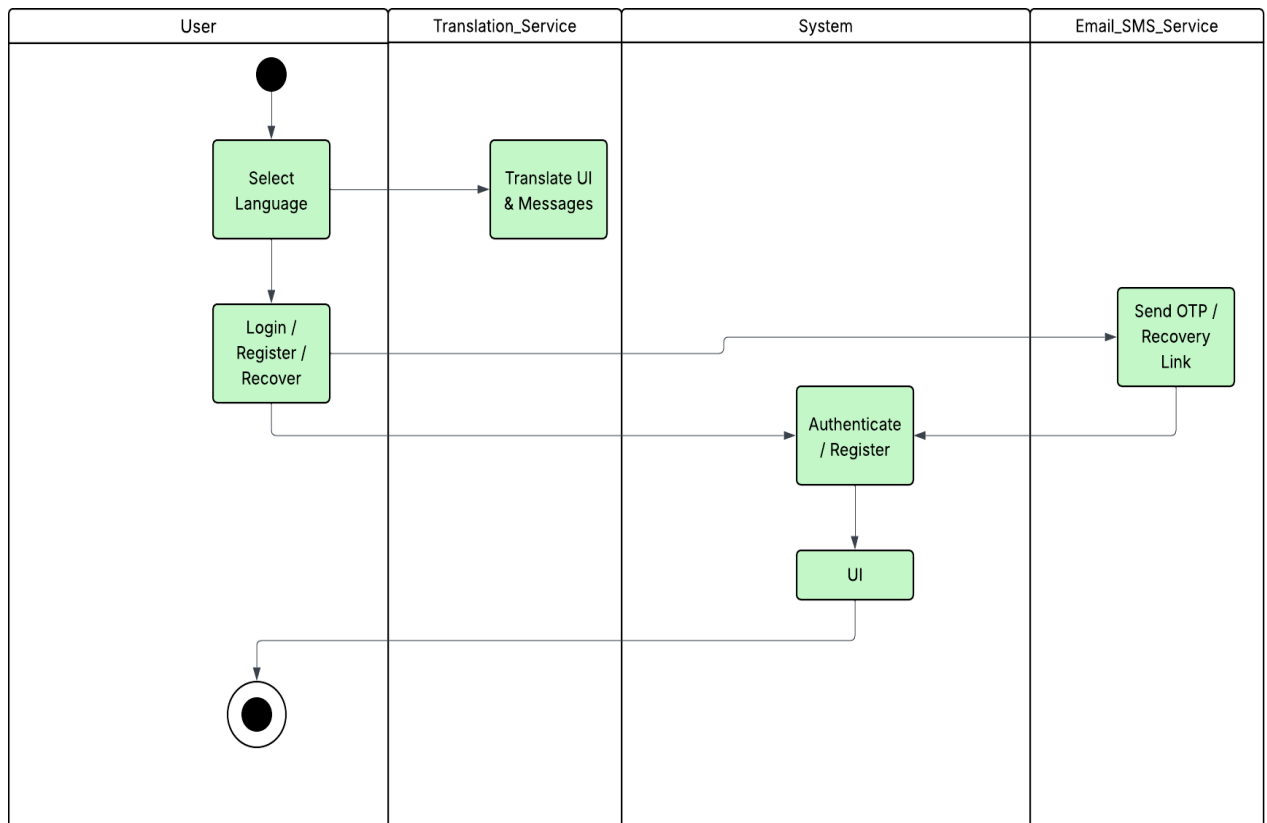
A swimlane diagram is a type of process flowchart that organizes activities into lanes, where each lane represents a role, department, or system responsible for those actions. It visually separates responsibilities while showing how tasks move across different actors or groups.

Its purpose is to clarify who does what in a process, reduce confusion about responsibilities, and highlight inefficiencies or handoffs between roles.

Swimlane Diagram Level: 1.1

Name: Digital Farming Assistant

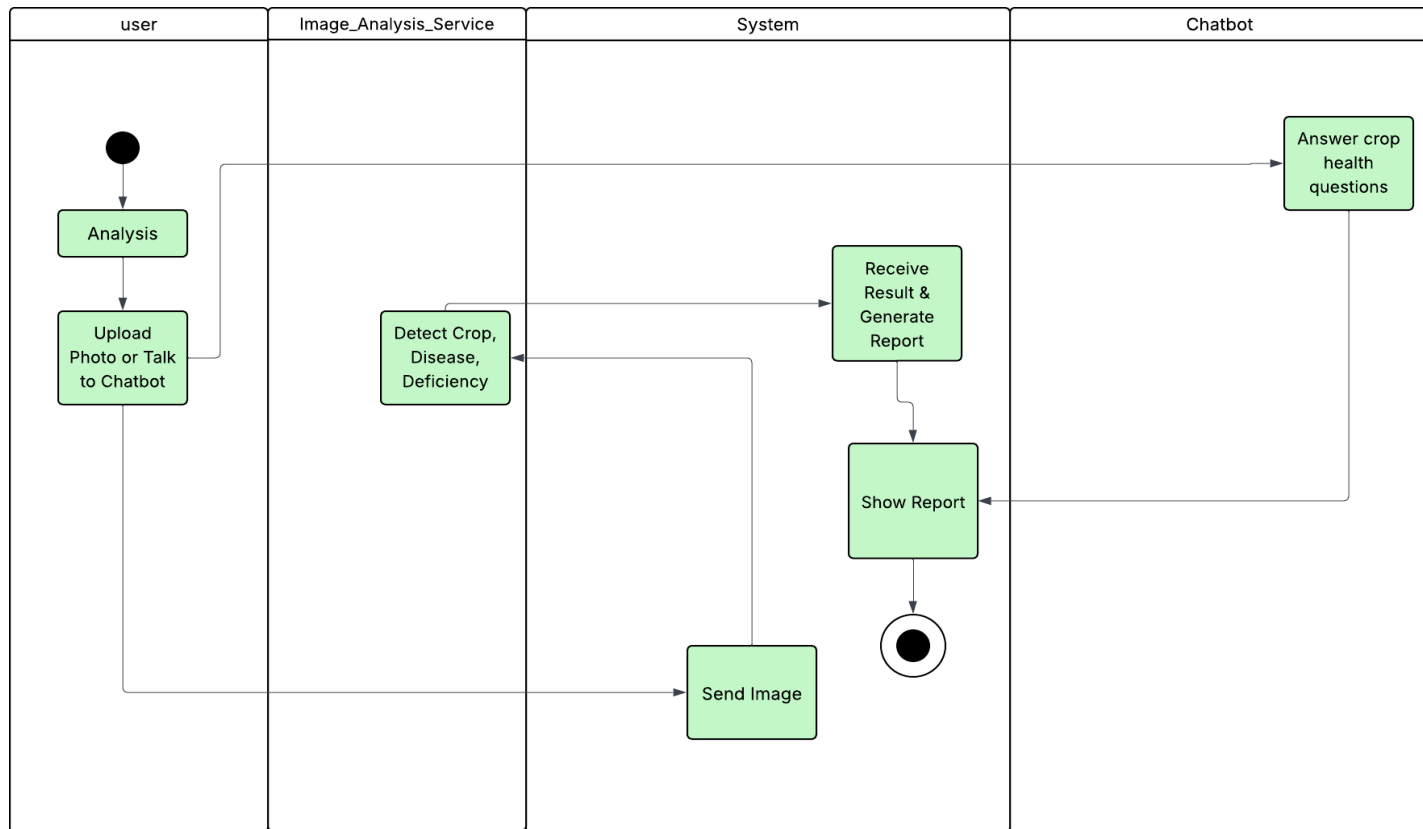
Reference: Activity Diagram Level – 1.1 (Account Management)



Swimlane Diagram Level: 1.2

Name: Digital Farming Assistant

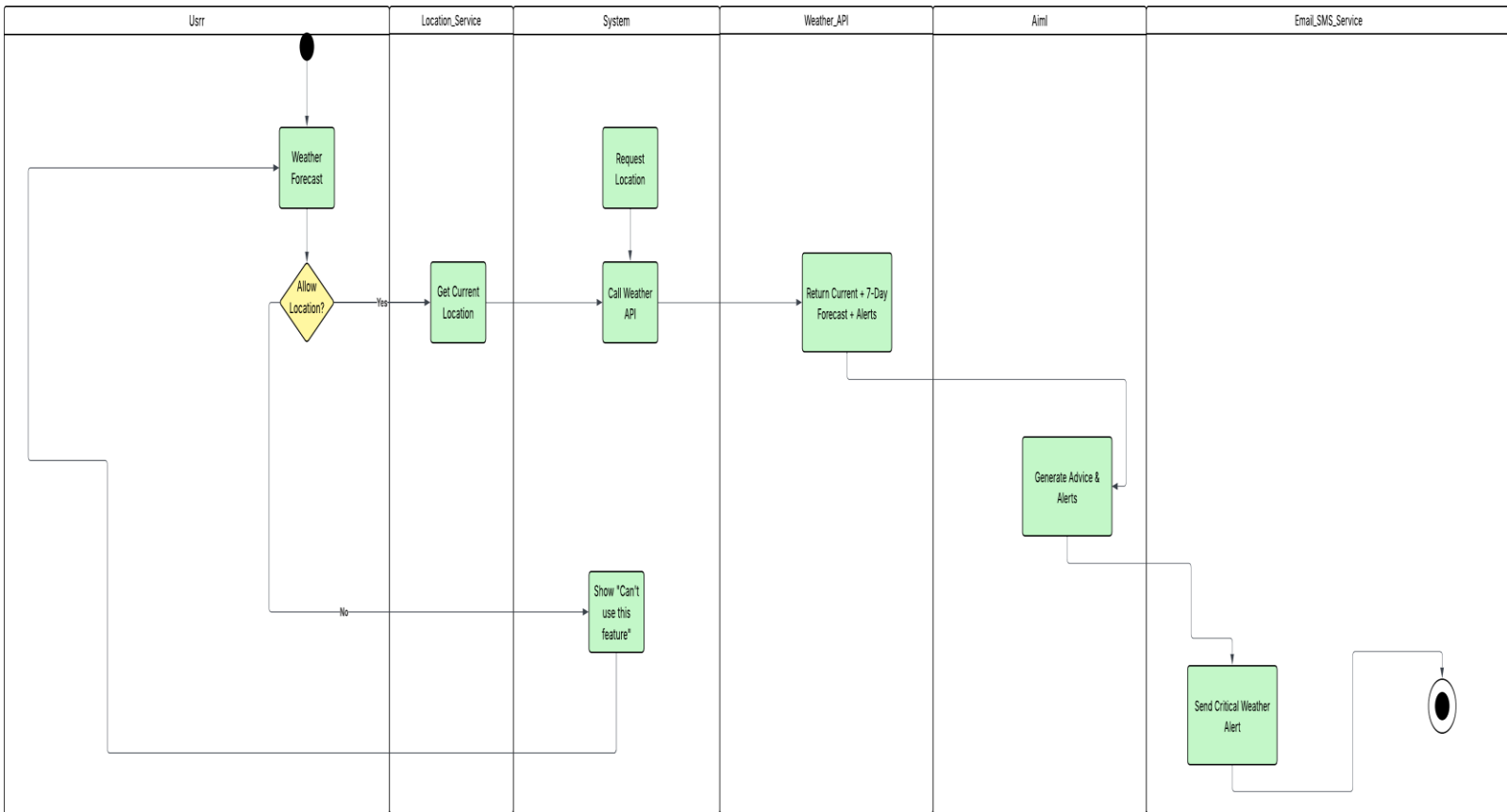
Reference: Activity Diagram Level – 1.2 (Crop Health Checker)



Swimlane Diagram Level: 1.3

Name: Digital Farming Assistant

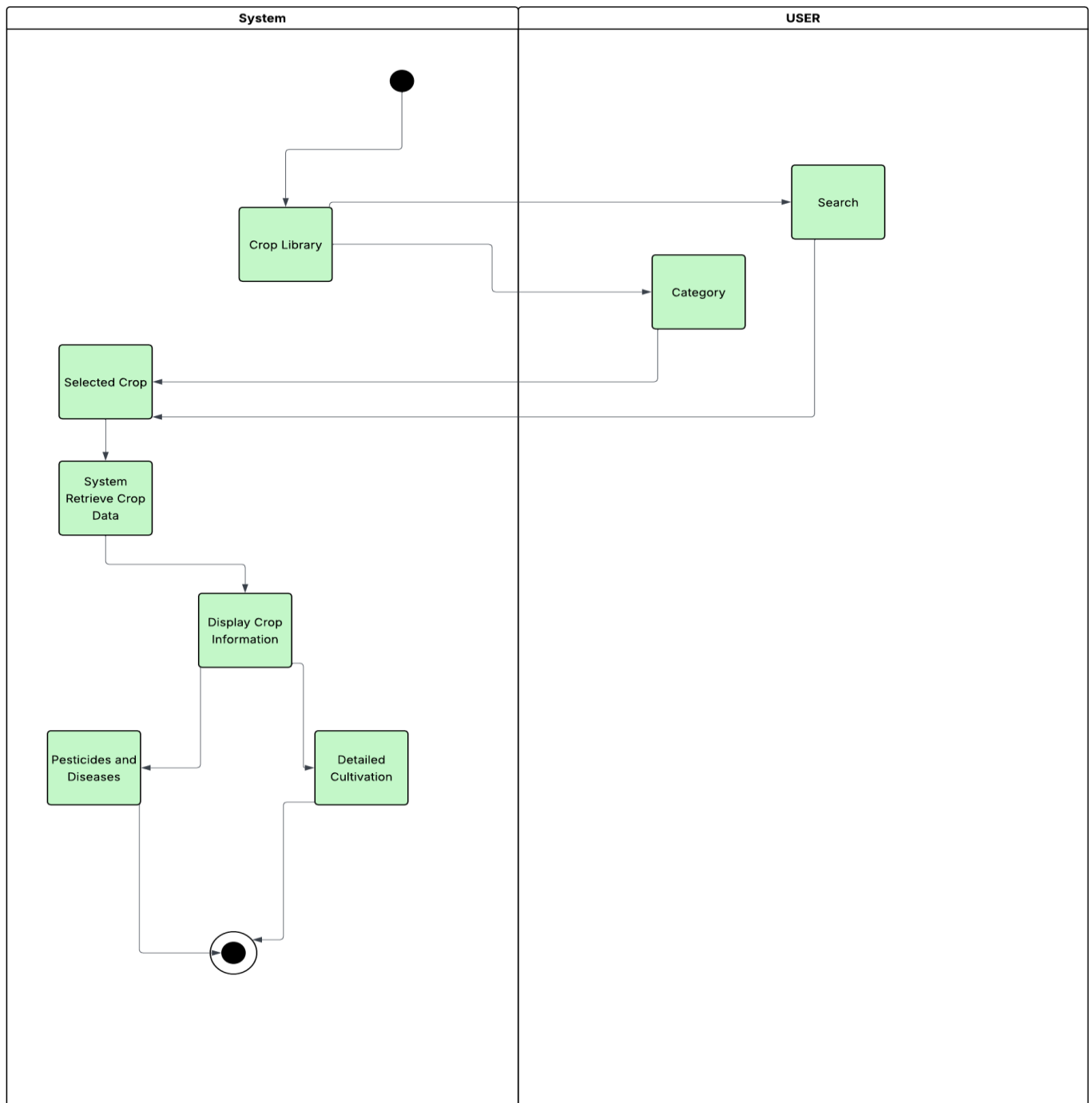
Reference: Activity Diagram Level – 1.3 (Weather Forecast)



Swimlane Diagram Level: 1.4

Name: Digital Farming Assistant

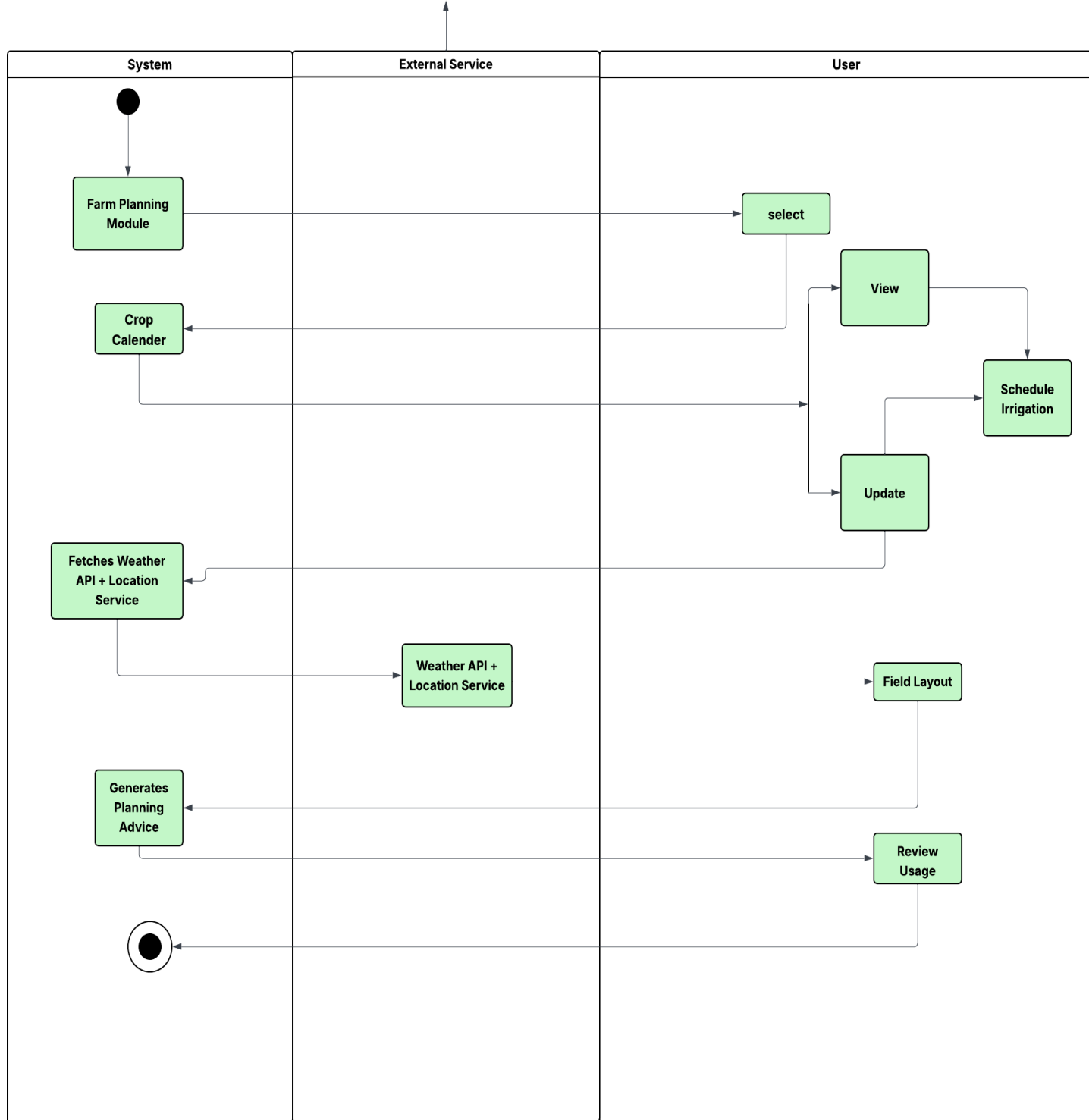
Reference: Activity Diagram Level – 1.4 (Crop Library)



Swimlane Diagram Level: 1.5

Name: Digital Farming Assistant

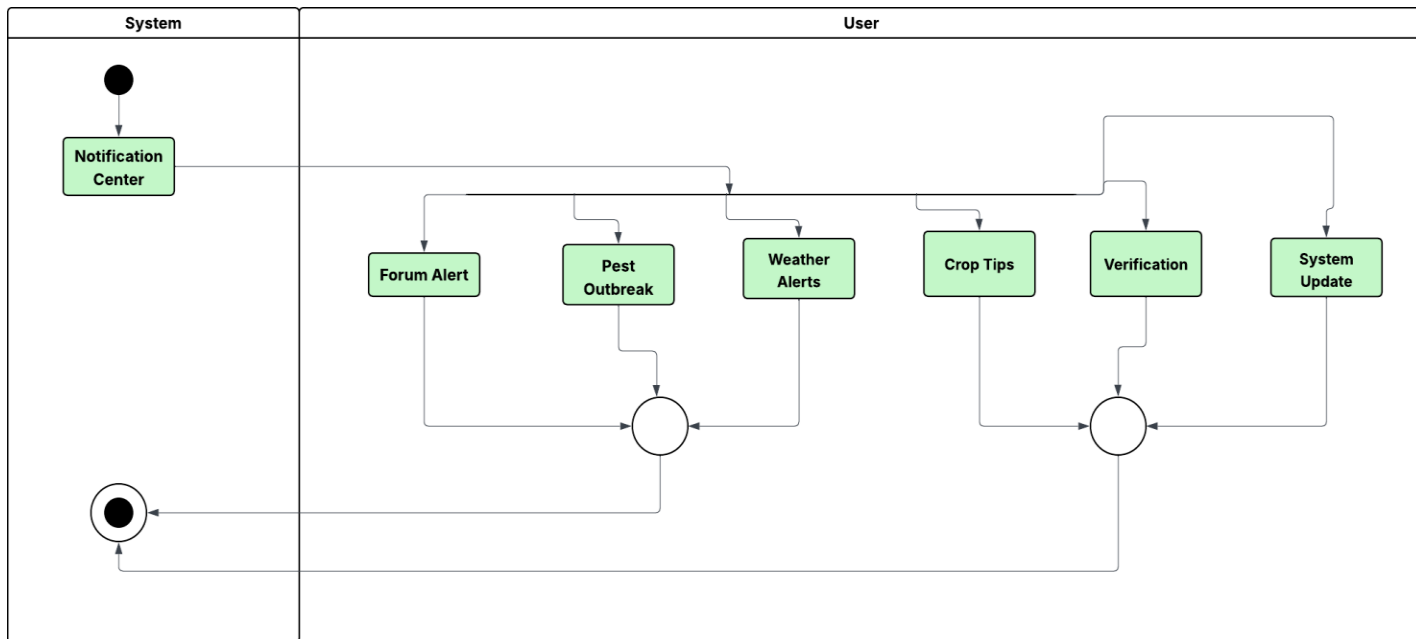
Reference: Activity Diagram Level – 1.5 (Farm Operations Management)



Swimlane Diagram Level: 1.6

Name: Digital Farming Assistant

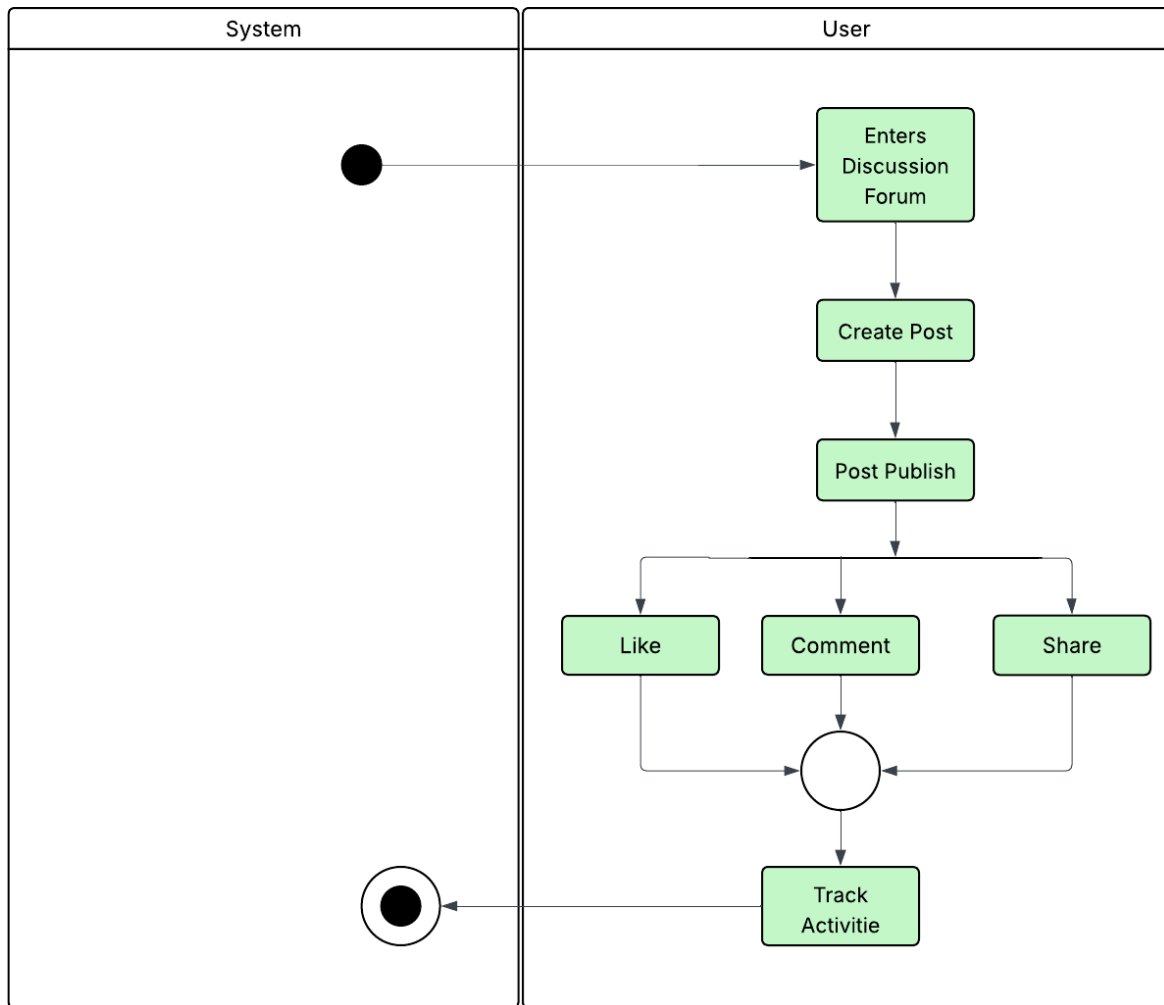
Reference: Activity Diagram Level – 1.6 (Notifications)



Swimlane Diagram Level: 1.7

Name: Digital Farming Assistant

Reference: Activity Diagram Level – 1.7 (Discussion Forum)



Data Based Modeling

Data Object Identification :

SL No.	Noun	Problem(p) / solution(s) space	attributes
1.	User	S	94,49,48,50
2.	Farmer	S	3,4,5,6,7,48,49,50
3.	Farmer ID	S	
4.	Farmer Name	S	
5.	Language Preference	P	
6.	Offline Accesibility	P	
7.	Location ID	S	
8.	Agronomist	S	9,94,10,49,48,92
9.	Agronomist ID	S	
10.	Experties Area	S	
11.	Registration Date	S	
12.	Crop ID	S	
13.	Crop Name	S	
14.	Image	S	16,17,15,96,12,3
15.	Upload Status	S	
16.	Image ID	S	
17.	File Path	S	
18.	Forecast Type	P	
19.	Weather Data	S	51,27,18,20,26,22,21,57,96
20.	Rain	S	
21.	Wind Speed	P	
22.	Temperature	P	
23.	Weather Alert	S	29,84
24.	Location	S	7
25.	Crop	S	12,13,66,31,63,41,42
26.	Humidity	S	
27.	Forecast Date	S	
28.	Crop Profile	S	92,32,26,41,42,89,74,96
29.	Planning Tips	S	
30.	Pesticide ID	S	
31.	Soil Type	S	
32.	Planting Method	P	
33.	Admin	S	34,94,35,48,49,50
34.	Admin ID	S	
35.	Role	S	
36.	Soil Requirement	S	75,76,
37.	Diagnosis	S	38,98,99,100,101,95,96
38.	Diagnosis ID	S	
39.	Result	S	
40.	Figure	S	

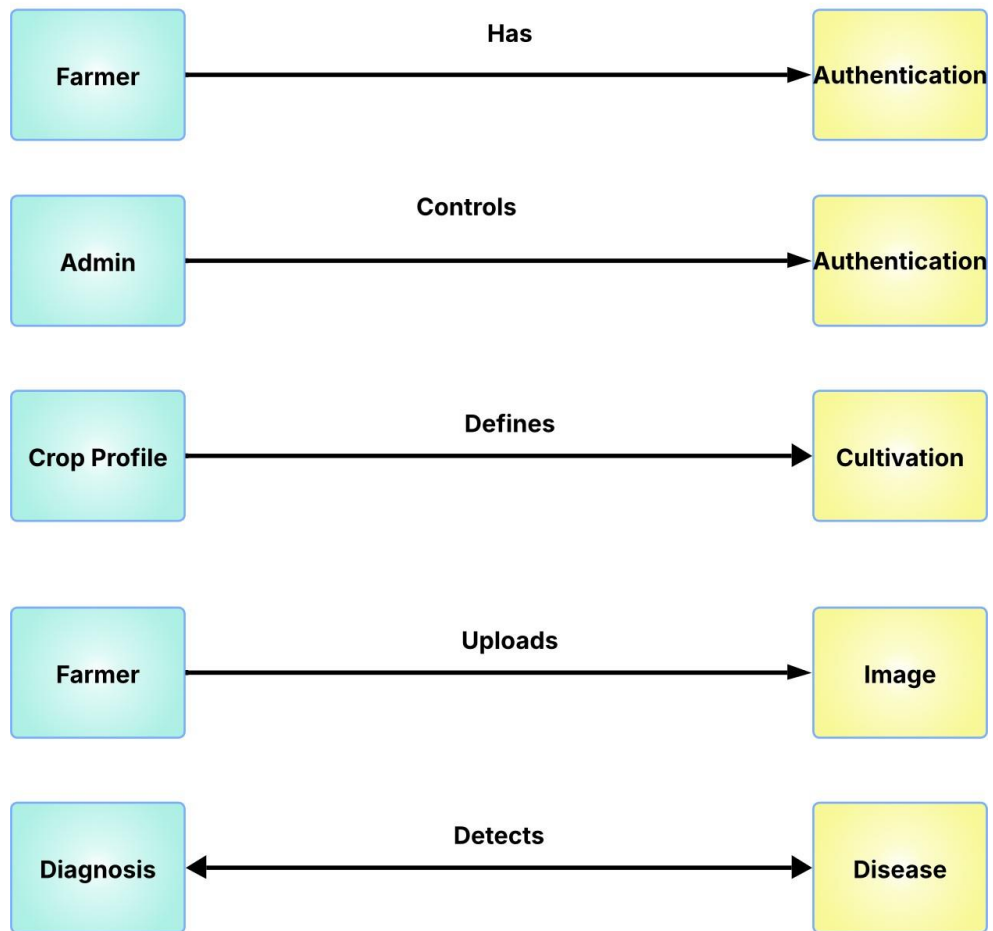
41.	Fertilizer	S	47,52
42.	Pesticide	S	47,52
43.	Post	S	72,14,96
44.	Comment	S	1,16
45.	Like	S	1,16
46.	Notification	S	46,47,102,96,61
47.	Message	S	
48.	Email	S	
49.	Phone Number	S	
50.	Password	S	
51.	ID	S	
52.	Authentication	S	51,54,53,55,93,96
53.	Login	S	
54.	Auth Method	S	
55.	Logout	S	
56.	Translation service	S	77
57.	Weather API	S	
58.	AI Assistant	S	51,105,60,101,104,103
59.	Chatbot	S	58
60.	Model Type	S	60,105
61.	Email/SMS	s	
62.	Image Analysis Service	S	105,60,101,38
63.	Disease	S	64,94,98,99,100,101,65,88,82
64.	Disease ID	S	
65.	Treatment Options	S	
66.	Crop Condition	P	47,30,66
67.	Weather Forecast	P	25,9
68.	Limitation	S	
69.	Irrigation	P	
70.	Internet	S	
71.	Community Forum	S	172,43,44,45,75,96
72.	Forum Post ID	S	
73.	Experience	S	
74.	Offline Access	S	
75.	Keyword Search	p	
76.	Database	S	
77.	Language	S	51,24
78.	AI/ML Model	S	68,69,67
79.	Schedule irrigation	S	
80.	Smartphone	S	
81.	Harvesting	S	
82.	Rot percentage	S	
83.	Crop library	P	107,106
84.	Guidance	P	
85.	User Interface	S	
86.	system update	S	96
87.	Crop tips	S	
88.	Pest outbreak	S	
89.	Field Layout	S	
90.	Cultivation	S	51,3,12,96,24,79,81,31,97

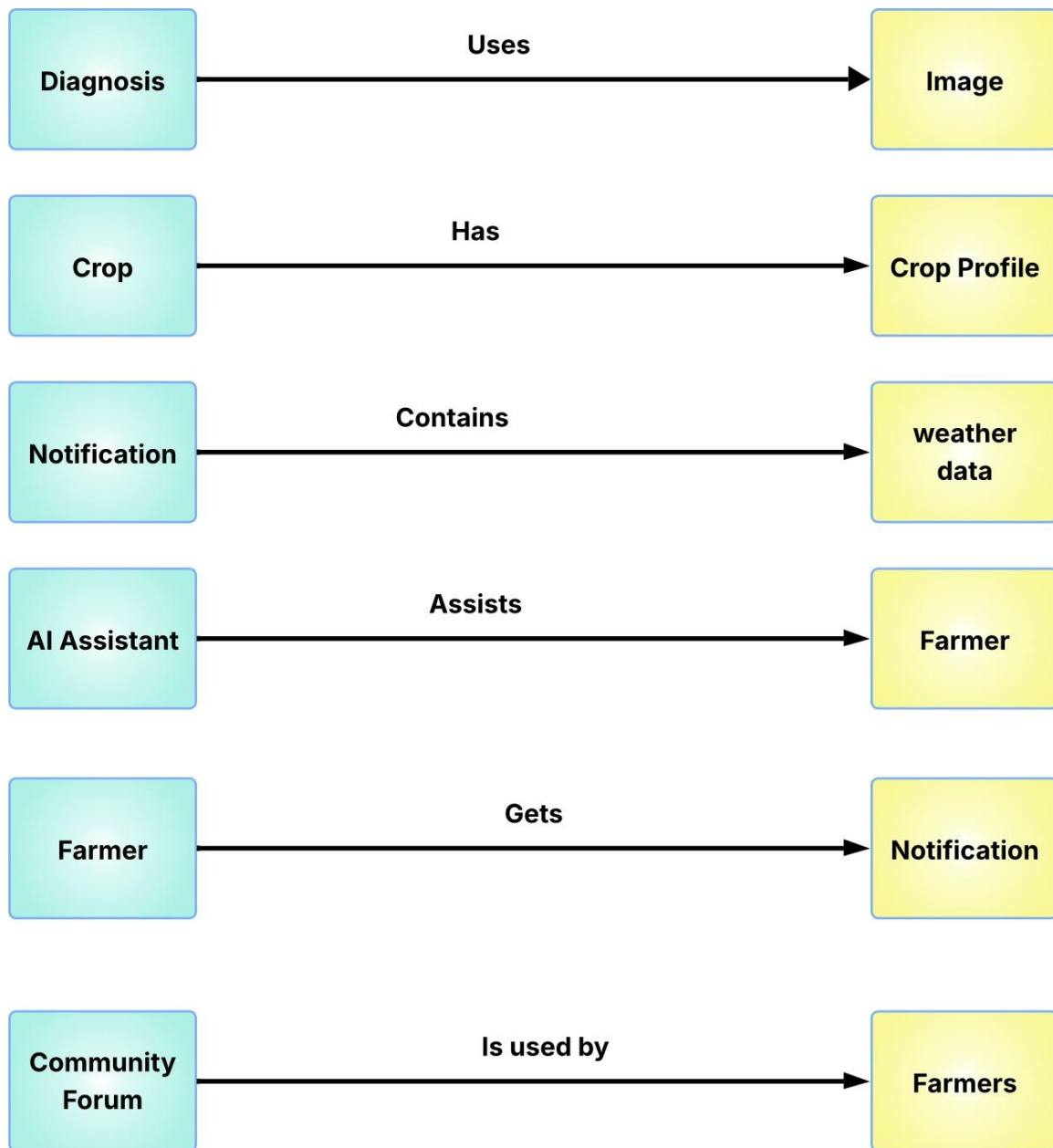
91.	Registration	S	
92.	Profile	S	94,49,50,51
93.	forgotten password	S	
94.	Name	S	
95.	recommendation	s	
96.	Timestamp	s	
97.	Yield Estimate		
98.	Symptoms	s	
99.	severity	s	
100.	description	p	
101.	accuracy	p	
102.	Alert type	s	
103.	Request	p	
104.	response	s	
105.	tools	s	
106.	Keyword Search	s	
107.	Disease	P	
108.	Crop List	S	

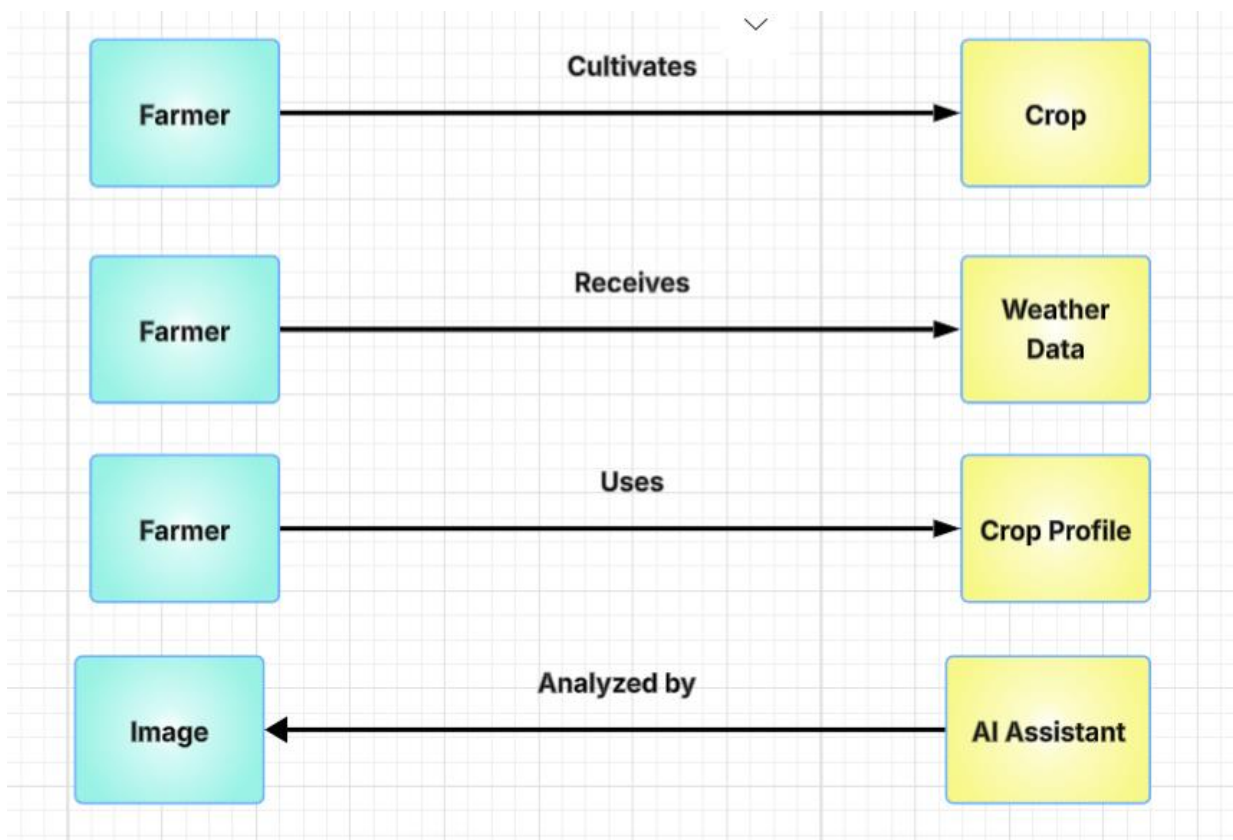
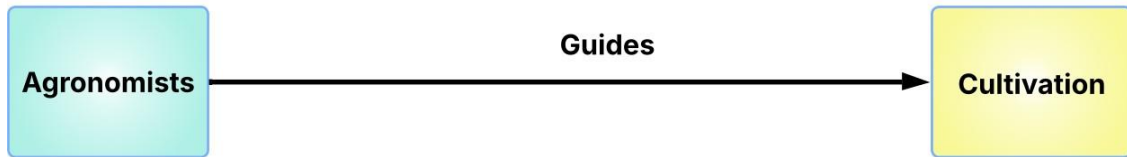
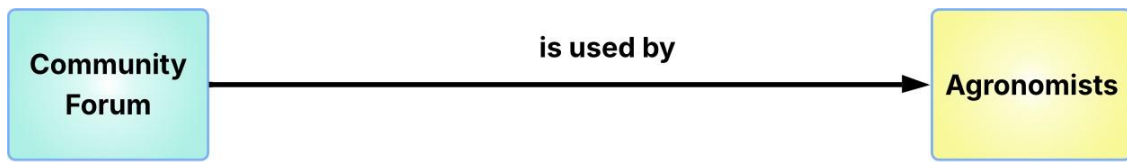
Final Data Objects:

1. Farmer
2. Agronomist
3. Image
4. Weather data
5. Crop
6. Crop Profile
7. Admin
8. Diagnosis
9. Notifications
10. Cultivation
11. Authentication
12. AI Assistant
13. Disease
14. Community Forum

Relationship Between Data objects







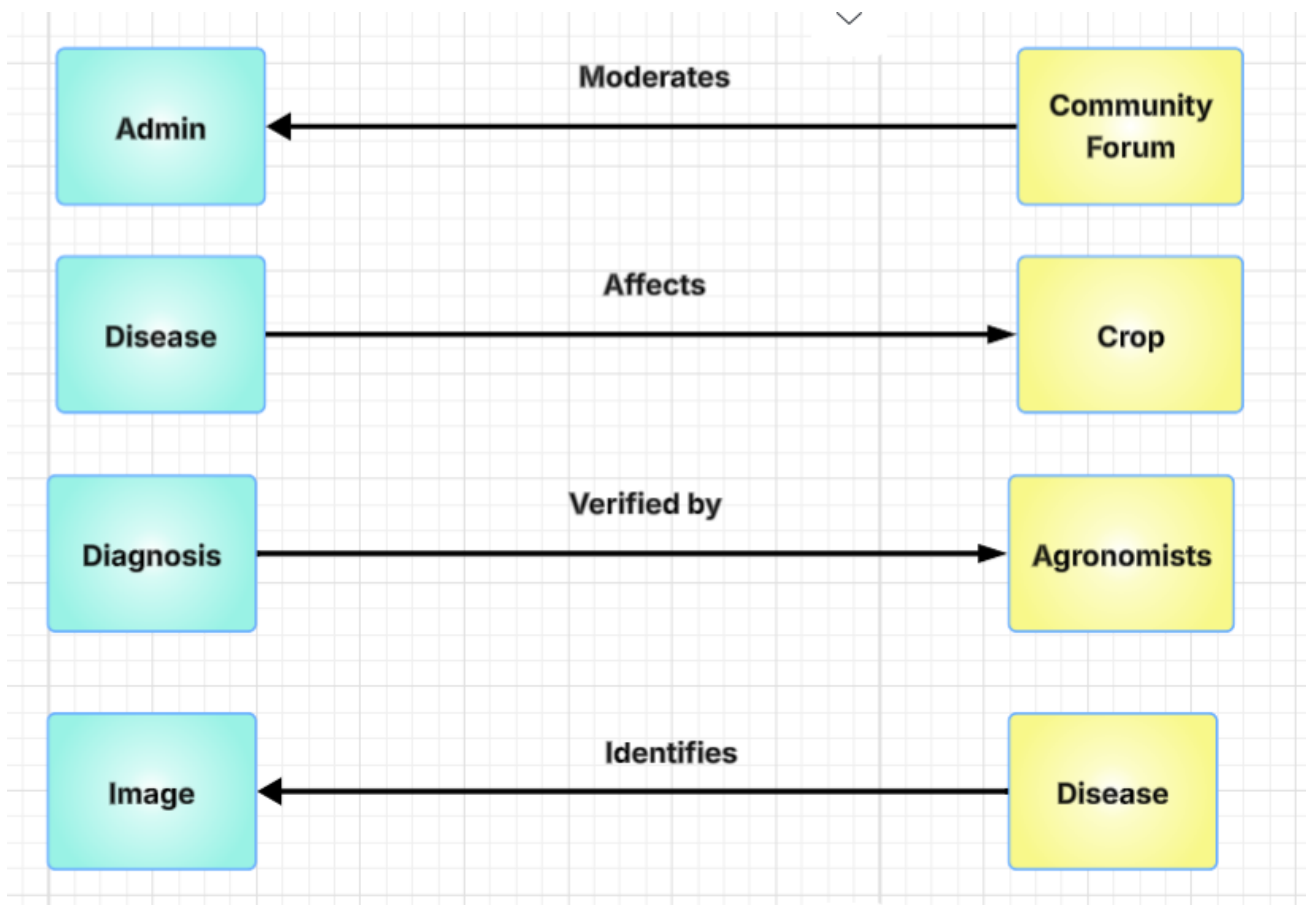


Fig : Relations between Data Objects

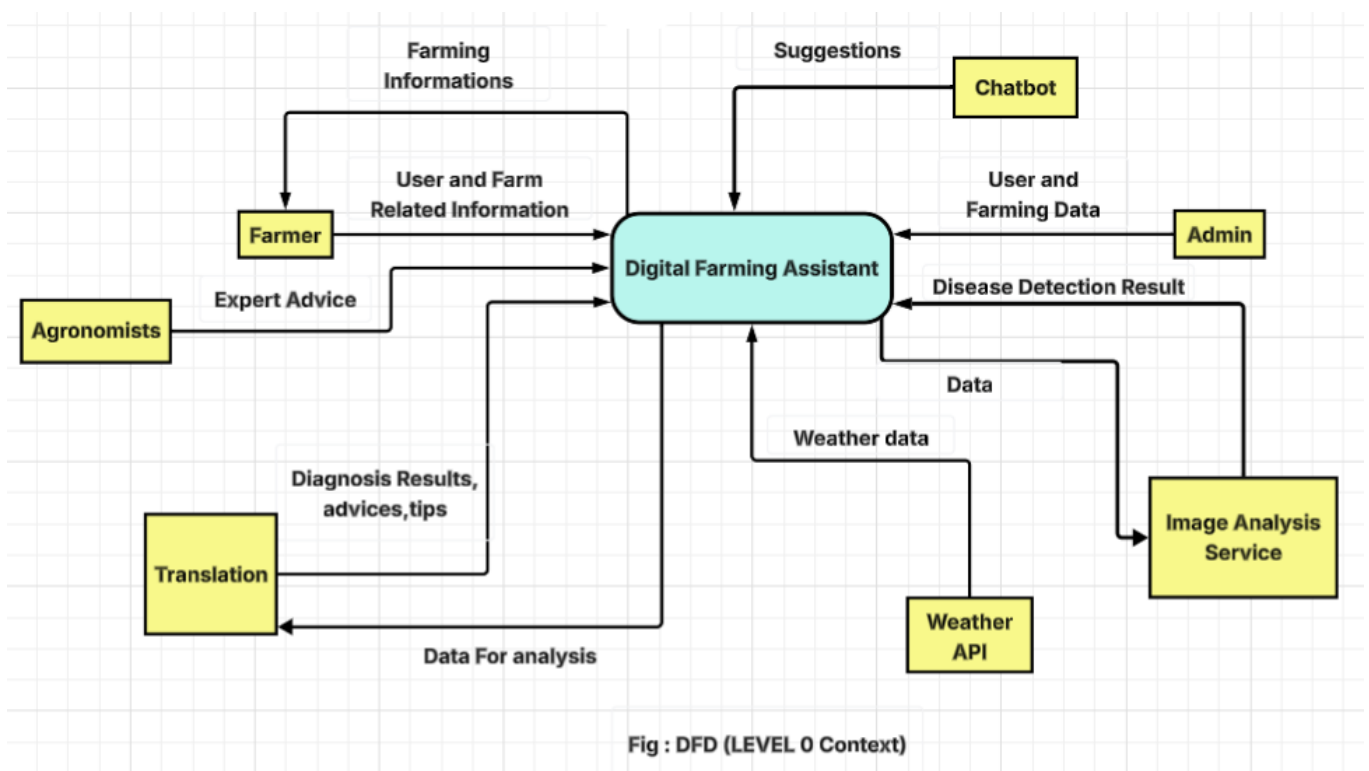
Data Flow Diagram :

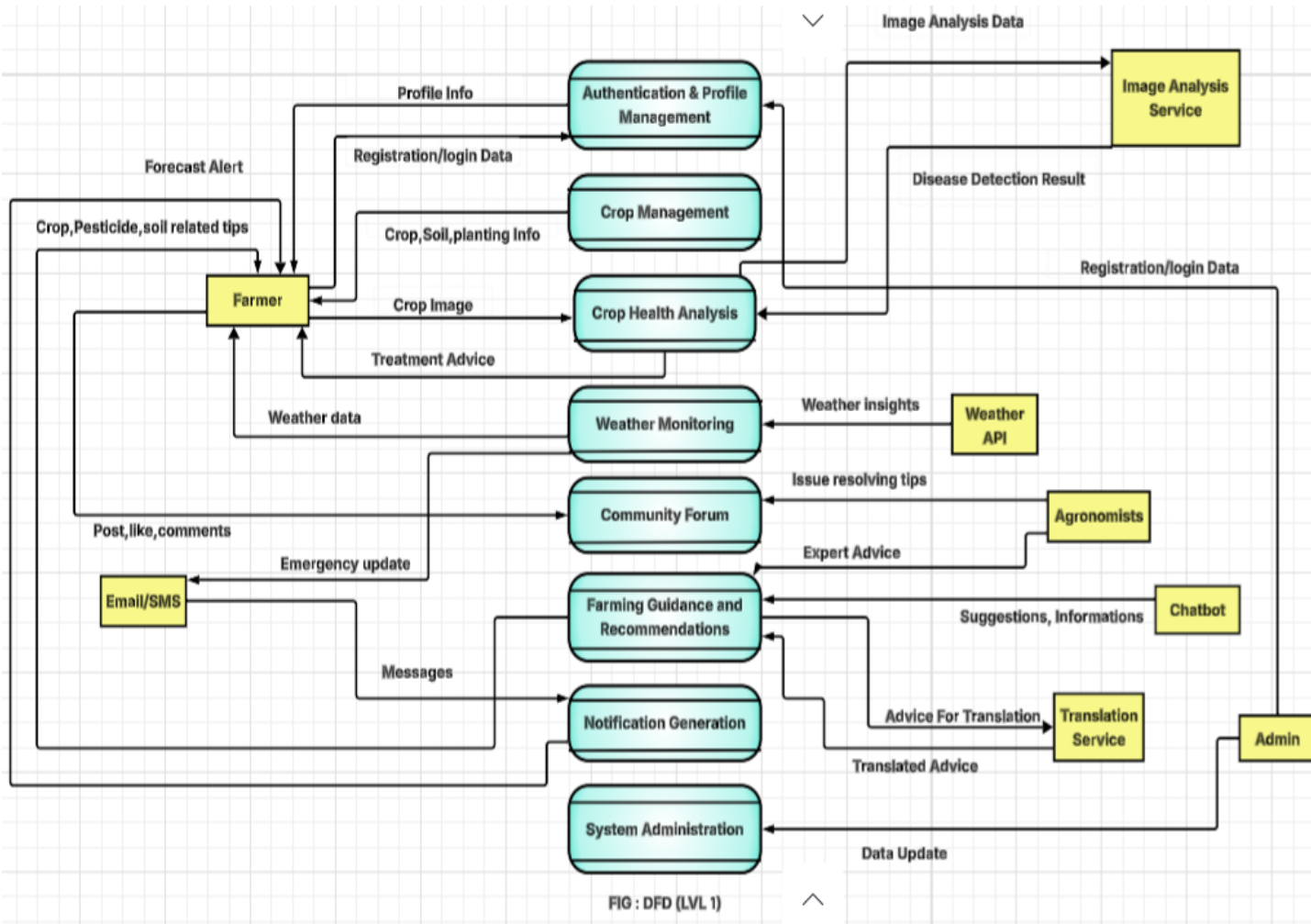
A data flow diagram, often abbreviated as DFD, is a graphical representation used in system analysis to show how data moves through a process or system. It focuses on the flow of information rather than the technical details of implementation, making it a powerful tool for understanding and communicating system functionality. In a DFD, processes are represented as circles or rounded rectangles, data stores as open-ended rectangles, external entities as squares, and arrows indicate the movement of data between these elements. By mapping out these flows, analysts can visualize how inputs are transformed into outputs and how information is stored or exchanged.

The purpose of a data flow diagram is to provide clarity about the way information travels within a system. It helps stakeholders, both technical and non-technical, to see the logical flow of data without being distracted by programming or hardware concerns. This makes it especially useful during the early stages of system design, when requirements are being gathered and validated. A DFD defines boundaries by showing which data originates outside the system and how it interacts with internal processes. It also highlights inefficiencies, redundancies, or potential risks in the flow of information, which can guide improvements in design.

Beyond communication, DFDs serve as a foundation for more detailed modeling. They can be expanded into multiple levels, starting with a high-level overview and then breaking down into more detailed diagrams that capture specific processes. This layered approach ensures that complex systems are understood step by step. In essence, a data flow diagram is both a map and a diagnostic tool: it shows how information travels, clarifies requirements, and supports the creation of efficient, well-structured systems.

Data Flow Diagram :





Class Based Modeling

Class-based modeling is part of the broader requirements modeling process, where analysts identify what a system must do to succeed. A **class** is essentially a blueprint: it defines attributes (data fields) and methods (functions or operations) that describe how objects of that class behave. Each object created from a class is an instance, carrying the same structure but potentially different values. By grouping related entities into classes, the system gains a clear, organized framework that reduces duplication and ensures consistency.

The purpose of class-based modeling is multifold. First, it helps translate real-world concepts into system components, making the design intuitive and aligned with business needs. Second, it ensures that data and operations are encapsulated, meaning they are bundled together in a way that promotes modularity and reusability. Third, it provides a foundation for building class diagrams in UML, which visually depict relationships among classes, their attributes, and their methods. This makes communication between stakeholders easier, as both technical and non-technical participants can understand how the system is structured.

Moreover, class-based modeling supports scalability and maintainability. By defining classes clearly, developers can extend or modify the system without disrupting unrelated parts. It also aids in database design, since classes often map directly to tables, ensuring that data is stored and accessed consistently. In short, class-based modeling is about capturing the essence of entities and their interactions, giving systems a logical backbone that guides both design and implementation.

List Of Nouns In Digital Farming Assistant

SL No.	Noun
1.	User
2.	Farmer
3.	Farmer ID
4.	Farmer Name
5.	Language Preference
6.	Offline Accesibility
7.	Location ID
8.	Agronomist
9.	Agronomist ID
10.	Experties Area
11.	Registration Date
12.	Crop ID
13.	Crop Name
14.	Image
15.	Upload Status
16.	Image ID
17.	File Path
18.	Forecast Type
19.	Weather Data
20.	Rain
21.	Wind Speed
22.	Temperature
23.	Weather Alert
24.	Location
25.	Crop
26.	Humidity
27.	Forecast Date
28.	Crop Profile
29.	Planning Tips
30.	Pesticide ID
31.	Soil Type
32.	Planting Method
33.	Admin
34.	Admin ID
35.	Role
36.	Soil Requirement
37.	Diagnosis
38.	Diagnosis ID
39.	Result
40.	Figure

41.	Fertilizer
42.	Pesticide
43.	Post
44.	Comment
45.	Like
46.	Notification
47.	Message
48.	Email
49.	Phone Number
50.	Password
51.	ID
52.	Authentication
53.	Login
54.	Auth Method
55.	Logout
56.	Translation service
57.	Weather API
58.	AI Assistant
59.	Chatbot
60.	Model Type
61.	Email/SMS
62.	Image Analysis Service
63.	Disease
64.	Disease ID
65.	Treatment Options
66.	Crop Condition
67.	Weather Forecast
68.	Limitation
69.	Irrigation
70.	Internet
71.	Community Forum
72.	Forum Post ID
73.	Experience
74.	Offline Access
75.	Keyword Search
76.	Database
77.	Language
78.	AI/ML Model
79.	Schedule irrigation
80.	Smartphone
81.	Harvesting
82.	Rot percentage
83.	Crop library
84.	Guidance
85.	User Interface
86.	system update

87.	Crop tips
88.	Pest outbreak
89.	Field Layout
90.	Cultivation
91.	Registration
92.	Profile
93.	forgotten password
94.	Name
95.	recommendation
96.	Timestamp
97.	Yield Estimate
98.	Symptoms
99.	severity
100.	description
101.	accuracy
102.	Alert type
103.	Request
104.	response
105.	tools
106.	Keyword Search
107.	Crop List

List Of Verbs In Digital Farming Assistant

SL No.	Noun
1.	Create Account
2.	Register
3.	Provide Information
4.	Provide User type
5.	Authenticate
6.	Login
7.	Confirm Signup
8.	Logout
9.	Capture Image
10.	Take Photo
11.	Upload
12.	Analyze
13.	Queue
14.	Detect
15.	Diagnose
16.	Choose

17.	Click
18.	Decide
19.	Estimate
20.	Explore
21.	Harvest
22.	Handle
23.	Guide
24.	Generate
25.	Recommend
26.	Verify
27.	Comment
28.	Post
29.	Like
30.	Search
31.	Filter
32.	View
33.	Save
34.	Track
35.	Notify
36.	Send
37.	Receive
38.	Schedule
39.	Forecast
40.	Advice
41.	Manage
42.	Update
43.	Reset
44.	Store
45.	Link
46.	Display
47.	Provide
48.	Enable
49.	Confirm
50.	Access
51.	Attach
52.	Reply
53.	Feedback
54.	Plan
55.	Record
56.	Reduce
57.	Track
58.	View
59.	Predict
60.	Irrigate

General Classification

Candidate classes are now characterized in seven general classes. The seven general characteristics are as follows :

1. External entities
2. Things
3. Events
4. Roles
5. Organizational units
6. Places
7. Structures

A candidate class is selected for special classification if it fulfills three or more characteristics.

SL No.	Solution Space Nouns	General Classification
1.	User	1,2,4
2.	Farmer	1,2,4
3.	Farmer ID	7
4.	Farmer Name	2
5.	Location ID	7
6.	Agronomist	1,2,4
7.	Agronomist ID	7
8.	Experties Area	2
9.	Registration Date	3,7
10.	Crop ID	7
11.	Crop Name	2
12.	Image	2,3,7
13.	Upload Status	2
14.	Image ID	7
15.	File Path	7
16.	Weather Data	2,3,7
17.	Wind Speed	2
18.	Location	2,6
19.	Crop	2,5,7
20.	Humidity	2
21.	Forecast Date	3,7
22.	Crop Profile	2,5,7
23.	Planning Tips	2
24.	Pesticide ID	7
25.	Soil Type	2
26.	Admin	2,4,5,7
27.	Admin ID	7
28.	Role	4,5
29.	Soil Requirement	2
30.	Diagnosis	2,3,7
31.	Diagnosis ID	7
32.	Result	2
33.	Figure	2
34.	Fertilizer	2
35.	Pesticide	2
36.	Post	2,7
37.	Comment	2
38.	Like	2
39.	Notification	2,3
40.	Message	2
41.	Email	2
42.	Phone Number	2
43.	Password	2

44.	ID	7
45.	Authentication	2,3,7
46.	Login	3
47.	Auth Method	2
48.	Logout	3
49.	Translation service	1,2
50.	Weather API	1,2
51.	AI Assistant	1,2,4
52.	Chatbot	1,2
53.	Model Type	2
54.	Email/SMS	2
55.	Image Analysis Service	2,7
56.	Disease	2,3,5
57.	Disease ID	7
58.	Treatment Options	2
59.	Internet	1,2
60.	Community Forum	2,5,7
61.	Forum Post ID	7
62.	Experience	2
63.	Offline Access	2
64.	Database	2,7
65.	AI/ML Model	2,7
66.	Smartphone	1,2
67.	Harvesting	3
68.	Crop library	2
69.	User Interface	2,7
70.	system update	3,7
71.	Crop tips	2
72.	Field Layout	2,6
73.	Cultivation	2,6,7
74.	Registration	3
75.	Profile	2,7
76.	Name	2
77.	recommendation	7
78.	Timestamp	7
79.	Yield Estimate	2
80.	description	2
81.	accuracy	2
82.	tools	2
83.	Keyword Search	2
84.	Crop List	2,7

Potential to be classes :

User

Farmer

Agronomist

Admin

Crop

Crop Profile

Image

Diagnosis

Disease

Weather Data

Notification

Authentication

AI Assistant

Community Forum

Cultivation

Selection Criteria

The candidate classes are then selected as classes by six Selection Criteria.

1. Retain information
2. Needed services
3. Multiple attributes
4. Common attributes
5. Common operations
6. Essential requirements

A candidate class generally becomes a class when it fulfills around Four or more characteristics.

SL No.	Solution Space Nouns	Selection Criteria
1.	User	1,2,3
2.	Farmer	1,2,3,4,5,6
3.	Agronomist	1,2,3,4,5
4.	Admin	1,2,3,4,5
5.	Crop	1,2,3,6
6.	Crop Profile	1,2,3,4,6
7.	Image	1,2,3,6
8.	Diagnosis	1,2,3,4
9.	Disease	1,2,3,6
10.	Weather Data	1,2,3,6
11.	Notification	1,2,3,5,6
12.	Authentication	1,2,3,4,5,6
13.	AI Assistant	1,2,3,5,6
14.	Community Forum	1,2,3,5
15.	Cultivation	1,2,3,6

Selected Classes

Farmer

Agronomist

Admin

Crop

Crop Profile

Image

Diagnosis

Disease

Weather Data

Notification

Authentication

AI Assistant

Community Forum

Cultivation

CRC Diagram

Class Cards

Class : Farmer	
Attributes	Methods:
- farmerID + name + phoneNumber - languagePreference - registrationDate - profileImage - locationID - offlineAccessEnabled - accountID	+ registerProfile() + uploadImage() + viewCropHistory() + postToForum() + saveCropProfile() + toggleOfflineAccess() + updateLocation()
Responsibility	Collaborator
• Manage personal profile and authentication • Upload crop images for diagnosis • View cultivation guidance and crop history • Participate in community forum • Receive weather alerts and notifications • Access crop library and profiles	- Authentication - Image - Diagnosis - Cultivation - Community Forum - Notification - Crop Profile

Class : Agronomist	
Attributes	Methods:
- agronomistID + name + phoneNumber + expertiseArea - verifiedStatus - profileImage - location - languagePreference - verificationDate	+ verifyDiagnosis() + addForumComment() + createCropProfile() + moderateForum() + updateExpertise() + viewFarmerCultivation()
Responsibility	Collaborator
• Verify and validate diagnoses • Provide expert recommendations • Advice in community forum • Share expertise in specific areas • View farmer cultivation data • Upload educational content	Diagnosis Community Forum Crop Profile Notification Authentication

Class : Admin	
Attributes	Methods:
- adminID + name + email + phoneNumber - expertiseArea - role - lastLogin - location - ProfileImage	+ manageUser() + moderateForum() + updateSystemConfig() + viewSystemLogs() + generateAnalytics report() + manageAuthenticationSettings()
Responsibility	Collaborator
• Manage System Users • Monitor and moderate Forum activities • Update System Configurations • View System logs and analytics • Manage authentication settings • Handle escalated issues	Authentication Community Forum Farmer Agronomist SystemLog

Class : Crop	
Attributes	Methods:
- cropID + name + season + soilType + growthDuration - commonDiseases - recommendedFertilizers - recommendedPesticides + tips - imageReference	+ getCropDetails() + findRelatedDiseases() + getGrowingSeason() + SuggestCultivationMethods() + filterBykeyword()
Responsibility	Collaborator
• Store basic crop information • Link to multiple crop profiles • Reference common diseases • Provide cultivation requirements • Support library search functionality	- Crop Profile - Disease - Cultivation - Image

Class : Crop Profile	
Attributes	Methods:
-profileID - cropID + plantingMethod + soilRequirement + fertilizerRecommendation + pesticideRecommendation + fieldLayoutGuide + OffilneSavedFlag - lastViewedDate - createdBy - createdDate	+ saveForOffline() + updateProfile() + getPlantingGuidance() + getFieldLayout() + trackUsage()
Responsibility	Collaborator
• Store detailed cultivation information • Provide planting and care guidance • Link specific field layouts • Support offline saving • Track usage and preferences	Crop Farmer Agronomist Cultivation

Class : Image	
Attributes	Methods:
- imageID - farmerID + timestamp - filePath + cropType + symtopsDetected + fieldLocation - OffilneFag - cropID	+ uploadToServer() + queueForOfflineUpload() + analyzeImage() + getImageMetadata() + updateStatus()
Responsibility	Collaborator
• Store and manage crop images • Track upload status • Link to farmer and crop • Provide metadata for analysis • Manage File storage paths	- Farmer - Diagnosis - Crop - AI Assistant

Class : Diagnosis	
Attributes	Methods:
- diseaseID - imageID - farmerID - cropID + diseaseDetected + severityLevel + accuracyScore + description + timestamp + recommendation - verifiedBy - verificationDate	+ createDiagnosis() + updateSeverity() + addAgronomistVerification() + getTreatmentOptions() + calculateAccuracy()
Responsibility	Collaborator
- Record disease detection results - Store analysis accuracy and confidence - Link to images and farmers - Provide treatment recommendations - Track verification status by agronomists	Image Farmer Disease Agronomist AI Assistant

Class : Disease	
Attributes	Methods:
- diseaseID + name + symptoms + severityLevel + affectedCrops + treatmentOptions - pestOutbreakFlag - rotPercentageRange - detectionAccuracy + preventionTips	+ getSymptoms() + SuggestTreatments() + checkAffectedCrops() + updateOutbreakStatus() + getPreventionGuidelines()
Responsibility	Collaborator
• Store disease characteristics • Provide symptom descriptions • List treatment options • Track affected crops • Provide prevention tips	Diagnosis Crop AI Assistant

Class : Weather Data	
Attributes	Methods:
- weatherDataID + location + date + temperature + humidity + rainfall + windSpeed + forecastType - source - timestamp - alertTriggered	+ fetchForecast() + checkForAlerts() + getHistoricalData() + updateWeatherData() + calculateGrowingConditions()
Responsibility	Collaborator
• Store current and forecast weather • Trigger weather alerts • Provide location-specific data • Track data sources and timestamps • Support cultivation planning	Notification Cultivation

Class : Notification	
Attributes	Methods:
- notificationID - recipientID + type + messageContent - timestamp - readStatus + deliveryMethod - priority - relatedEntityID	+ sendNotification() + markAsRead() + resendFailed() + filterByType() + getUnreadCount()
Responsibility	Collaborator
• Deliver alerts and messages • Track delivery status • Support multiple delivery methods • Manage recipient lists • Store message content and timing	- Farmer - Agronomist - Admin - Weather Data - Authentication

Class : Authentication	
Attributes	Methods:
- authID - farmerID - adminID - AgronomistID + phoneNumber + authMethod - lastLoginTimestamp - deviceInfo + languagePreference - forgotPasswordFlag - accountStatus	+ login() + logout() + resetPassword() + resendFailed() + updateAuthMethod() + getUnreadCount() + lockAccount()
Responsibility	Collaborator
• Responsibilities • Manage user login/logout • Handle password resets • Store authentication methods • Track login history • Manage session tokens • Support multiple auth types	- Farmer - Agronomist - Admin - Notification

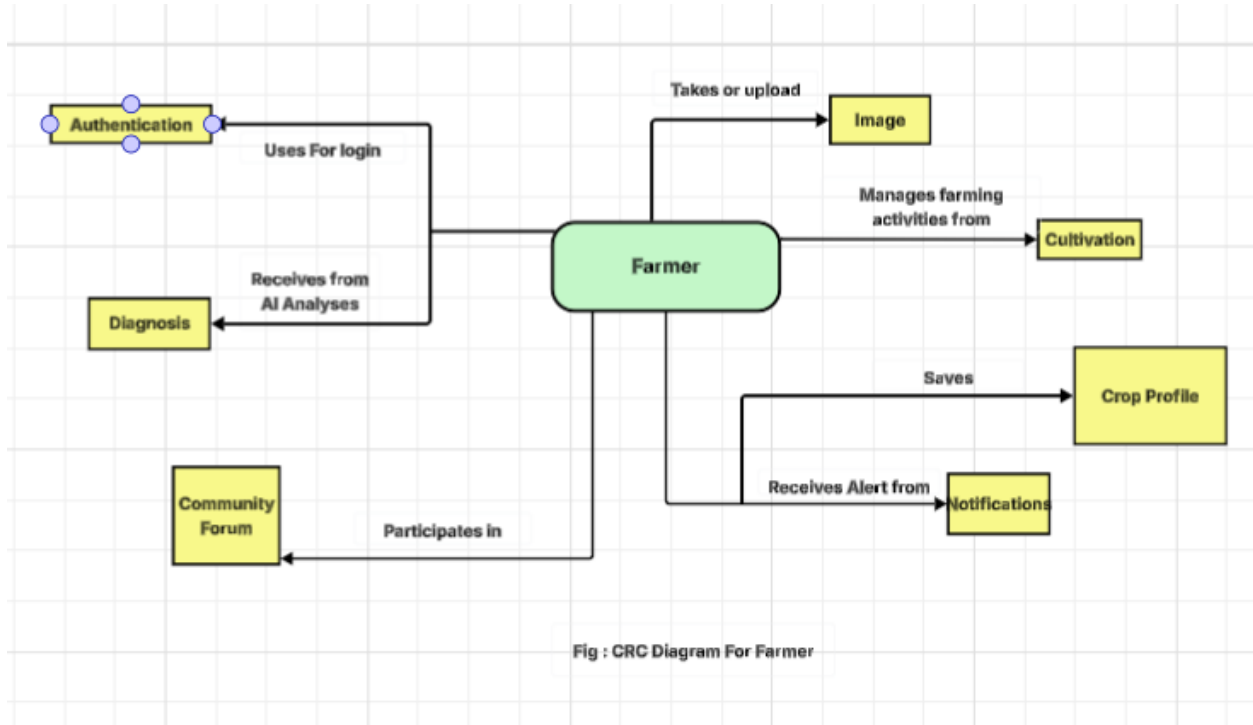
Class : AI Assistant	
Attributes	Methods
- assistantID - toolUsed + AccuracyScore + responseContent + serviceType - requestID - linkedDiagnosisID - modelType - analysisTimestamp	+ analyzeImage(imageID) + provideGuidance(topic, context) + calculateConfidence(metrics) + updateModel(newModel) + logRequestResponse(request, response) + getRecommendations(context)
Responsibility	Collaborator
• Analyze images for diseases • Provide cultivation guidance • Generate recommendations • Track confidence scores • Manage analysis tools/models • Link to service requests	Image Diagnosis Cultivation Crop Profile Weather Data

Class : Community Forum	
Attributes	Methods:
- forumID + name + description + category -moderatorIDs +totalPosts +totalComments +searchableFlag -creationDate -lastActivity	+ createPost(authorID, content) + addComment(postID, authorID, content) + likePost(postID, userID) + searchPosts(keyword) + moderateContent(postID, action) + getForumStats()
Responsibility	Collaborator
• Manage discussion posts • Handle comments and likes • Support search functionality • Moderate content • Track user engagement • Categorize topics	Farmer Agronomist Admin

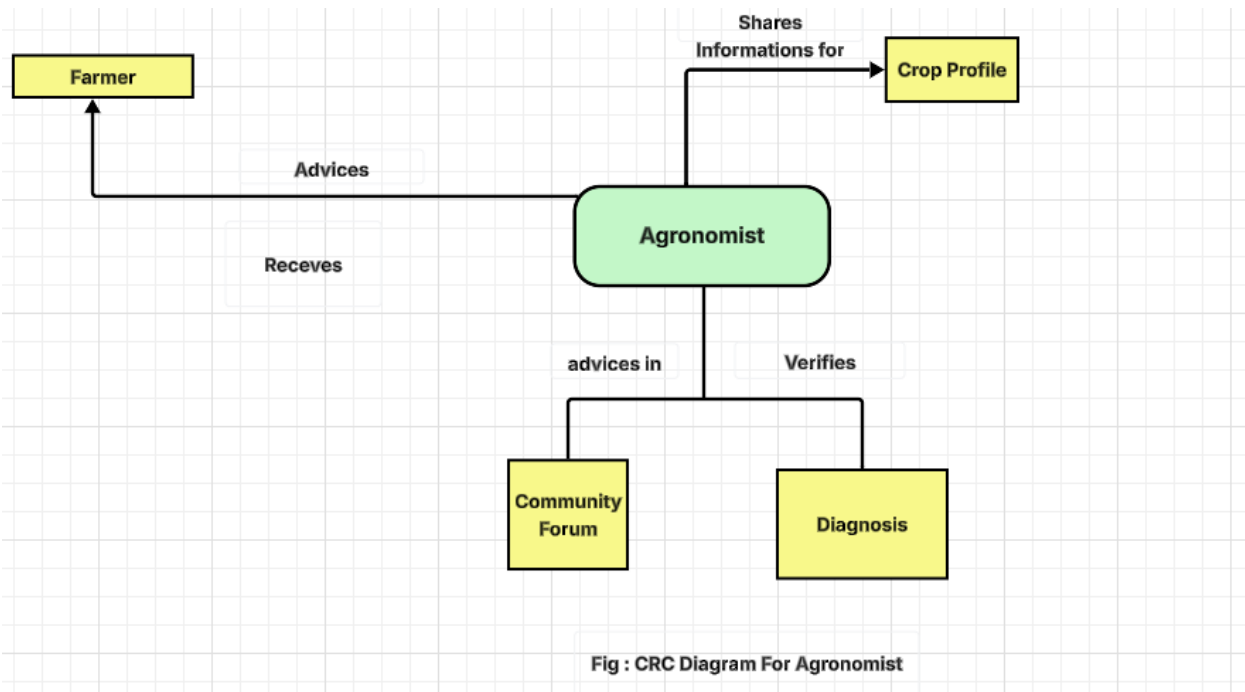
Class : Cultivation	
Attributes	Methods:
-cultivationID -farmerID - cropID +startDate +endDate +fieldLocation +irrigationSchedule +harvestingDate +soilType -yieldEstimate - actualYield - status	+ scheduleIrrigation(dates, amounts) + updateGrowthStage(stage) + recordHarvest(yield, date) + calculateYieldEstimate() + getWeatherImpact() + adjustSchedule(weatherData)
Responsibility	Collaborator
- Track farming activities - Schedule irrigation - Monitor growth stages - Estimate yields - Record harvest data - Link to weather conditions	Farmer Crop Weather Data AI Assistant Crop Profile

CRC Diagram

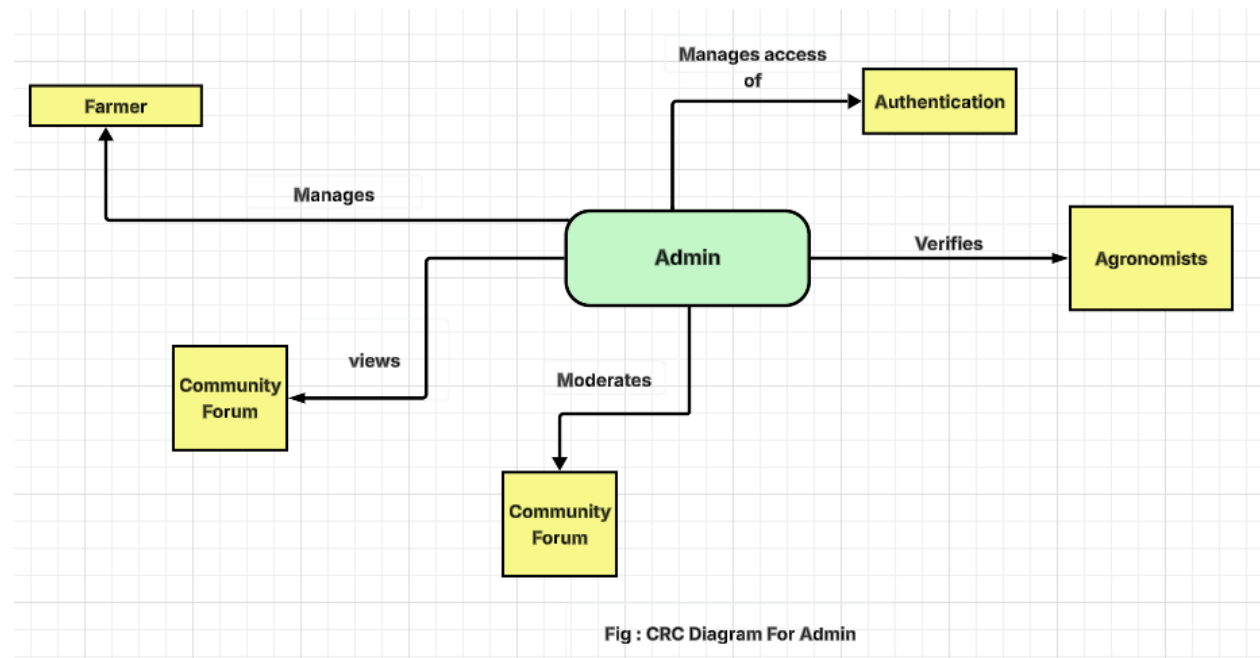
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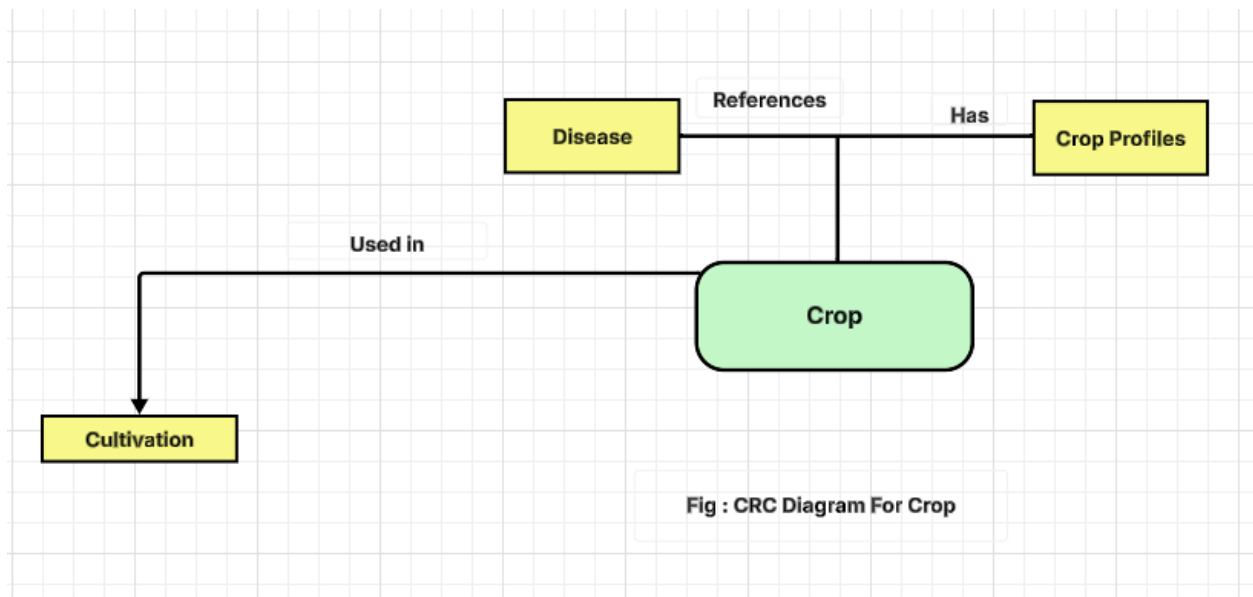
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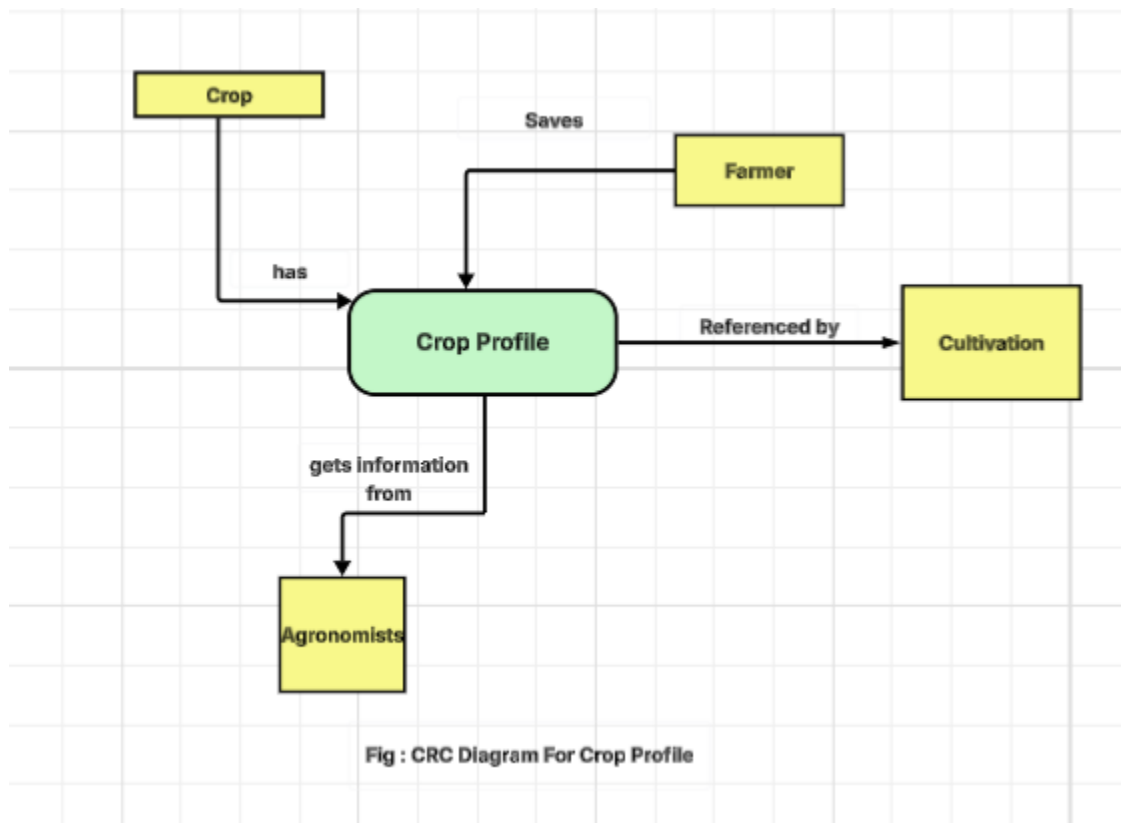
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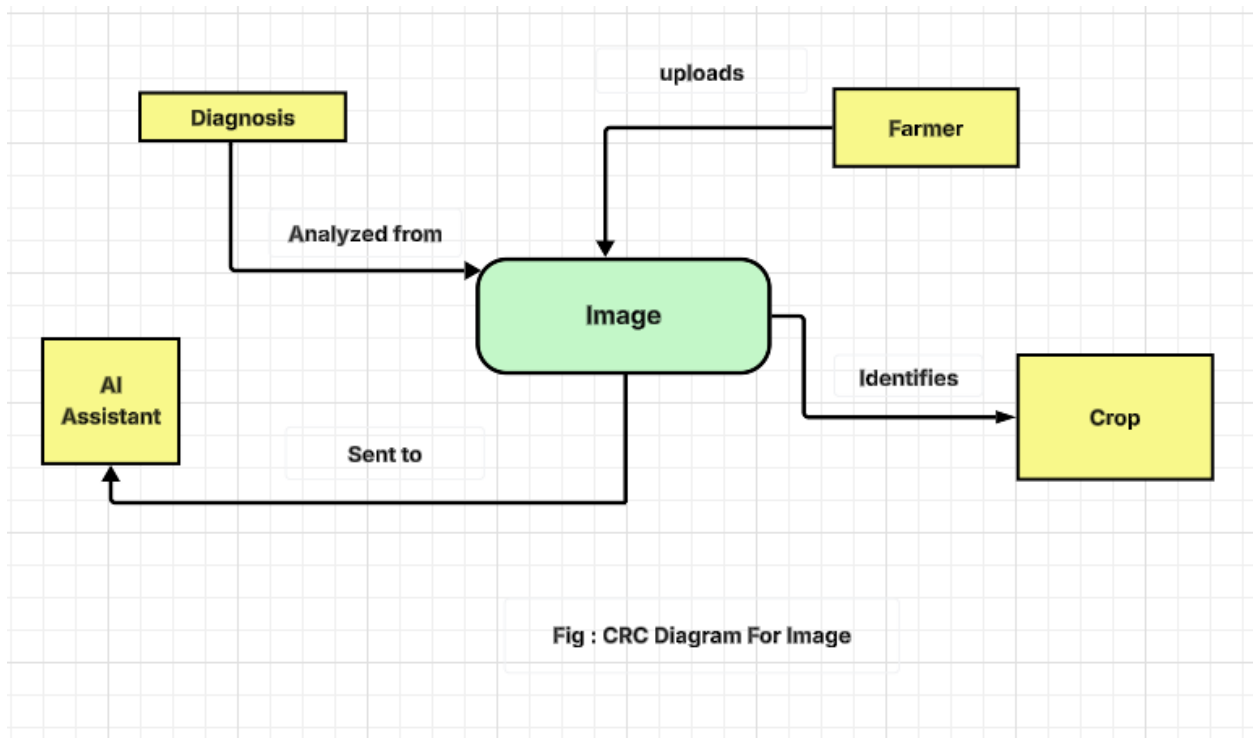
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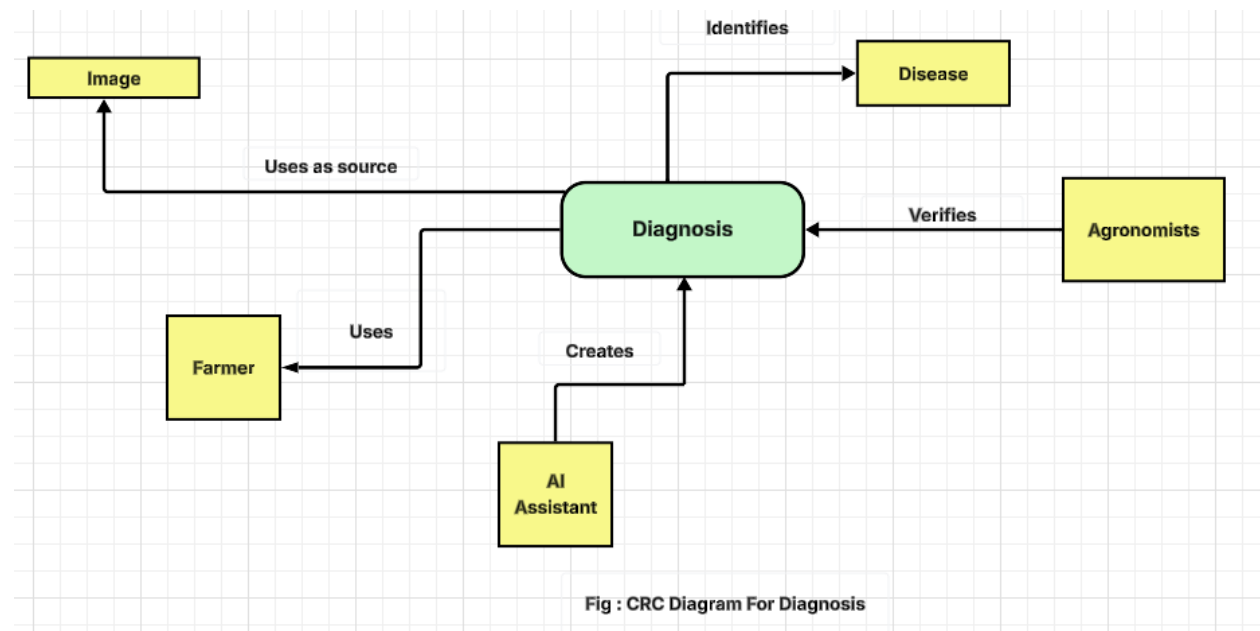
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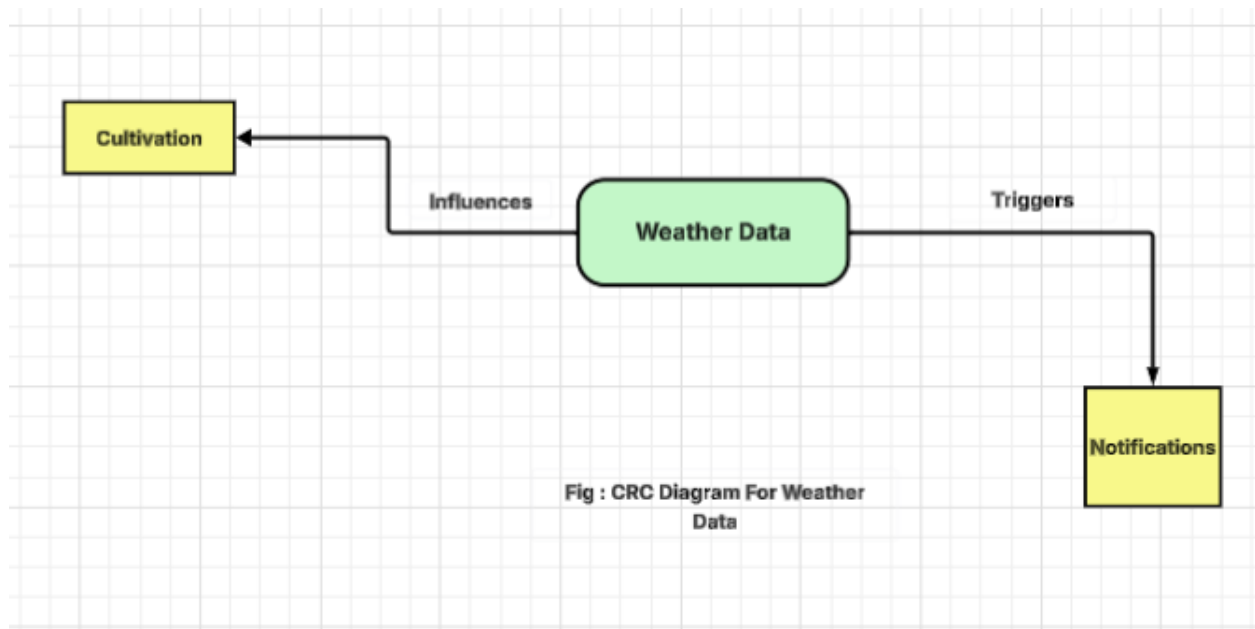
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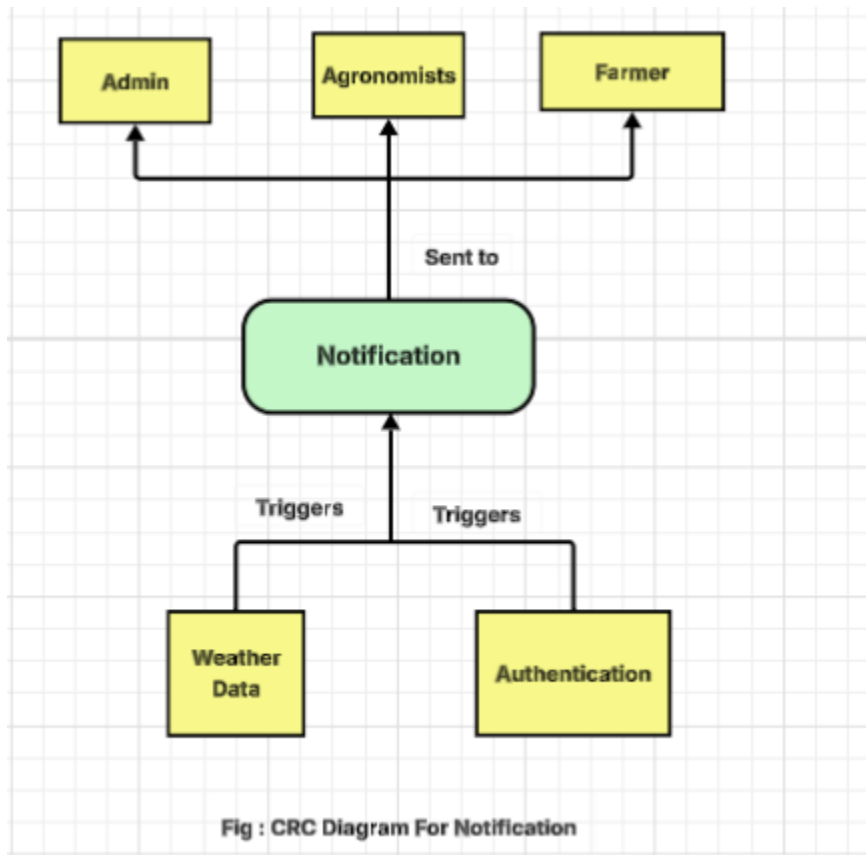
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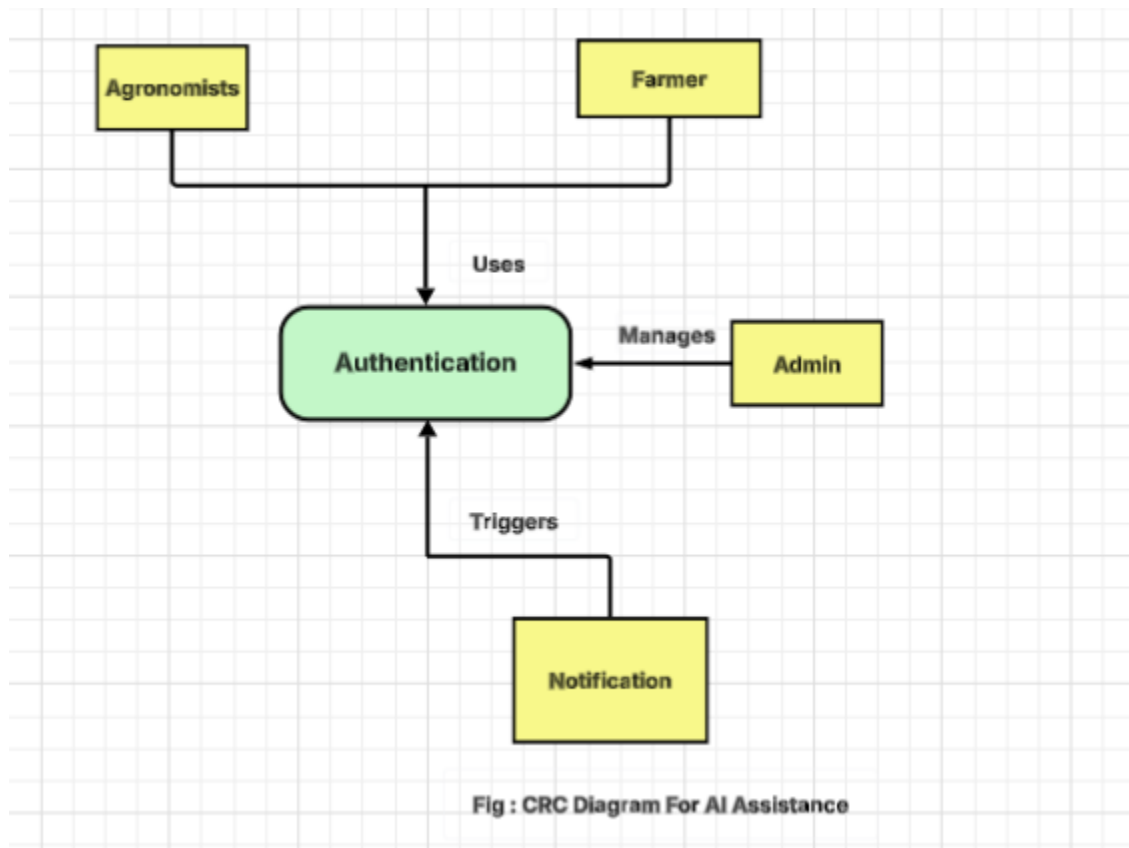


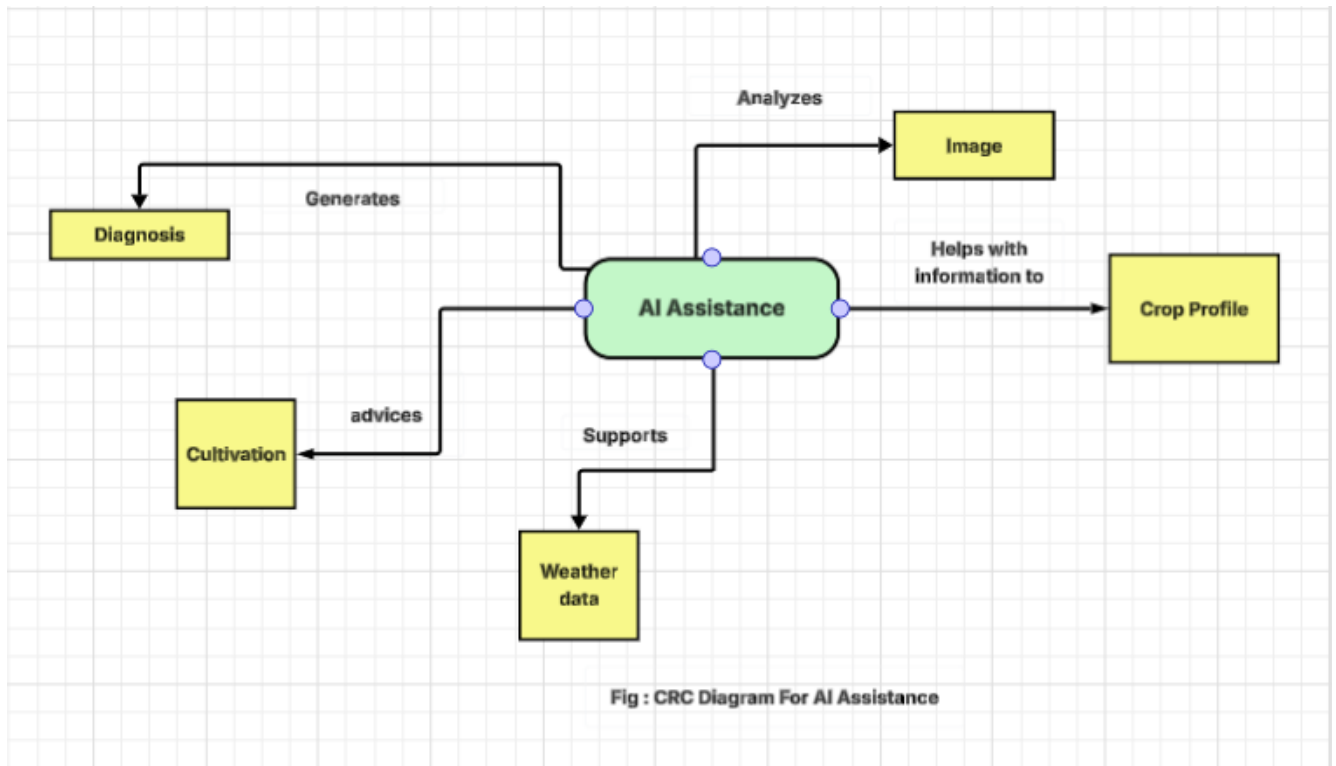
ID – 8



ID – 10







ID – 13

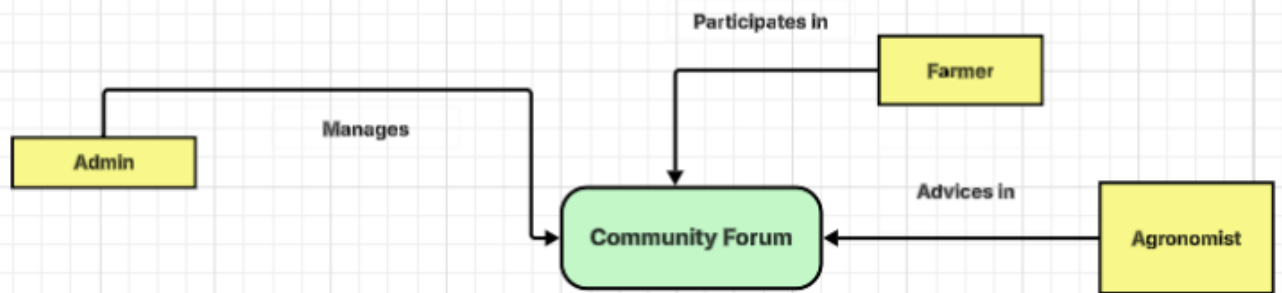
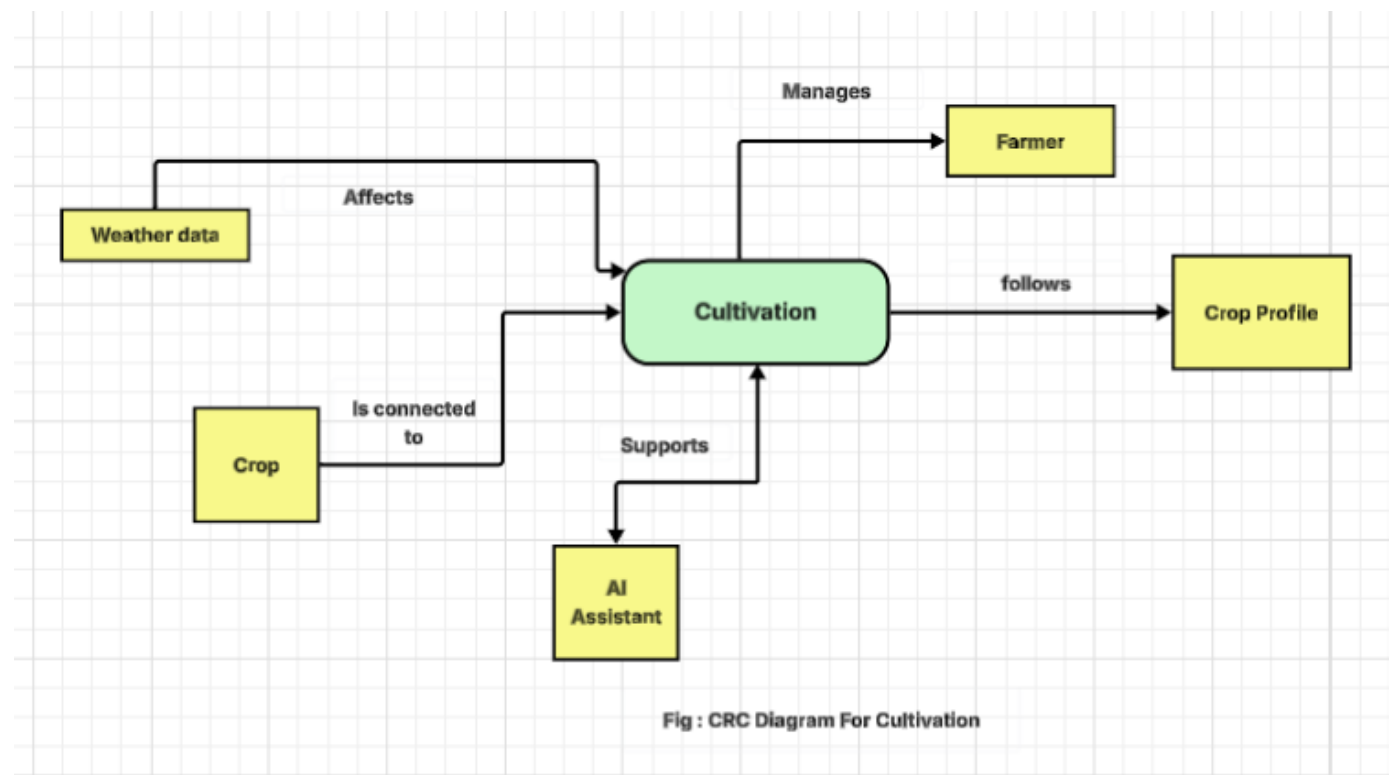


Fig : CRC Diagram For Community Forum



Main CRC Diagram

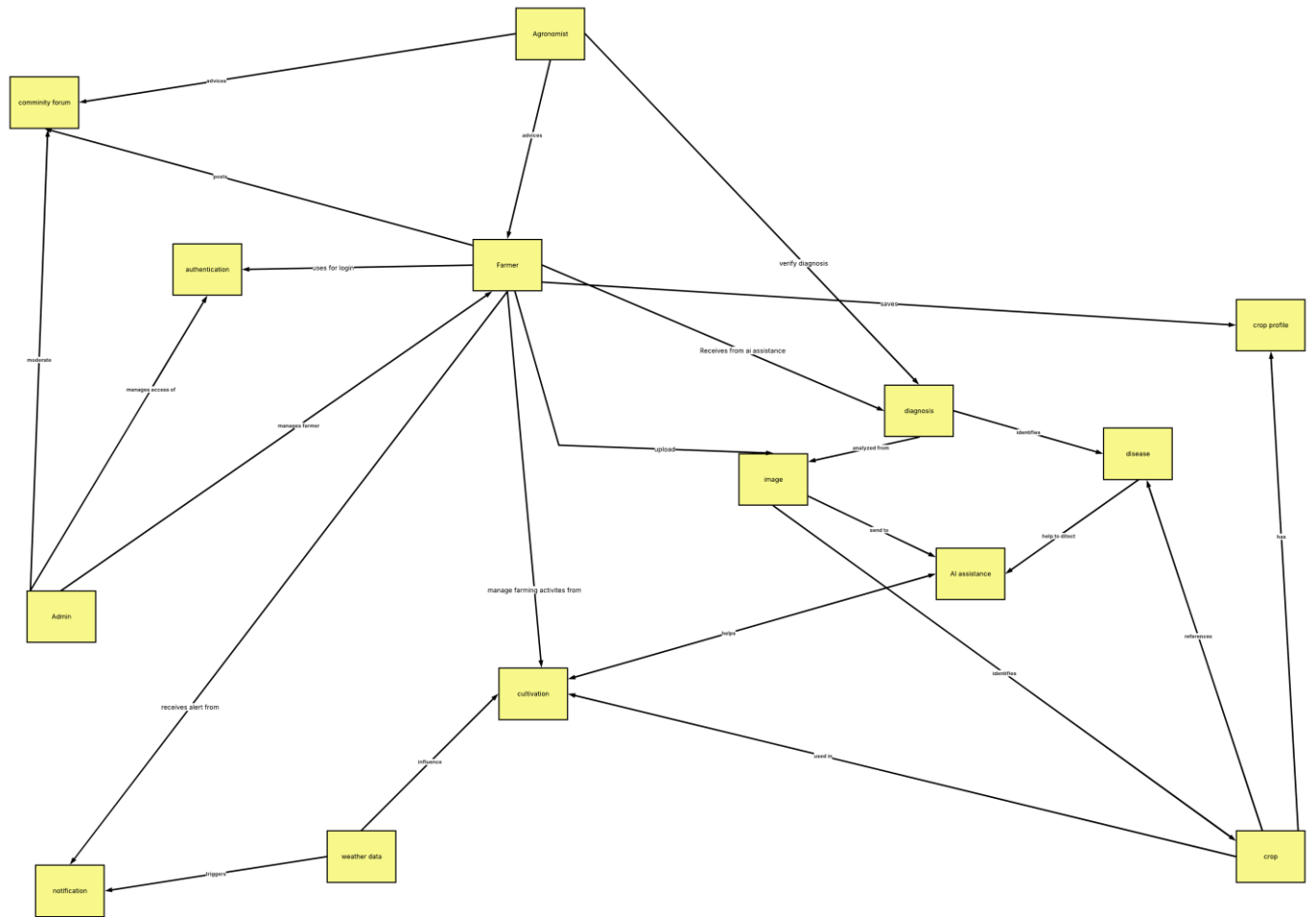


Fig : CRC Diagram For all classes

Sequence Diagram

